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A FINAL YEAR PROJECT REPORT

ON

"OPTIMAL LOCATION AND SIZING OF BATTERY ENERGY STORAGE SYSTEM IN DISTRIBUTION FEEDER FOR POWER QUALITY IMPROVEMENT"

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A project progress report submitted to the department of Electrical Engineering in the partial fulfillment of the requirements for degree of Bachelor of Engineering in Electrical Engineering

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ABSTRACT

This project presents a comprehensive study on the optimal location and sizing of Battery Energy Storage Systems within a distribution feeder to effectively mitigate power quality issues such as voltage fluctuations and high energy losses, which are further intensified by the increasing penetration of intermittent renewable energy sources. The analysis is carried out using the IEEE 33-bus radial distribution system as a standard benchmark network. Initially, conventional manual "hit and trial" approaches were considered; however, these methods were found to be inefficient and time-consuming for identifying optimal solutions. Therefore, the study adopts a more advanced and reliable optimization technique using Particle Swarm Optimization (PSO), implemented through Python and integrated with OpenDSS for accurate power flow analysis. The proposed methodology evaluates five different scenarios, ranging from a base case without any energy storage to configurations incorporating up to four optimally placed and sized BESS units. Each scenario is analysed in detail to assess its impact on key performance indicators such as voltage profile, power losses, and overall system efficiency under varying daily load conditions.

The system's operational logic is centered on a peak-shaving strategy: the BESS charges during off-peak periods (typically hours 0–7) and discharges during peak demand (hours 15–21) to offset grid stress. In the base case, the system exhibits a total power loss of 5035.29 kWh/day and a minimum bus voltage of 0.9004 pu, indicating poor voltage stability and higher losses. Upon integrating BESS, significant improvements are observed. With a single BESS placed at bus 16, losses are reduced by 17.77% and voltage profile improves. Further enhancement is achieved with two BESS units placed at buses 32 and 16, resulting in a maximum loss reduction of 36.41% and improved voltage stability. Although the three BESS configurations (at buses 16, 31, and 33) provide the highest loss reduction (36.46%) and best voltage profile (Vmin ≈ 0.946 pu), it comes with a higher payback period (1.94 years). The four BESS case shows slightly reduced performance compared to the three BESS configuration, indicating diminishing returns with increased installations.

Economic analysis reveals that the single BESS configuration has the shortest payback period (0.98 years), making it the most cost-effective solution, while the two BESS configurations offer the best trade-off between technical performance and economic feasibility.

Ultimately, the study concludes that PSO-based BESS optimization is a reliable, scalable, and effective solution for improving the reliability, efficiency, and stability of modern smart grids. By integrating PSO with power flow analysis, the method effectively identifies the most suitable bus location and optimal BESS capacity. The results demonstrate significant improvements in voltage profile and reduction in energy losses compared to conventional approaches. Due to its simplicity, fast convergence, and ability to handle nonlinear constraints, PSO proves to be a reliable and robust optimization tool for enhancing power quality and operational efficiency in modern distribution systems.

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LIST OF ABBREVIATIONS

AC	Alternating Current
BCBV	Branch Current to Bus Voltage
BESS	Battery Energy Storage System
BIBC	Bus Injection to Branch Current
CNRS	National Centre for Scientific Research
COA	Crayfish Optimization Algorithm
CSIRO	Commonwealth Scientific and Industrial Research Organization
DC	Direct Current
DG	Distributed Generation
DLF	Distribution Load Flow
DOD	Depth of Discharge
GA	Genetic Algorithms
KW	Kilowatt
KWh	Kilowatt-hour
MILP	Mixed Integer Linear Programming
MW	Megawatt
MWh	Megawatt-hour
NEA	Nepal Electricity Authority

OpenDSS	Open Distribution System Simulator
PCS	Power Conversion System
PSO	Particle Swarm Optimization
PVDG	Photovoltaic Distributed Generation
SCADA	Supervisory Control and Data Acquisition
SOC	State of Charge
SOCP	Second Order Cone Programming
SOH	State of Health
SSA	Salp Swarm Optimization
WOA	Whale Optimization Algorithm
ZEBRA	Zero Emission Battery Research Activity

CHAPTER ONE: INTRODUCTION

1.1 Background

During the day, peak hourly load profile variations will possess variations in load burden at grid side. To meet such high demand there is always a risk for power system instability during peak hours of the day. From the grid stability point of view, the utility generally installed extra energy sources such as diesel generators, pump storage power plants etc. to meet such high peak demand variation for only a few hours of day [1]. Renewable energy power generation has become an important part for China's power supply. By June 2016, the grid's wind-power capacity had been 124 GW, and the photovoltaic (PV) capacity had been 61 GW. The rapid development and implementation of renewable power generation pose great challenges to the operation, control, and security of the Chinese power grid. In China, there are several large-scale BESS demonstration projects currently underway. For example, in Zhangbei, a large-scale BESS, which includes a 14 MW/63 MWh lithium-ion BESS and a 2 MW/8 MWh vanadium redox flow BESS, has been put into operation (flow BESS is still in the site commissioning stage). Further, Guodian Longyuan Woniushi windfarm energy storage power station, using total vanadium flow batteries (5 MW/10 MWh), is adopted mainly to resolve wind-curtailment and brownout issues arising at the Woniushi wind farm. The BESS of Southern Power Grid Shenzhen Baoqing adopted the lithium-ion battery (planned capacity is 10 MW and completed capacity is 4 MW/16 MWh) to achieve peak load shaving, frequency regulation, and voltage regulation, etc. [2].

Battery Energy Storage Systems (BESS) have become an important solution in distribution feeders to tackle growing power quality issues caused by increased load variability, renewable energy integration, and sensitive electronic devices. Traditionally, distribution networks were set up for one-way power flow with limited control over voltage and frequency. However, the rapid rise of nonlinear loads, electric vehicles, rooftop solar PV, and wind energy has resulted in problems like voltage fluctuations, voltage sags and swells, flicker, harmonics, and poor power factors. These issues have lowered the reliability and quality of power supplied to users, making conventional equipment such as capacitor banks and voltage regulators inadequate in many situations.

In this context, BESS offers a fast, flexible, and controllable source of active and reactive power when placed at key points along the distribution feeder. With power electronic interfaces, BESS can respond in milliseconds to sudden disturbances, either injecting or absorbing power to maintain voltage stability and reduce power quality issues. This ability makes BESS particularly suitable for modern distribution feeders with high renewable penetration, where power output is often intermittent and unpredictable. As distribution systems move towards smart grids, BESS has become a key technology for improving power quality while also boosting feeder reliability and operational efficiency. By enhancing voltage profiles, cutting losses, and protecting sensitive loads, BESS helps create a more stable and resilient distribution network. Therefore, the development of BESS in distribution feeders is closely linked to the rising demand for high-quality power and the shift towards cleaner, decentralized energy systems.

Power utility companies are being compelled to upgrade their existing power systems due to the rising demand for load by residential, commercial, and industrial customers on radial distribution networks. However, the cost of implementing these system upgrades may not be justifiable considering the short duration of peak loads, resulting in a lower utility of the affected distribution assets. The use of a distribution-level battery energy storage system (BESS) is an advanced solution to tackle this challenge of managing electricity demand. Charging a BESS during off-peak periods and discharging it during peak periods can decrease the peak demand on the power grid. For example, since July 2021, the Moss Landing Energy Storage Facility has been operational in California, USA, serving as a large BESS with a 400 MW/1,600 MWh capacity [3].

Battery Energy Storage System (BESS) has proven its strong potential in maintaining the stability of electrical energy in the power grid with its primary ancillary service function of frequency regulation, voltage support, and active and reactive power control. The rapid charging and discharging function of BESS mitigates well the variability of electrical energy from renewable energy sources and electrical vehicle charging. Although there has been major progress made in improving BESS location and sizing optimization and cost-effectiveness, currently available solutions are still based mostly on static optimization methods that do not possess flexibility and dynamic adaptation capabilities concerning real-time changes in electrical demand and supply. This fact underlines another area of

research focus and interest in developing dynamic and hybrid optimization models, which integrate various models and methods such as analytical, mathematical, probabilistic, artificial models of neural networks, and meta-heuristics. Implementing many energy storage devices such as energy storage batteries in new and future distributed networks of electrical energy faces various challenges, such as uncertainty of electrical energy demand and supply variability from renewable energy sources and electrical vehicle charging demand, and variability of electrical spot prices. To tackle such concerns, strategic models of energy storage batteries and BESS have been developed with enhanced electric power quality and support services, and secure operation in new and future deep digitalized electrical power networks. Moreover, an increased number of electrical smart devices in such new and future electrical energy networks enlarge vulnerability threats from electrical smart devices and other threats such as data falsification in potential cascaded failure events. To address such concerns and threats, new and secure models of electrical energy storage devices such as energy storage batteries and smart devices have also been developed.

1.2 BESS Topology

BESS topology refers to the structural arrangement and electrical configuration through which the battery is integrated with power systems, converters, and the grid. It defines how energy flows between the battery, renewable energy sources, and loads, as well as how power conversion and control are managed. The selection of an appropriate BESS topology plays a crucial role in determining system efficiency, flexibility, cost, and overall performance. Common topologies of BESS are explained below:

A. DC-Coupled BESS Topology

In a DC-coupled topology, the battery energy storage system is connected directly to the DC bus, typically alongside renewable sources such as solar PV. Both the battery and the generation source share a common inverter to interface with the grid. This configuration reduces the number of power conversion stages, resulting in higher efficiency and lower energy losses. It is especially suitable for new installations where solar and storage are integrated from the beginning. However, it offers less flexibility for retrofitting into existing systems and requires more complex control strategies.

B. AC-Coupled BESS Topology

In an AC-coupled topology, the battery is connected to the AC grid through its own dedicated inverter, separate from other generation sources. This allows the battery system to operate independently and makes it easier to integrate into existing grid-connected systems. Although this topology involves multiple conversion stages (AC–DC–AC), which increases energy losses slightly, it provides greater flexibility and modularity. It is widely used in grid applications and for retrofitting storage into already deployed renewable energy systems.

C. Hybrid Topology

The hybrid topology combines both AC and DC coupling methods, allowing the battery to interact with both the DC bus and the AC grid. This setup uses multiple converters and provides maximum operational flexibility, enabling efficient power flow management between renewable sources, storage, and the grid. It is particularly useful in complex energy systems where multiple energy sources and loads are involved. However, the system design is more complex and costly due to the additional components and control requirements.

D. Centralized BESS Topology

In a centralized topology, a large-scale battery energy storage system is installed at a single location, such as a substation or a central node in the network. This setup simplifies system monitoring, control, and maintenance while providing bulk energy storage and grid support. It is commonly used in utility-scale applications for load leveling, frequency regulation, and renewable energy integration. However, it may not effectively address localized issues such as voltage drops or losses in distant parts of the distribution network.

E. Distributed BESS Topology

In a distributed topology, multiple smaller battery systems are installed at different locations across the distribution network, such as various buses in a feeder system. This approach enhances local voltage support, reduces line losses, and improves overall power quality more effectively than a centralized system. It is particularly suitable for applications like the IEEE 33 bus system, where optimization techniques are used to determine the best

locations and sizes of BESS units. However, it involves higher installation and coordination complexity due to the presence of multiple units.

1.2.1 Interfacing with distribution system

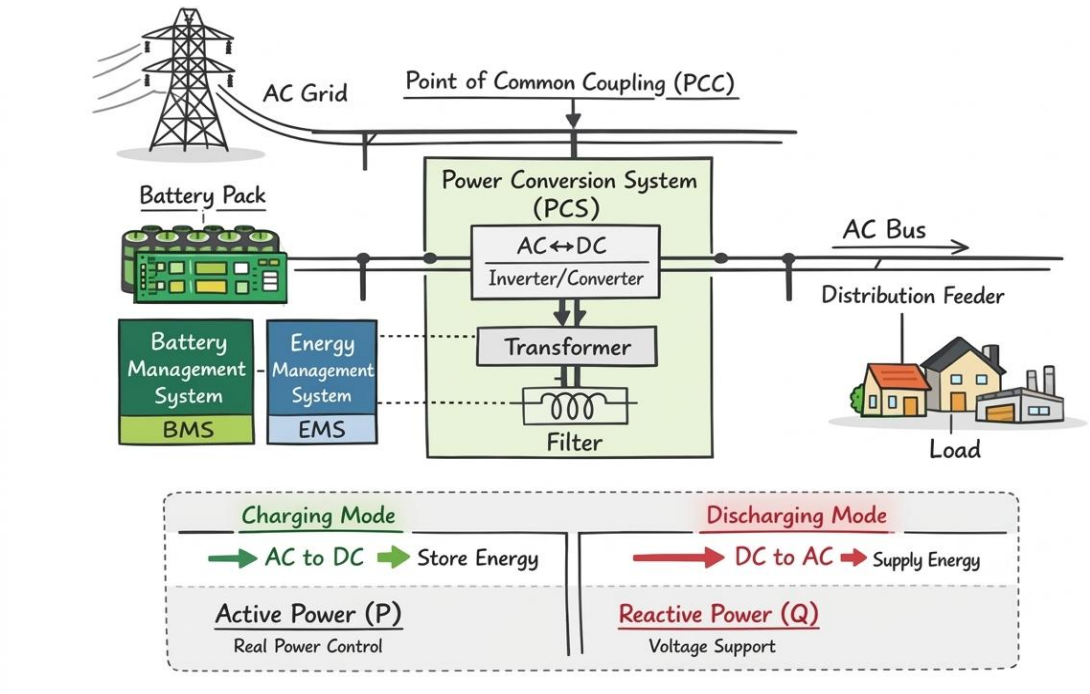


Figure 1: Interfacing of BESS with distribution system

A Battery Energy Storage System is interfaced with the distribution system at a Point of Common Coupling (PCC), typically through a power electronic converter (inverter) and a coupling transformer. The battery operates on DC power, while the distribution network is AC; therefore, an inverter is required to convert DC to AC during discharge and AC to DC during charging. Depending on the topology (AC or DC coupled), the BESS can be connected either directly to the AC bus or through a shared DC link. The transformer ensures proper voltage matching and isolation, while filters are used to reduce harmonics. This interface allows the BESS to exchange power with the grid for applications such as voltage support, peak shaving, and loss minimization. A typical BESS for a distribution network includes:

a. Battery Modules

Energy stored in DC form. It can be arranged as centralized or modular.

b. Inverter/ Power conversion system

It converts DC to AC for the grid and AC to DC for charging. It is used to controls voltage, current and power flow.

c. Transformer

It is used to steps voltage up or down to match the distribution system.

d. Protective Devices

To prevent faults we used circuit breakers, fuses and isolators.

e. Control System

Control system manages charging/discharging, monitoring grid stability. It also implements strategies like peak shaving, load levelling and voltage support.

1.2.2 Power Flow Control in BESS

Power flow in a Battery Energy Storage System is controlled through a bidirectional power conversion system that enables energy exchange between the battery and the distribution network. During charging mode, excess AC power from the grid is converted into DC by the inverter and stored in the battery, while during discharging mode, the stored DC energy is converted back into AC and supplied to the load or grid. The Energy Management System (EMS) determines the timing and magnitude of charging or discharging based on system conditions, while the Battery Management System (BMS) ensures safe operation by monitoring battery parameters. Additionally, the inverter regulates both active power (P) for energy balance and reactive power (Q) for voltage support, thereby improving overall system efficiency, stability, and power quality. [4]

1.3 Problem Statements

Electrical distribution systems play a vital role in delivering power from substations to end users, but they face several techniques and operational challenges. One of the primary problems is voltage fluctuations or drop, particularly in long radial feeders, where voltage drops occur at distant nodes, resulting in poor power quality. Another major issue is high power losses due to line resistance and uneven load distribution, which reduces overall

system efficiency. These problems negatively impact both utilities and consumers. Voltage instability can damage electrical equipment, while high losses increase energy wastage and operational costs.

In modern electrical distribution systems, Battery Energy Storage Systems (BESS) have emerged as an important solution to enhance performance, improve reliability, and manage power quality. However, the benefits of BESS are significantly affected by how and where they are placed in the network. When BESS is placed in a poor location or without proper planning, it can result in low system performance and increased operational and investment costs. This is because the battery may not be able to support the areas where it is needed most, leading to inefficient usage.

The overall effectiveness of a BESS depends heavily on choosing the right location and size. If the battery is too small, it may not be able to handle the energy requirements of the system. On the other hand, if it is too large, it can lead to unnecessary expenses. Similarly, placing it in the wrong part of the distribution system will limit its ability to reduce power losses or regulate voltage effectively. Therefore, for BESS to be truly beneficial, both its size and position must be carefully optimized.

Another issue in many distribution networks is the presence of long feeder lines, which are used to transmit power from substations to distant consumers. These long lines often result in high power losses, especially during peak demand periods. A properly placed BESS can help minimize these losses by storing energy closer to the load centers and releasing it when needed. But again, this is only possible if the BESS is correctly planned and integrated into the network.

1.4 Objectives and scope of the Study

The main objective of this project is to improve power quality in the distribution feeder by optimally locating Battery Energy Storage Systems (BESS) for enhanced voltage regulation, reduced power losses, and increased system efficiency. To achieve the aim, a few sub-objectives have been defined, which are listed below.

- To model IEEE-33 bus distribution feeder using OpenDSS and Python

- To implement PSO optimization algorithm for optimal sizing and location of BESS

1.5 Limitation of Study

This study shows that BESS can improve power quality and reduce system losses, but it has several limitations.

1. The study is based on a standard test system, IEEE-33 bus system, a small radial network that does not reflect the complexity of larger, looped grids with bidirectional power flow.

2. Load conditions are assumed to be static or predefined, whereas actual systems have dynamic and time varying loads.

3. The optimization is only a fixed 24-hour cycle, ignoring the unpredictable nature of renewable generation and season-based demand charges.

4. The study does not extensively account for battery degradation, aging, and lifecycle costs, which are critical for practical implementation.

5. The BESS is often assumed to operate under ideal conditions, ignoring practical issues such as energy losses, temperature effects, and control delays.

6. The results are based on simulation studies and may not directly reflect real-time system performance due to communication delays and control system limitations.

1.6 Report Organization

Chapter 1: This chapter contains the introduction, problem statement, objectives, scope of study and limitations of the study.

Chapter 2: This chapter gives an overview of the different types of batteries and their life cycle, energy density, charging/discharging efficiency, self-discharge rate.

Chapter 3: This chapter provides methodology used to fulfill the objectives of the study. It contains information related to the software used in implementation of the Battery Energy Storage System and the also information about the used algorithm particle swarm optimization.

Chapter 4: This chapter contains results obtained from the simulation against different abnormal conditions and its waveform. This also includes results of Battery energy storage system in different buses.

Chapter 5: Conclusions from research work are summarized in this chapter.

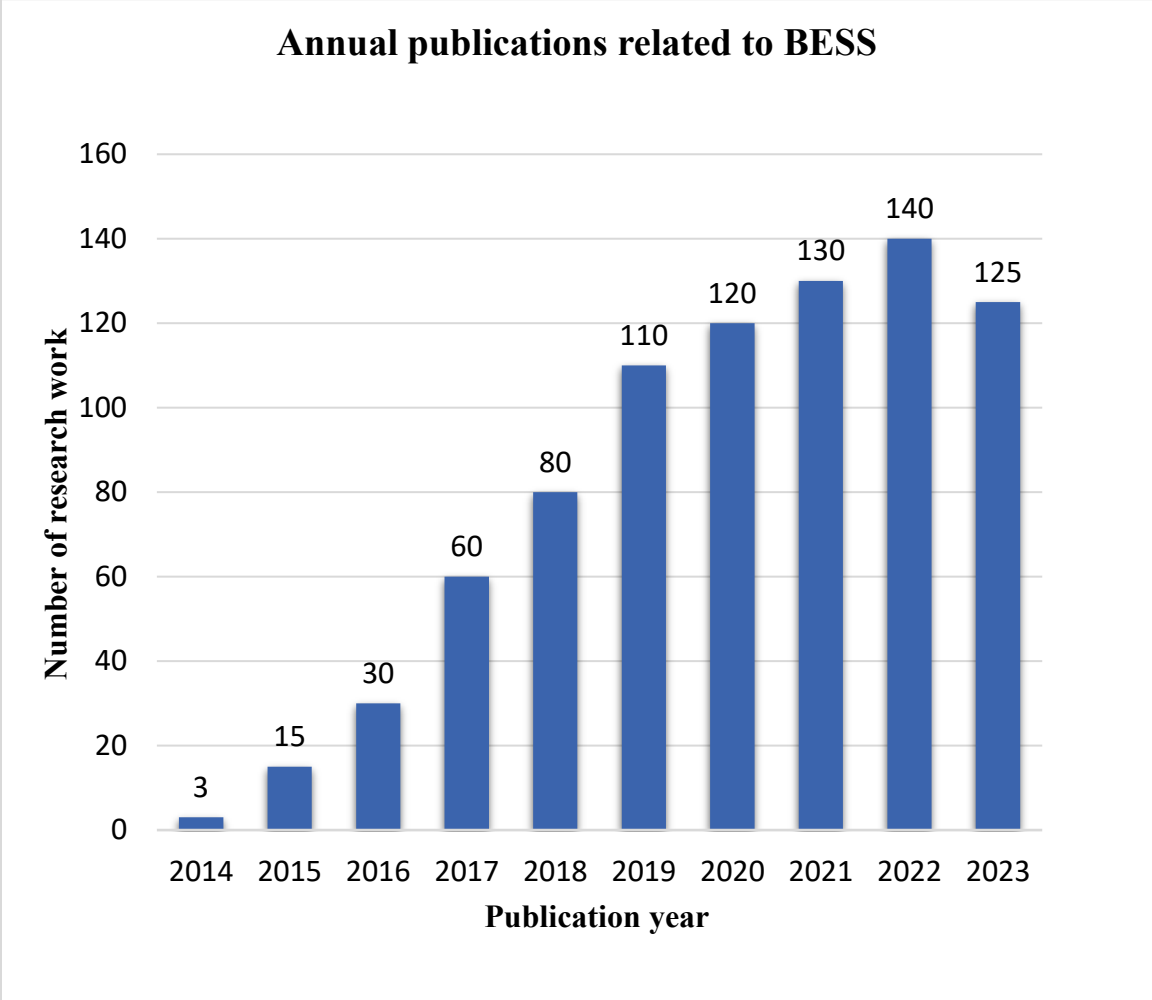


Figure 2: Year-wise distribution of published research works on Battery Energy Storage Systems (2014–2023)

(Source: IEEE data portal)

Figure 2 illustrates the year-wise trend of published research works related to Battery Energy Storage Systems (BESS) from 2014 to 2023. The figure shows a consistent growth in research interest in BESS, with a gradual increase from 2014 and a peak around 2022. This trend highlights the growing importance of BESS in modern power systems due to increased renewable energy integration and power quality concerns.

2.1 General Literature Review

The research paper titled on “Optimal Allocation of Battery Energy Storage Systems for Peak Shaving and Reliability Enhancement in Distribution Systems” is presented in [5]. This paper focus on the simultaneous sizing and siting of battery energy storage systems (BESS) is commonly formulated as a mixed-integer nonlinear programming (MINLP) problem, which is computationally challenging due to the presence of discrete variables and limited scalability. To address this complexity, some studies adopt heuristic methods, such as voltage sensitivity analysis, to identify suitable BESS locations by using network-based indices to assess voltage volatility and mitigate voltage violations. However, due to limitations in accuracy and optimality associated with heuristic approaches, subsequent research proposes mixed-integer linear programming (MILP) formulations that enable more efficient and reliable optimal sizing and placement of BESS, particularly for improving system reliability and supporting energy arbitrage.

The research paper titled on “Optimal control and management of a large-scale battery energy storage system to mitigate fluctuation and intermittence of renewable generations” is presented in [2]. This paper focuses on the SCADA system is the central component of a battery energy storage system power station, enabling centralized access, real-time control, and operation scheduling. It communicates with key BESS equipment, such as local control and monitoring systems and power conversion systems (PCS), to exchange information, perform real-time monitoring, and manage control functions. Large-scale BESS can operate flexibly as either a power source or a load, and unlike traditional power generation systems, they can rapidly inject or absorb both active and reactive power to support grid operation.

The research paper titled on “Optimal Location and Sizing of BESS for Performance Improvement of Distribution Systems with High DG Penetration” is presented in [6]. This paper investigates the optimal sizing and placement of battery energy storage systems (BESS) in distribution networks with high penetration of distributed generation (DG), such as photovoltaic and wind turbine units, to minimize system costs and enhance overall performance. Daily variations in load demand and DG output are considered, and three optimization algorithms genetic algorithm (GA), particle swarm optimization (PSO), and

the slap (small, transparent, jellyfish like sea animal) swarm algorithm (SSA) are applied and compared. The IEEE 33- and 69-bus distribution systems are used as test cases to evaluate cost reduction and performance improvement after BESS installation. The main contributions include minimizing system costs related to power losses, voltage deviation, and peak power, as well as improving voltage profiles and reducing losses and peak demand. In addition, the study emphasizes the importance of battery operating parameters, such as state of charge (SOC) and state of health (SOH), in optimizing BESS operation to extend battery lifetime and enhance overall system effectiveness.

The research paper titled on "Optimal Placement and Sizing of Battery Energy Storage System Considering the Duck Curve Phenomenon" is presented in [7]. This study proposes a novel method for optimal sizing and placement of battery energy storage systems (BESS) that explicitly incorporates duck curve mitigation into the optimization process, enabling simultaneous reduction of total system losses and alleviation of steep ramping and over-generation issues. The duck curve is model based on Malaysia's photovoltaic distributed generation (PVDG) and load patterns, demonstrating the effectiveness of BESS in addressing these challenges through optimal deployment. The analysis is conducted in a dynamic time-domain framework using hourly data over a 28-day period, making the results more operationally relevant than prior studies. Additionally, the proposed approach is validated through performance comparisons with other well-known optimization algorithms.

The research paper titled on "Optimal Sizing and Placement of BESS in Distribution Grid with High PV Penetration Considering BESS Optimal Operation" is presented in [8]. This paper presents a new strategy for voltage regulation in distribution systems with high penetration of renewable energy sources by optimally sizing and locating battery energy storage systems (BESS). The approach aims to prevent overvoltage by determining the minimum required BESS capacity and its optimal placement based on the size and location of integrated solar generation. The problem is formulated using grid sensitivity coefficient analysis combined with second-order cone programming (SOCP) for power flow modelling. Instead of relying on reactive power control, which may be ineffective in networks with high R/X ratios, the method focuses on regulating voltage through optimal management of active power, including solar power curtailment via BESS. Simulation

results on a modified IEEE 33-bus system demonstrate the effectiveness of the proposed strategy.

A brief review on “Battery Energy Storage System Allocation in the IEEE 33 Bus Test System for enhanced system performance” is presented on [9]. Battery Energy Storage Systems (BESS) work by storing electrical energy in batteries during periods of surplus or low demand and releasing it when demand is high or when additional power is needed. The primary components of a BESS include batteries, power, conversion systems, and control systems. Here’s an overview of how BESS typically works.

1. Battery Components: Battery Cells:

BESS usually consists of multiple battery cells connected in series or parallel. These cells can be based on various chemistries such as lithium-ion, lead-acid, or flow batteries, depending on the application and requirements.

2. Charging Process:

During periods of low demand or when excess renewable energy is available, the BESS charges its batteries. This process involves converting electrical energy from the grid or renewable sources into chemical energy stored in the batteries.

3. Energy Storage:

The charged batteries store electrical energy in the form of chemical potential. The amount of energy stored depends on the battery capacity, which is measured in kilowatt-hours (kWh) or megawatt-hours (MWh).

4. Discharging Process:

When demand for electricity increases or during peak hours, the BESS discharges stored energy. The power conversion system converts the chemical energy back into electrical energy, making it available for use in the electrical grid.

5. Power Conversion System:

BESS includes power electronics that manage the conversion of electrical energy between the grid, the batteries, and the load. This system ensures efficient and controlled energy transfer in both charging and discharging modes.

6. Inverters and Converters:

Inverters convert direct current (DC) from the batteries into alternating current (AC) for use in the grid or loads. Converters may also be used to match voltage levels and frequencies between the BESS and the electrical system.

7. Control Systems:

Sophisticated control systems manage the operation of the BESS. These systems monitor grid conditions, battery status, and demand patterns. Advanced algorithms optimize charging and discharging cycles to maximize the efficiency and lifespan of the batteries.

8. Grid Integration:

BESS can be integrated into the electrical grid at various points, such as distribution substations or directly connected to renewable energy sources. Grid operators or control systems can dispatch stored energy based on real- time needs, contributing to grid stability and reliability.

9. Frequency Regulation and Ancillary Services:

BESS can provide rapid response to fluctuations in the grid frequency. By adjusting the charging or discharging rate, they contribute to frequency regulation, enhancing grid stability. Additionally, BESS may participate in providing ancillary services, supporting grid reliability.

10. Applications:

BESS finds applications in various sectors, including renewable energy integration, grid stabilization, peak shaving, load balancing, and backup power. They are crucial for supporting the integration of intermittent renewable energy sources, such as solar and wind, into the grid.

By efficiently managing the storage and release of electrical energy, BESS contributes to a more flexible, resilient, and sustainable electrical grid. The control systems play a crucial role in optimizing the performance of BESS and ensuring seamless integration with the broader power infrastructure.

The research paper based on "A Comprehensive Review of the Integration of Battery Energy Storage Systems Into Distribution Networks" is presented in [10] gives clear ideas about different types of batteries with theirs charging efficiency, discharging efficiency, life span, DOD, nominal voltage as well as self-discharge rate.

The research paper based on "Optimal Placement and Capacity of Battery Energy Storage System in Distribution Networks Integrated with PV and EVs Using Metaheuristic Algorithms" is presented on [11].This study shows the lifespan of lithium-ion batteries in BESS depends on temperature, cycling frequency, and depth of discharge, with an 80% DOD commonly recommended to prolong battery life.

The research paper based on "Modeling of a Battery Storage System on a Distribution Feeder" is presented on [12]. This shows that during summer, high cooling demand increases peak loads and makes power systems more vulnerable to fluctuations, while renewables may be less reliable. Battery energy storage systems (BESS) can rapidly supply power to stabilize the grid and reduce customer interruptions.

The research paper based on "Review on the optimal placement, sizing and control of an energy storage system in the distribution network" is presented on [13]. This explains BESS stores energy in electrochemical form, converting chemical energy into electrical energy during discharging and reversing the process during charging. Its main components include batteries, control systems, and power conversion systems (PCS). BESS is a highly valuable technology for distribution networks due to its wide range of applications such as power system regulation, protection, spinning reserve, and power factor correction.

The research paper based on "Applications of Battery Storage to Improve Performance of Distribution Systems" is presented on [14]. This illustrates BESS improve power quality by providing fast, high-power responses for a few seconds, helping in flicker reduction, voltage sag correction, and harmonic compensation. For energy management, BESS are designed for longer discharge durations (hours) with high energy capacity, supporting

applications such as load leveling, peak shaving, energy trading, renewable integration, and islanded operation.

2.2 Types of Batteries

A battery is an electrochemical device that stores energy in chemical form and converts it into electrical energy when required. Batteries are widely used in portable electronics, electric vehicles, renewable energy systems, and power backup applications. They play a crucial role in modern energy systems by enabling energy storage, load balancing, power quality improvement, and reliability enhancement. For battery storage applications a variety of types are available, including lead acid, Lithium-ion (Li-ion), Sodium Sulphur (Na-S), Nickel Cadmium (Ni-Cd),Flow battery, Nickel Metal Hydride (Ni-MH), Sodium Nickel Chloride (NaNiCl2), and the ultra-battery (also known as the advanced lead acid battery), each variant possesses unique features and applications.

2.2.1 Lead acid battery

A French physicist, Gaston Plante, invented the lead acid battery in the 1850s. These batteries were the first rechargeable battery technology and are now considered one of the most mature technologies existing. They are the first batteries used for energy storage applications. It exhibits many specific attributes like robust operation, easy control, low cost, easy maintenance, and less response time. These specifications make it suitable for stationary energy storage applications. A 12-volt lead–acid car battery typically delivers 35–40Wh/kg of specific energy and 80–90Wh /L of energy density, with a specific power of about 180 W/kg. Its charge/discharge efficiency ranges from 50–95%**,** and it experiences 3–20% self-discharge per month. The battery generally lasts for fewer than 350 cycles. It provides 7–18Wh per US dollar depending on the type. Each cell has a nominal voltage of 2.1 V**,** and the battery can be charged within a temperature range of –35 °C to 45 °C**.** [3]

2.2.2 Lithium ion (li-ion) battery

The first commercialization of lithium-ion batteries was done in the year 1991. These batteries possess several properties, such as high round-trip efficiency, high energy and power density, and long life. Due to their high energy power density properties, these

batteries have transformed the energy storage and EV industry. It exhibits high specific energy density compared to other existing battery technologies, making it lighter and suitable to be used for mobile energy storage applications. However, safety is the main concern for these types of batteries. Many cases have occurred over the period regarding its flammability problem. The reserves of lithium in the earth's crust are limited, resulting in a difficult and costly extraction process compared to other battery technologies. To further enhance its energy density and to mitigate its flammability apprehension, many types of lithium-ion batteries are developed like lithium-ion phosphate (LFP), lithium manganese oxide (LMO), Nickel Manganese Cobalt (NMC), and lithium titanate oxide (LTO). The battery exhibits a charge/discharge efficiency of 80–90% and supports 400–1,200 charge cycles, indicating moderate to high durability. Its energy density ranges from 250 to 693Wh/L, while the specific energy varies widely from 1 to 270Wh/kg, reflecting different chemistries or configurations. The battery can deliver specific power between 1 and 10,000 W/kg, showing the potential for both low- and extremely high-power applications. Cost wise, it provides about 8.7Wh per US dollar. The self-discharge rate is low, ranging from 0.35% to 2.5% per month, depending on its state of charge. [3]

2.2.3 Sodium sulphur (Na-S) battery

Na-S battery was invented in the year 1996 by Ford Company, but it was first commercialized by Tokyo Power Corporation and NGK Insulators in the 1990s. They are also known as high-temperature batteries. From an energy storage point of view, these batteries play a very crucial role as they parade high efficiency, energy density, and high depth of discharge. The only issue associated with these batteries is that they require very high temperatures for their operation, which is difficult to achieve. This increases the overall cost of the system and poses a threat to the atmosphere if not operated properly.

2.2.4 Nickel cadmium (Ni-Cd) battery

The nickel Cadmium battery was invented in the year1899 by Swedish inventor Waldemar Jungner. The battery is constructed by using nickel oxide as the positive terminal and cadmium as the negative electrode, resulting in the creation of are chargeable batteries possessing stable voltage output and high energy density. These batteries offer very similar

properties to lead acid batteries but have a superior depth of discharge and energy density. It is used for various electronic applications like lead acid battery. The primary drawback of these batteries is they are very expensive due to the high cost involved in the manufacturing process. These batteries also suffer from a term called the memory effect, which usually happens when recharging a battery before it is fully discharged. This results in less capacity than it does, and it will be discharged more quickly than anticipated. The battery offers a charge/discharge efficiency of 70–90% and supports around 2,000 cycles of durability. Its energy density is 50–150Wh/L**,** and the nominal cell voltage is 1.2 V**.** It has a self-discharge rate of about 10% per month. The specific energy ranges from 40–60Wh/kg, while the specific power is about 150 W/kg**.** [3]

2.2.5 Sodium nickel chloride battery

NaNiCl2 battery was developed by a group of researchers from the French National Centre for Scientific Research (CNRS) in the 1980s. They are also known as ZEBRA (Zero Emission Battery Research Activity) and are a type of high-temperature battery. It uses two types of electrolytes, namely sodium chloride (NaCl) and nickel chloride (NiCl2). They are recognized because of their high operating temperature, high energy density, long life, and very low environmental impact. Due to these properties, it is applicable to use for electric transportation, stationery grid systems, and energy storage. However, it suffers from low power density, resulting in a longer charging time, limiting its practice. It is high temperature, rechargeable molten salt battery that uses molten sodium as the cathode and nickel chlorides as the anode. Temperature range between 270℃-300℃.

2.2.6 Nickel metal hydride (Ni-MH) battery

Ni-MH batteries were developed by a team of researchers in an institute in Ohio, United States, led by Stanford Ovshinsky in the 1980s. The Ni-MH battery consists of an anode of hydrogen-absorbing alloy (MH), a cathode of nickel hydroxide (Ni (OH)2), and a potassium hydroxide (KOH) electrolyte. These batteries were recently used for many electric and hybrid electric vehicle applications, as they do not have oxidizing properties and provide a safer environment compared to lithium-ion and nickel-cadmium batteries. The main disadvantages of these batteries are their high self-discharging rate, low energy

density, and reduced voltage per cell. The battery has a charge/discharge efficiency of 66–92% and a cycle durability ranging from 180–2,000 cycles. Its energy density is between 140–300Wh/L. The self-discharge rate varies widely: 13.9–70.6% per month at room temperature, 36.4–97.8% per month at 45 °C, while low-self-discharge types fall between 0.08–2.9%. The specific energy is 60–120Wh/kg (0.22–0.43 MJ/kg), and the specific power ranges from 250–1,000 W/kg. [3]

2.2.7 Flow batteries

The concept of flow batteries was proposed by German scientist Friedrich H. Karl in the 1930s, while the first practical battery was developed by a team of NASA scientists in the 1970s. These batteries are constructed by dissolving two chemical components in a liquid separated by a membrane. The main advantages of these batteries are they can be packed in immense volumes for storing a large amount of energy, fast recharging by changing the liquid electrolyte, quick response time, and no harmful emission into the environment. The only limitation of these batteries is the capital investment required due to the high cost of materials used for their construction. Hybrid Flow batteries (HFBs), Redox Flow batteries, and Membrane-less flow batteries are the three types of flow batteries used for energy storage and other renewable applications. The redox flow and hybrid flow batteries are based on the principle of the flow of electrolytes, a process through which energy is stored and released in the battery. It involves the pumping of two different electrolyte solutions in the cell, where they endure different chemical reactions that generate electricity. During the charging process, one electrolyte gets oxidized while the other electrode reduces, storing the electrical energy in the form of chemical energy. During discharge, the process gets reversed. The main difference between the two-flow batteries is the active material used and the chemical reactions they undergo, whereas membrane-less batteries use an ion-selective membrane instead of the traditional membrane used by redox and hybrid flow batteries.

2.2.8 Ultra battery

The Ultra-battery was developed jointly by Commonwealth Scientific and Industrial Research Organization (CSIRO) of Australia and Furukawa Battery Company of Japan. It

is a type of advanced battery that combines the features of a lead-acid battery and a supercapacitor. The two components are connected in parallel, in which the positive terminal of the lead acid battery is made up of lead oxide, while the negative terminal is constructed of porous lead material. During high charging and discharging rates, the supercapacitor can provide high power to limit the current and prevent the lead-acid battery from any damage. Ultra-battery has several advantages, such as a long-life cycle, high efficiency, and a safe operating environment compared to the traditional lead acid battery.

Among all the battery types compared, lithium-ion batteries emerged as the most efficient option. They offer higher energy density, better charge–discharge efficiency, and longer cycle life compared to traditional lead-acid and other common battery technologies. Lithium-ion batteries also have lower self-discharge rates, require less maintenance, and provide more stable performance under varying load conditions. Due to their lightweight design, improved durability, and superior overall efficiency, lithium-ion batteries are considered the most reliable and advanced choice for modern energy storage applications [2].

2.2.9 Comparative Analysis of Different Types of Batteries

Table 1: Comparison of Different Types of Batteries

Battery Type	Typical Energy Density	Life Cycle	Advantages	Disadvantages
Lead-Acid	About 30-50 Wh/kg	500-1000 cycles	low cost, reliable	heavy, shorter life
Lithium-Ion (Li-ion)	About 100-265 Wh/kg	1000-4000 + cycles	high energy density, lightweight	higher cost
Sodium-Sulphur (NaS)	About 150-240 Wh/kg	2500-4500 cycles	high efficiency	high temperature
Nickel Cadmium	About 40-60 Wh/kg	1000-2000 cycles	very durable	high self-discharge
Nickel -Metal Hydride	About 60-120 Wh/kg	500-1000 cycles	improved energy density	sensitive to overcharging
Flow Batteries	About 20-70 Wh/kg	10000 + cycles	long life cycle	low energy density

2.3 Optimization Problem Formulation

Optimization is the process of finding the best possible solution among several feasible solutions while satisfying system constraints. In this study, optimization is used to determine the optimal placement and capacity of the Battery Energy Storage System (BESS) in a distribution network.

The main objective of the optimization problem is to minimize power loss, improve voltage profile, and enhance overall system performance.

The total power loss in the system is given by:

1. Minimize total power loss

Minimize $F = \sum_{t=1}^{T} p_{loss}(t)$

$P_{loss}(t)$ denotes the power loss at a particular time interval t=1, and T represents the total number of time intervals considered in the study.

Constraints related to objectives are:

1. Power balance constraints

$$p_{grid}(t) + p_{dis}(t) - p_{ch}(t) = p_{load}(t)$$

2. State of Charge constraints

$$SOC_{min} \leq SOC\ (t) \leq SOC_{max}$$

3. Charging/Discharging limits

$$0 \leq\ p_{ch}(t) \leq\ p_{ch}^{max}$$

$$0 \leq\ \ p_{dis}(t) \leq\ p_{dis}^{max}$$

4. SOC update equation

$$SOC\ (t+1) = SOC(t) + n_{ch}\ \frac{p_{ch(t)\Delta t}}{E_{bess}}\ -\ \frac{1}{n_{dis}}\ \frac{p_{dis(t)\Delta t}}{E_{bess}}$$

This is achieved by selecting the optimal bus location and appropriate size of BESS while considering system constraints such as voltage limits, power balance, and charging–discharging limits of the battery.

2.3.1 Ways of Solving Optimization Problems

Optimization problems in power systems can be solved using different techniques, broadly classified into classical optimization methods and metaheuristic optimization methods. Classical methods use mathematical models and derivatives to find optimal solutions; however, they are not suitable for complex, nonlinear, and multi-objective problems.

Therefore, metaheuristic algorithms are widely used as they provide better solutions with less computational complexity and are effective for large-scale power system optimization problems.

Example: Solve the linear equation

$2x + 3=11$

First, Identify the equation

$2x + 3=11$

Subtract 3 from both sides.

$2x +3-3 =11-3$

$2x =8$

Divide both sides by 2.

$X =8/2 =4$

Verify the solution

$2 *4 +3 +11$

But the optimization of power systems is non-linear. That's why we need various of algorithms.

2.3.2 Various Algorithms for Solving Optimization Problem

2.3.2.1 Metaheuristic Algorithms

To determine the optimal placement and capacity of BESS in distribution systems connected with PV and EVs, three metaheuristic algorithms including PSO, SSA and COA can also be used. These three algorithms are described in this section.

A. Salp Swarm Algorithm (SSA)

SSA is an optimization algorithm inspired by salps which are types of living organisms like a jellyfish, with a cylindrical transparent body. In the simulation process of SSA, salps normally form a chain which is divided into two groups consisting of leader and followers.

The leader is the salp located in front of the chain while the remaining salps are the followers. The positions of all salps are stored in a two-dimensional matrix called x. To update the leader's position, the following equation is calculated.

$$x_{1j} = \begin{cases} F_j + c_{1((ub_j - lb_j)c_2} + lb_j, c_3 \geq 0 \\ F_j - c_{1((ub_j - lb_j)c_2} + lb_j, c_3 < 0 \end{cases}$$

where x_j^1 j indicates the leader's position in dimension j, F_j represents the position of the food source in dimension j, ub_j is the upper boundary of dimension j,lb_j is the lower boundary of dimension j, and c_2 and c_3 are random numbers between 0 and 1. The variable c_1 can be computed by the equation below.

$$c_1 = 2e^{-\left(\frac{4l}{L}\right)^2}$$

where l represents the current computation iteration, and L denotes the total computation iterations. To update the positions of the follower group, the following equation will be utilized based on the principles of Newton's motion law.

$$x_j^i = \frac{1}{2}\left(x_j^i + x_j^{i-1}\right), i \geq 2$$

Where $i \geq 2$, x_j^i represents the index j of the follower group member i, and x_j^{i-1} j is the index j of the follower group member i-1.

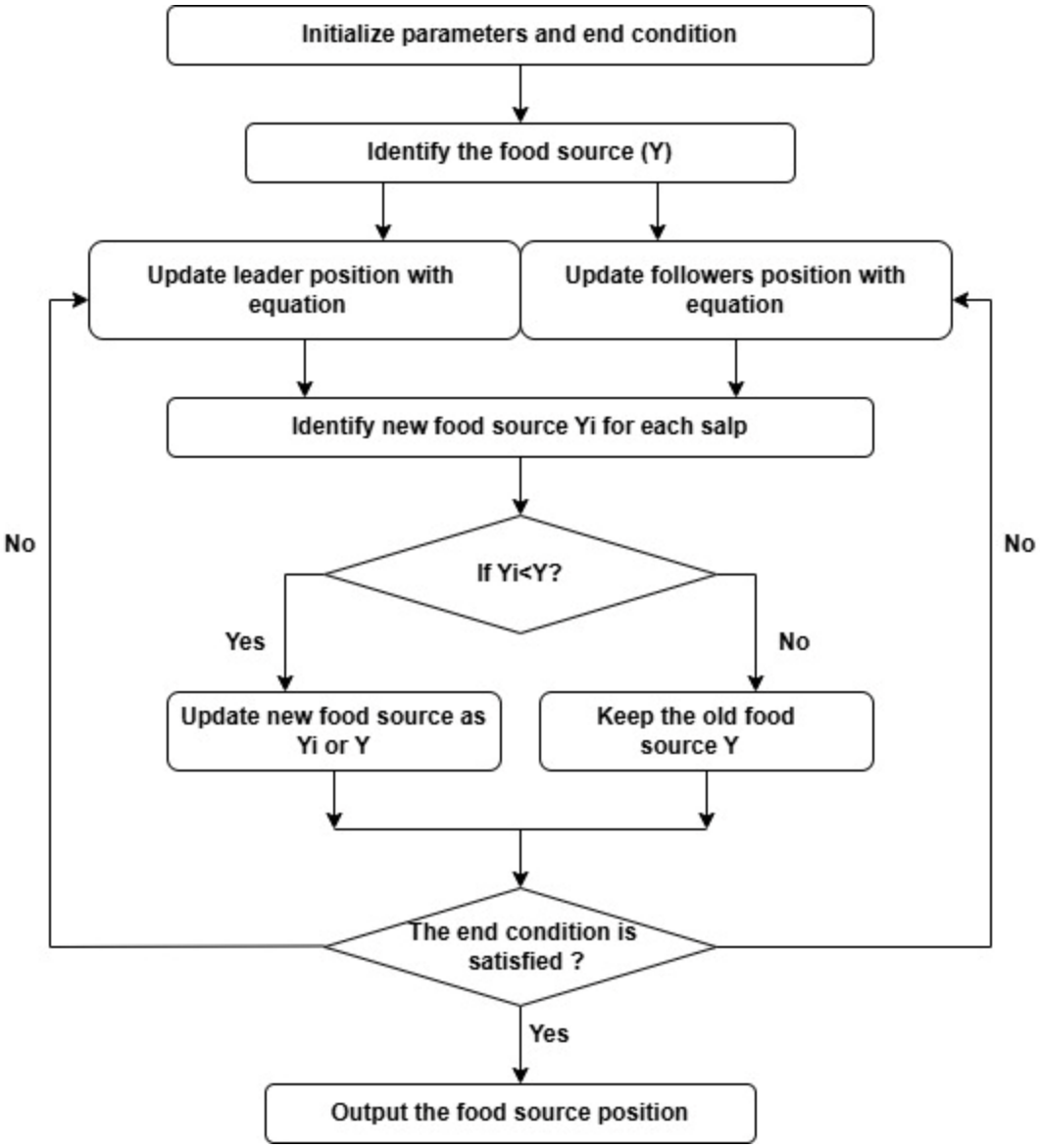


Figure 3: Flowchart of Salp Swarm Algorithm (SSA)

B. Crayfish Optimization Algorithm (COA)

Crayfish Optimization Algorithm (COA) is inspired by the natural behaviours of crayfish, mainly sheltering in caves during high temperatures, competing for habitat, and foraging for food. The algorithm models these behaviours under different temperature conditions to balance exploration and exploitation. Food size represents search step length, guiding how solutions are explored. This process is repeated iteratively until the algorithm converges to the best feasible solution. The simulation begins with each crayfish denoted as X_i rep

resenting the position of crayfish i. The crayfish position is initialized by the following equation.

$$x_{i,j} = lb_j + \left(ub_j - lb_j\right) \times rand$$

where j represents the number of variables of the problem, ub_j is the upper boundary of dimension j, lb_j is the lower boundary of dimension j, and rand represents a random variable between 0 and 1.

The behavior and stage of each crayfish is based on the temperature where the temperature can be calculated by the given equation.

$$\text{temp} = \text{rand} \times 15 + 20$$

Where temp is the temperature of the environment around the crayfish.

When the temperature exceeds 30◦C, crayfish will explore the caves for habitation, waiting for the temperature to become suitable where this phase is the exploration phase of the optimization process. Here, X_{shade} represents the position of the cave, and its value is determined by the equation provided below.

$$X_{shade} = \frac{(X_G + X_L)}{2}$$

where X_G is the value of the best position generated so far, and X_L is the value of the current position.

If the temperature is lower than 30◦C, the crayfish will engage in foraging behavior considered as exploitation phase. Upon finding food, the crayfish will consider the size of the food. If the food is too large, it will divide the food into smaller portions and use its second and third legs alternately to assist in consuming the food. If the food is excessively large, it will adjust its position using equation . However, if the food is at an appropriate size, it will adjust its position using below equation.

$$X_{i,j}^{t+1} = X_{i,j}^{t} + X_{food} p(\cos(2 \times \pi \times rand) - \sin(2 \times \pi \times rand))$$

$$X_{i,j}^{t+1} = \left(X_{i,j}^{t} - X_{food}\right)p + p \times rand \times X_{i,j}^{t}$$

where X_{food} is the position of the food which is calculated by the following equation.

$$X_{food} = \exp\left(-\frac{1}{Q}X_{food}\right)$$

where Q is the size of food, which is a randomly generated value between 0 and 1.

This process iterates repeatedly until obtaining the best solution for everyone. The final step involves determining the best solution overall by evaluating the objective function using the obtained position values.

2.3.2.2 Whale Optimization Algorithm

This algorithm consists of two main elements, namely, exploration and exploitation processes.

A. Exploitation Process

Bubble-net feeding is one behavior of the humpback whales which happens in groups. Hence, this behavior inspired the mathematical modeling of the exploitation process. The exploitation process can be further divided into two parts, which are the shrinking encircling pattern and the spiral position update.

a. Shrinking Encircling Pattern

After detecting the prey, the humpback whales encircle their prey in which each whale is denoted as a search agent. In the current context, however, no prior knowledge of the optimal location (solution) in the search space is available. Hence, the location obtained by the current best search agent will be considered as the target prey. Other search agents will then update their locations according to the location of the current best agent. This pattern can be modelled as:

$$\underset{Y}{\rightarrow} = |\overline{C2}.\overline{X*}\,(t) - \underset{X}{\rightarrow}(t)$$

$$\underset{X}{\rightarrow}(t+1) = \overline{X*}\,(t) - \overline{C1}\,.\underset{Y}{\rightarrow}$$

Where t denotes the current iteration, $\overline{C1}$ and $C2$ are the coefficient vectors, $\overline{X*}$ is the position vector for the current best solution and $\underset{X}{\rightarrow}$is the position vector.

The expressions for vectors $\overline{C1}$ and $\overline{C2}$ are as follows:

$\overline{C1} = 2\underset{c}{\rightarrow} . \overline{rn} - \underset{c}{\rightarrow}$

$\overline{C2} = 2 . \overline{rn}$

where $\overline{rn}$ is a random vector within the interval [0,1], while $\underset{c}{\rightarrow}$ is a vector decreasing linearly from two to zero along with the iteration.

$\overline{C1}$ is a random number within the range [−c, c], where the shrinking encircling pattern is demonstrated through the reduction of the variation for $\overline{C1}$ during the iteration process. When the magnitude of $\overline{C1}$ is less or equal to one, the updated location for the subsequent search agent will fall between the current location of the agent and the location of the current best agent.

b. Spiral Position Update

The spiral position update equation was modelled based on the spiral movement of humpback whales as illustrated in:

$\underset{X}{\rightarrow}(t + 1) = \overline{Y^t} . e^{bl} . \cos(2\pi l) + \overline{X *}(t)$

where $\overline{Y^t}$ represents the distance between the whale and the prey at jth iteration with the formula,

$\overline{Y^t} = |\overline{X *}(t) - \underset{X}{\rightarrow}(t)$

while b denotes a logarithmic spiral shape constant and l is a random number between the interval [−1, 1]. However, during the location update process, the same probability is assumed for the whale to employ either the shrinking encircling pattern or the spiral model.

The overall equation to update the location of the humpback whales in the process of exploitation is presented as:

$\underset{X}{\rightarrow}(t + 1) = \{ \; X * (t) - C1 \cdot Y \qquad if \; p < 0.5$

$Y t \cdot e \; bl \cdot cos(2\pi l) + X * (t) \qquad if \; p \geq 0.5$

B. Exploration Process

The humpback whales search for their prey randomly based on their relative locations from other agents during the exploration process. This process employs the same approach based on the variation of C1 vector. When the value of C1 is randomly allocated to be larger than 1, the search agents are driven to move away from a reference whale. The updated locations are then decided according to a randomly chosen search agent. The global search process deployed by WOA can be illustrated using the mathematical expressions below:

$$\vec{Y} = | \overline{C2} \cdot \overline{X_{rnd}}(t) - \vec{X}(t)$$

$$\vec{X}(t+1) = \overline{X_{rnd}}(t) - \overline{C1} \cdot \vec{Y} \quad [15]$$

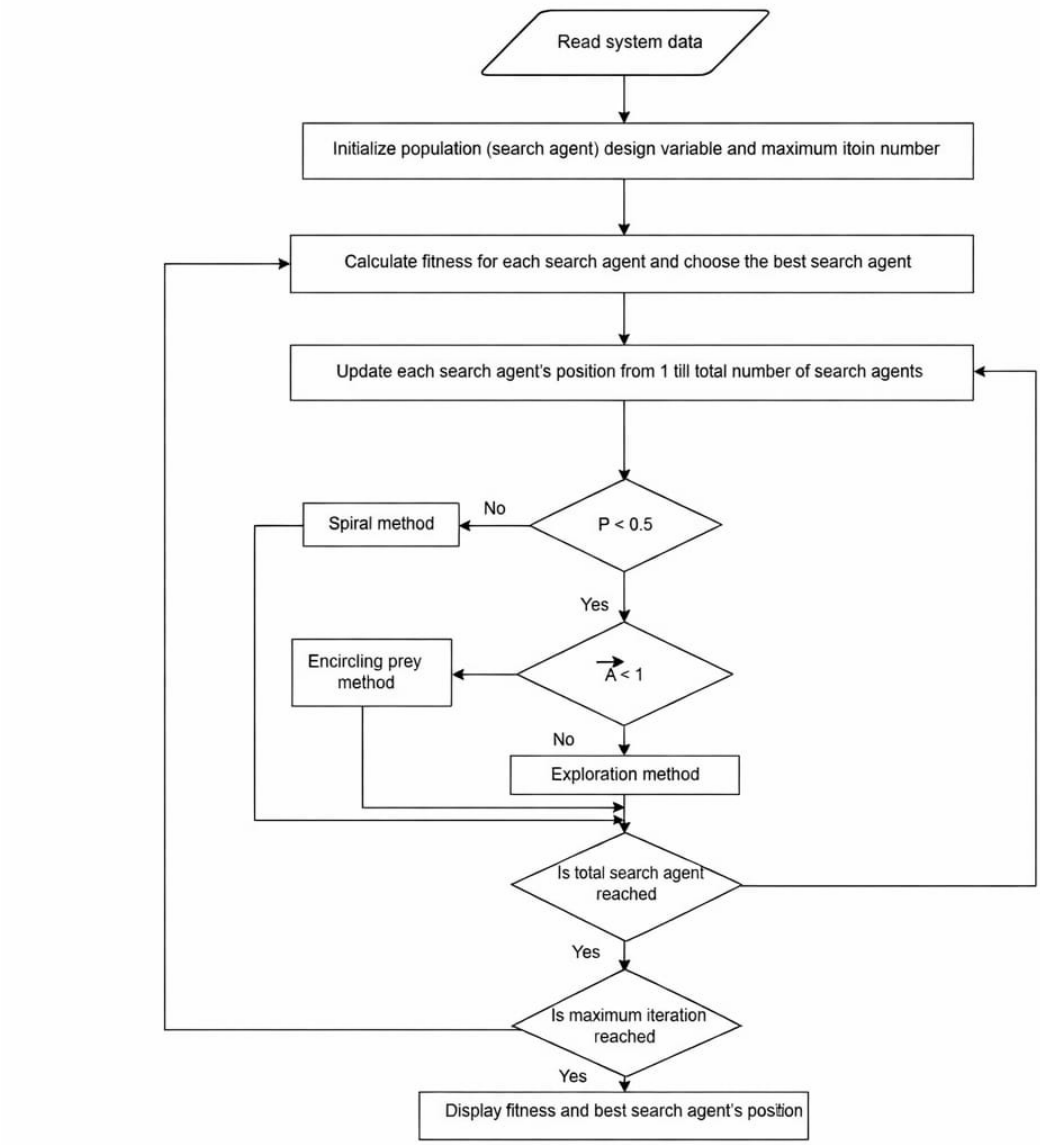


Figure 4: Flowchart of Whale Optimization Algorithm

2.3.2.3 Mixed-Integer Linear Programming (MILP)

Mixed-Integer Linear Programming (MILP) is a powerful mathematical optimization technique used to find the most efficient and cost-effective ways to manage energy flow. It helps make optimal decisions about when to charge the battery, when to discharge it, and how to size the system, while adhering to real-world physical and economic constraints. MILP involves linear objective functions and continuous value. It is called mixed because it uses two types of variables: Integer variables (whole numbers), binary variables (0 or 1), and continuous variables (real numbers). MILP is particularly well-suited for the

operational scheduling of Battery Energy Storage Systems, as it can accurately represent complex real-world constraints, system dynamics, and interdependencies. In this study, the MILP model incorporates not only technical constraints such as battery SoE limits, charging/discharging rates, and efficiency, but also market-specific requirements. MILP is widely used to minimize losses, operational cost, and voltage deviation, or to maximize profit, efficiency, and load served while satisfying system constraints. MILP is widely applied in fields such as operations research, power systems, energy management, supply chain optimization, transportation, production planning, telecommunications, and finance because it can accurately model real-world logical constraints and discrete decisions while maintaining mathematical rigor. Despite its modeling power, MILP can be computationally expensive for large-scale problems, and solution time depends heavily on problem size, structure, and the number of integer variables; however, advances in solvers and computing power have made MILP one of the most important and practical optimization techniques in modern engineering and decision science.

MILP minimizes total system losses and voltage deviation by optimally placing and sizing BESS. It improves system efficiency, voltage stability and overall power quality.

mint=1∑T(α·power losst+β·voltage deviationt)

where α and β are weighting factors that balance the relative importance of power loss minimization and voltage deviation minimization.

T represents the total number of time intervals.

Sequentially, higher α more focus is on reducing losses and higher β more focus on voltage deviations.

Min$\sum_{t=1}^{T}(\alpha.power\ loss_t + \beta.voltage\ deviation_t)$

The objective function minimizes total real power loss and voltage deviation simultaneously. The weighting factors α and β control the relative importance of power loss reduction and voltage profile improvement.

Voltage deviation= $\sum_{t=1}^{T}\sum_{i=1}^{N}|V_i(t)\text{-}V_{ref}(t)|$

Voltage deviation is calculated as the sum of absolute differences between the actual bus voltage V_i(t) and the reference voltage V_{ref}(t) (usually 1 p.u.) over all buses and time intervals.

Some of the terms associated with BESS are explained below:

1. Energy Balance (State of Charge Equation)

$$E_{i,t} = E_{i,t-1} + \eta_{ch}.P_{i,t}^{in}.\Delta t - \frac{1}{\eta dis}.P_{i,t}^{dis}.\Delta t$$

This equation represents the state of charge (SOC) of the BESS. During charging, energy increases based on charging efficiency, while during discharging energy decreases considering discharging efficiency.

Also define symbols:

$E_{i,t}$ → energy at time t

P_{ch},P_{dis}→ charging & discharging power

n_{ch},n_{dis}→ efficiencies

Δt → time interval

2. Power limits

$$0 \leq P_{i,t}^{ch} \leq P_{max}.x_i, 0 \leq P_{i,t}^{dis} \leq P_{max}.x_i$$

These inequalities restrict the charging and discharging power of the BESS within its rated capacity. The inclusion of the binary variable ensures that power flow is allowed only if a BESS is installed at the bus.

3. Energy limits

$$E_{min}.x_i \leq E_{i,t} \leq E_{max}.x_i$$

This constraint ensures that the energy stored in the battery remains within safe operational limits, preventing overcharging and deep discharging, both of which can damage the battery. If x_i=0, then $E_{i,t}$=0, it means no battery exists on that bus.

4. Voltage limits

$$\vartheta_{min} \leq \vartheta_{i,t} \leq \vartheta_{max}$$

This constraint ensures that the bus voltage magnitude remains within permissible limits to maintain power quality and system stability. BESS helps regulate voltage by injecting or absorbing power locally.

5. One BESS per Location limit / Maximum Number of BESS

$$\sum_{i} x_i \leq N_{max}$$

This constraint limits the total number of BESS units installed in the network, reflecting economic, technical, or planning restrictions. This avoids excessive installations and ensures realistic planning.

2.3.2.4 BIBC/BCBV Power Flow

The BIBC/BCBV power flow method is a computationally efficient technique designed for the analysis of radial distribution systems, where traditional load flow methods often struggle due to the high resistance-to-reactance ratios of distribution lines. This method establishes the relationship between bus power injections, branch currents, and bus voltage drops through two constant matrices that depend only on the network topology. In each iteration, bus current injections are calculated from the specified active and reactive power demands using the corresponding bus voltages. These currents are then transformed into branch currents using the Bus Injection to Branch Current (BIBC) matrix, which captures the contribution of each bus to the upstream branch currents. The resulting branch currents are subsequently used in the Branch Current to Bus Voltage (BCBV) matrix to determine voltage drops by accounting for line impedances. Bus voltages are updated by subtracting these drops from the slack bus voltage, and the procedure is repeated until the solution converges. Because the BIBC and BCBV matrices remain unchanged for a fixed network configuration, the method provides fast convergence and numerical stability, making it well suited for time-series studies. In this work, the BIBC/BCBV approach is applied to assess voltage profiles, power losses, and the effectiveness of integrating a Battery Energy Storage System with reactive power support in improving distribution network performance under varying operating conditions.

2.3.2.4.1 BIBC matrix and BCBV matrix

There are two matrices that can be used to improve computational efficiency, which takes advantage of the topological characteristics of distribution systems and solve the distribution load flow. Bus Injection to Bus Current (BIBC) matrix: relationship between the bus current injections and branch currents. Branch current to Bus Voltage (BCBV) matrix: relationship between the branch currents and bus voltages.

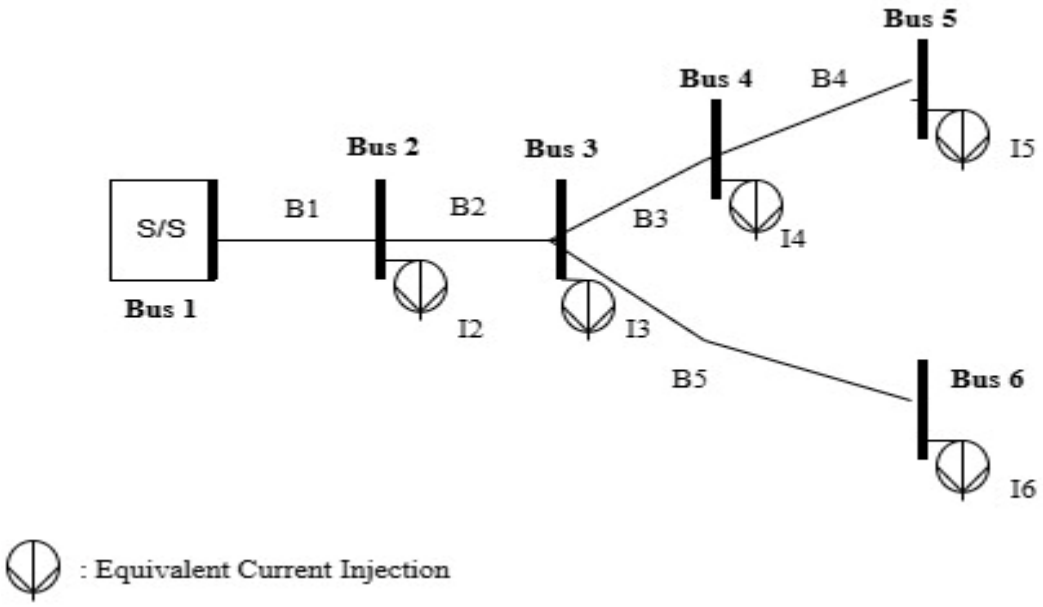


Figure 5: Equivalent Current Injection based Model of Distribution Network

B is branch current and I is bus current injection.

1. By using KCL,

$B_1 = I_2 + I_3 + I_4 + I_5 + I_6$

$B_2 = I_3 + I_4 + I_5 + I_6$

$B_3 = I_4 + I_5$

$B_4 = I_5$

$B_5 = I_6$

Relationship between the bus current injections and branch currents

$$\begin{bmatrix} B_1 \\ B_2 \\ B_3 \\ B_4 \\ B_5 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} I_2 \\ I_3 \\ I_4 \\ I_5 \\ I_6 \end{bmatrix}$$

[B] = [BIBC] [I]

The constant BIBC matrix is an upper triangular matrix and contains values of 0 and 1 only.

2. By using the KVL, V is bus voltage and Z is line impedance.

$V_2 = V_1 - B_1 Z_{12}$

$V_3 = V_2 - B_2 Z_{23}$

$V_4 = V_3 - B_3 Z_{34}$

$V_4 = V_1 - B_1 Z_{12} - B_2 Z_{23} - B_3 Z_{34}$

Relationship between branch currents and bus voltages

$$\begin{bmatrix} V_1 \\ V_1 \\ V_1 \\ V_1 \\ V_1 \end{bmatrix} - \begin{bmatrix} V_2 \\ V_3 \\ V_4 \\ V_5 \\ V_6 \end{bmatrix} = \begin{bmatrix} Z_{12} & 0 & 0 & 0 & 0 \\ Z_{12} & Z_{23} & 0 & 0 & 0 \\ Z_{12} & Z_{23} & Z_{34} & 0 & 0 \\ Z_{12} & Z_{23} & Z_{34} & Z_{45} & 0 \\ Z_{12} & Z_{23} & 0 & 0 & Z_{36} \end{bmatrix} \begin{bmatrix} B_1 \\ B_2 \\ B_3 \\ B_4 \\ B_5 \end{bmatrix}$$

$[\Delta V]$ = [BCBV] [B]

The constant BIBC matrix is a lower triangular matrix and contains values of 0 and line impedance only.

3. Distribution Load Flow (DLF) matrix is a multiplication matrix of BCBV and BIBC matrices.

[DLF] = [BIBC] [BCBV]

The voltage drop on each branch is computed using DLF and load currents.

The node voltages are computed by using the source bus voltage and voltage drops. [6]

CHAPTER THREE: METHODOLOGY

This research focuses on identifying the optimal location and sizing of Battery Energy Storage System (BESS) within a distribution feeder to enhance power quality. The methodology adopted in this general methodology. The study begins with defining the power quality issues in the distribution such as voltage fluctuations.

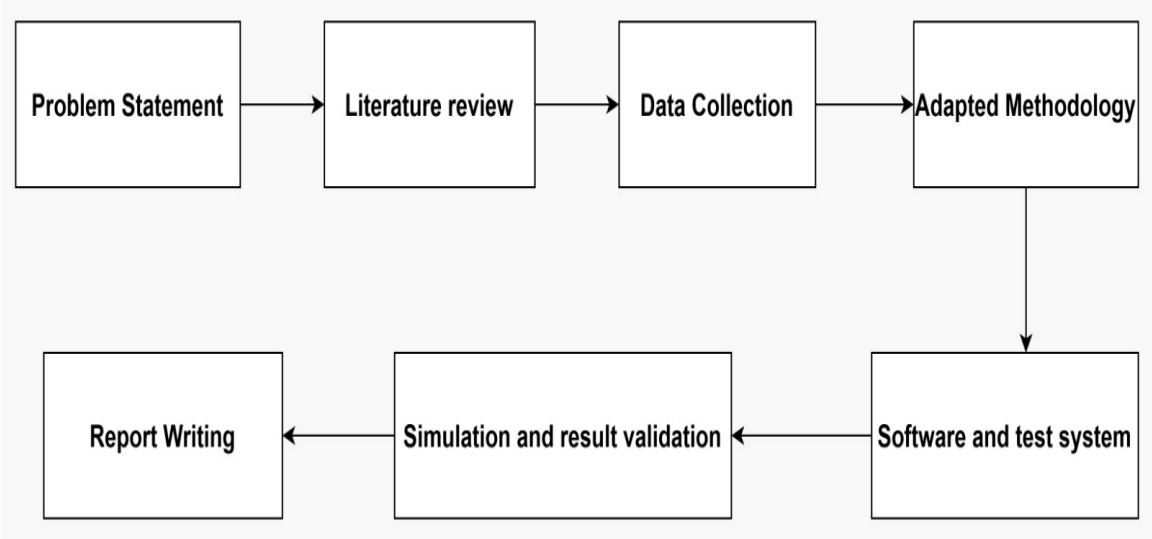


Figure 6: Block Diagram of Research Methodology

3.1 Data Collection

System data includes load profiles, feeder topology, voltage levels and other distribution system parameters are collected from Nepal Electricity Authority (NEA). Battery characteristics such as maximum and minimum energy levels (Emax B and Emin B), Depth of Discharge (DOD) and life cycle are obtained from manufacturer data sheets.

3.2 Adopted Methodology

The adopted methodology focuses on optimizing the location and sizing of the Battery Energy Storage System in a distribution network to improve voltage profile and reduce power losses. The IEEE 33-bus system is used as the test system, and load flow analysis is carried out for the base case. A Particle Swarm Optimization (PSO) algorithm, implemented in Python and integrated with OpenDSS, is used to determine the optimal placement and capacity of the BESS.

3.2.1 Particle Swarm Optimization (PSO)

The optimization of Battery Energy Storage System (BESS) placement and sizing in this study is carried out using the Particle Swarm Optimization (PSO) algorithm, implemented in Python. PSO is a population-based metaheuristic technique in which a group of particles represents potential solutions, each encoding the location, power rating, energy capacity, and inverter size of BESS units. The algorithm iteratively updates the position of each particle based on its own best-known solution and the global best solution found by the swarm, enabling efficient exploration of the search space. In this work, Python is used to integrate the PSO algorithm with a distribution load flow model based on the BIBC-BCBV method for the IEEE 33-bus system. For each candidate solution, a 24-hour simulation is performed, incorporating load variation, time-of-use electricity pricing, and BESS charging–discharging behavior. The objective function is formulated to minimize the annual net cost, which includes power loss cost, capital investment, and operational costs, while satisfying system constraints such as voltage limits, thermal limits, inverter capacity, and battery cycling constraints. The iterative PSO process continues until convergence is achieved, and the optimal configuration is selected based on the best fitness value, demonstrating improved voltage profile, reduced losses, and enhanced economic performance.

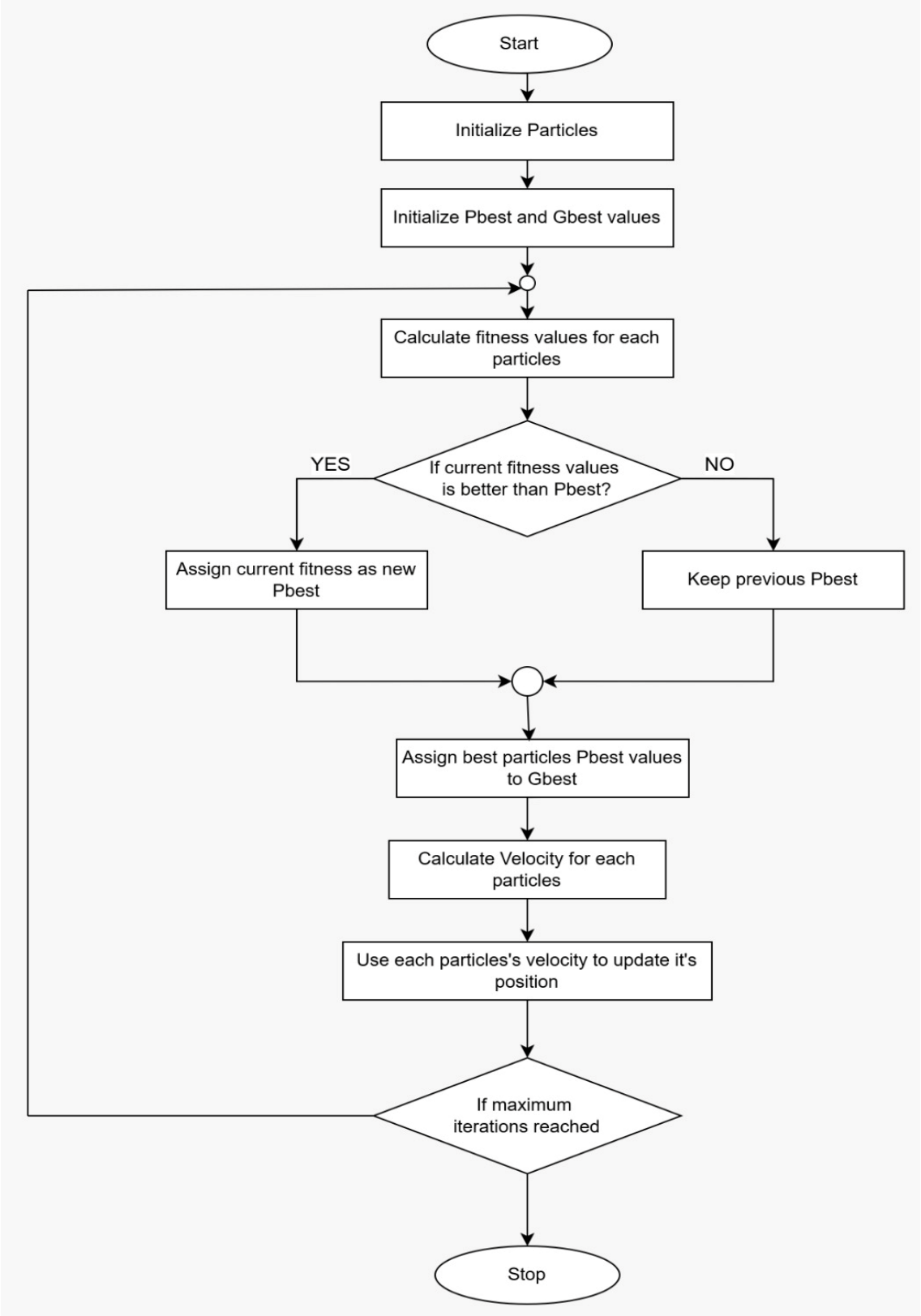


Figure 7: Flowchart of Particle Swarm Optimization (PSO)

PSO is one of the most widely used techniques for solving optimization problems because of its simplicity and high efficiency in searching for solutions. Although PSO is a traditional method, it could overcome newly proposed optimization algorithms in some works. So, it is possible that PSO can be better than modern algorithms in this work. PSO was proposed by Kennedy and Eberhart inspired by the social behavior of a group of fish or birds looking for food. The velocity of the ith particle is updated by the equation below:

$$v_i^{k+1} = w_i^k \times v_i^k + c_1 r_1 (p_{best,i}^k - x_i^k) + c_2 r_2 (g_{best}^k - x_i^k)$$

The position of the ith particle is updated by the equation below:

$$x_i^{k+1} = x_i^k + v_i^{k+1}$$

Where the velocities of the ith particle at iteration k+1 and k are denoted by v_i^{k+1} and v_i^k respectively, $x_i^{k+1}$and x_i^k represent the positions of the ith particle at iterations k+1 and k respectively. Similarly, $c_1$and $c_2$are both positive acceleration coefficients, $r_1$and r_2 are two uniformly and randomly produced numbers. $p_{best,i}^k$ is the personal best position of the ith particle at iteration k , g_{best}^k refers to the global best position of all particles during the kth iteration and w is the inertia weight at the k^{th}iteration calculated by the equation below:

$$w = w_{max} - \left(\frac{w_{max} - w_{min}}{iter_{max}} \times iter\right)$$

where iter is the current iteration number and $iter_{max}$ is the maximum iteration number.

Battery Energy Storage Systems (BESS) are critical for enhancing modern power grids, offering benefits such as loss reduction, voltage support, and renewable integration. However, maximizing these benefits requires precise determination of the unit's installation site and capacity, a complex optimization challenge. Particle Swarm Optimization (PSO) is particularly effective for this task due to its ability to handle both discrete variables (bus locations) and continuous variables (battery sizing) simultaneously. In this framework, each particle encodes a candidate solution representing specific location and capacity parameters. The optimization process evaluates these configurations through iterative power flow analyses, measuring fitness against objectives like cost minimization or voltage deviation reduction. To ensure practical feasibility, the algorithm strictly

enforces system constraints such as voltage limits and state of charge bounds by penalizing or correcting non-compliant particles. As the swarm updates its personal and global bests, it converges toward an optimal BESS configuration that maximizes system efficiency while satisfying all operational constraints. [7]

3.2.1.1 Mathematical Formulation of the Optimization Problem

The optimal placement and sizing of the Battery Energy Storage System (BESS) is formulated as an optimization problem with defined objective functions and system constraints. The aim is to enhance power quality by minimizing power loss and improving voltage profile in the distribution network.

3.2.1.2 Objective Function

The primary objective of this work is to minimize the total power loss in the distribution system. Power loss mainly occurs due to resistance in the distribution lines, and it increases with higher current flow. Therefore, minimizing power loss improves system efficiency.

The total power loss in the system is given by:

1. Minimize total power loss

$$\min F = \sum_{t=1}^{T} p_{loss}(\mathrm{t})$$

P_{loss}(t) denotes the power loss at a particular time interval t=1, and T represents the total number of time intervals considered in the study. By summing the power loss at each time step, the total power loss over the whole period is obtained.

3.2.1.3 Constraints

Constraints define the operating limits within which the optimization problem must be solved. These constraints ensure safe, reliable, and practical operation of the system.

1. Power balance constraint

$$p_{grid}(\mathrm{t}) + p_{dis}(\mathrm{t}) - p_{ch}(\mathrm{t}) = p_{load}(\mathrm{t})$$

This equation ensures that power supply and demand are balanced at every time interval t. p_{grid}(t)is the power supplied by the grid, p_{dis}(t) is the power discharged by the BESS, and p_{ch} (t) is the power absorbed by the BESS during charging. The right-hand side, p_{load}(t) represents the total load demand.

When the battery is charging, it consumes power and therefore is subtracted from the total supply. When it is discharged, it supplies power to the system and is added. This constraint ensures that the total power generated and supplied always meets the load demand.

2. SOC (state of charge) constraint

$$SOC_{min} \leq SOC\ (t) \leq\ SOC_{max}$$

This constraint limits the operating range of the battery's state of charge. The SOC indicates the amount of energy stored in the battery at time t. The SOC is restricted between a minimum and maximum value to prevent overcharging or deep discharging of the battery. Maintaining the SOC within these limits ensures safe operation, improves battery lifetime, and maintains reliable system performance.

3. Charging/Discharging limits

$$0 \leq\ p_{ch}\,(\mathrm{t}) \leq\ p_{ch}^{max}$$

$$0 \leq\ \ p_{dis}\,(\mathrm{t}) \leq\ p_{dis}^{max}$$

These equations define the charging and discharging limits of the Battery Energy Storage System (BESS). The charging power p_{ch} (t) represents the amount of power absorbed by the battery at time t, while the discharging power p_{dis} (t) represents the power supplied by the battery to the system. The lower limit ensures that charging and discharging power cannot be negative. The upper limits p_{ch}^{max} and p_{dis}^{max} represent the maximum allowable charging and discharging power of the battery. These limits are imposed to protect the battery from excessive current, prevent overheating, and ensure safe and reliable operation of the BESS.

4. SOC update equation

$$SOC\ (t+1) = SOC(t) + n_{ch}\ \frac{p_{ch(t)\Delta t}}{E_{bess}} - \frac{1}{n_{dis}}\ \frac{p_{dis(t)\Delta t}}{E_{bess}}$$

This equation describes how the state of charge (SOC) of the battery changes from one time step to the next. The SOC at time t+1 depends on the SOC at time t, the charging power, and the discharging power during the time interval Δt.When the battery is charging, the SOC increases by an amount proportional to the charging power p_{ch} (t), charging efficiency n_{ch} , and the time interval. When the battery is discharging, the SOC decreases based on the discharging power p_{dis} (t) and discharging efficiency n_{dis} .The term E_{bess} represents the total energy capacity of the battery. Including efficiency terms in the equation accounts for energy losses during charging and discharging, making the SOC calculation more realistic and practical.

Using the minimum system loss as an objective function and giving constraints the optimal size and locations of distribution feeder are calculated. The PSO also includes the personal best (Pbest) and group best (Gbest) result where after the completion of maximum iteration, the Gbest is considered the best result. PSO also includes fitness function and penalty when the constraint is not fulfilled and includes different variables which are parameters manually. The parameters and the fitness function and penalty are given as:

Objectives function: Minimum system losses

Constraints: Voltage limits: 0.95≤V≤1.05

Current limits: I/Imax ≤1

Battery cycle: Cycles/day ≤1

Inverter:

Apparent power limit: S≥1.2P

C-rate limit: P≤ Crate·E

penalty voltage = 5×10^{13}

penalty thermal = 2×10^{12}

penalty cycles = 5×10^{10}

penalty inverter = 5×10^{10}

Total penalty = Pv+Pt+Pc+Pi

The objective is to minimize total annual cost:

$$J = w_1 \cdot C_{\text{annual}} + w_2 \cdot C_{\text{capex}} + \text{Penalties}$$

3.3 Software and Test System

A proper and realistic distribution feeder model is selected for analysis. The test system is modeled using the OpenDSS, Python software. Load flow studies and power quality analysis are conducted to assess the effect of BESS at various locations. For the test system IEEE-33 bus is used.

3.3.1 IEEE-33 Bus System

The IEEE 33-bus distribution system is one of the most widely used benchmark test systems in research related to electrical power distribution networks. It is a standardized and well-defined radial system consisting of 33 buses and 32 lines, which allows researchers to test, validate, and compare different algorithms, techniques, and technologies in a consistent environment. The system closely models a real-world distribution feeder with long line lengths and unbalanced loads, making it representative of practical distribution networks. It experiences noticeable voltage drops and power losses, which makes it particularly suitable for studies on power quality improvement, loss reduction, and the integration of Battery Energy Storage Systems (BESS). Additionally, the IEEE 33-bus system is easy to analyse and can be readily simulated using OpenDSS software. Its widespread use in the literature enables easy comparison of results across different studies, while enough buses make optimization problems such as placement and sizing of BESS meaningful and realistic.

The IEEE 33-bus distribution system also includes detailed line and load data, such as bus power demands and line impedances, which makes it suitable for accurate power flow and loss analysis. It operates at a medium voltage level and is designed as a radial network, a common structure in practical distribution systems. The system is often used to study load flow methods, especially backward–forward sweep algorithms, due to its radial nature. In addition, it serves as a standard test case for evaluating distributed generation (DG) integration, capacitor placement, network reconfiguration, and renewable energy penetration. Because of its moderate size, the IEEE 33-bus system strikes a balance between computational simplicity and realistic complexity, making it ideal for testing optimization techniques such as genetic algorithms, particle swarm optimization, and mixed-integer linear programming (MILP).

Beyond these advantages, the IEEE 33-bus system provides a controlled and repeatable environment in which complex distribution network behaviours can be examined with high confidence. The availability of consistent benchmark data allows researchers to isolate the effects of new control strategies, storage technologies, and optimization frameworks without interference from modelling inconsistencies. Furthermore, its compatibility with modern simulation platforms such as OpenDSS enables time-series analysis, allowing the study of dynamic operating conditions, load variations, and storage charging–discharging cycles. This is particularly important for modern distribution systems where renewable energy penetration and flexible resources such as BESS require continuous operational assessment rather than static evaluation. Consequently, the IEEE 33-bus test system continues to remain a foundational tool for advancing research in smart grid development and modern distribution system planning.

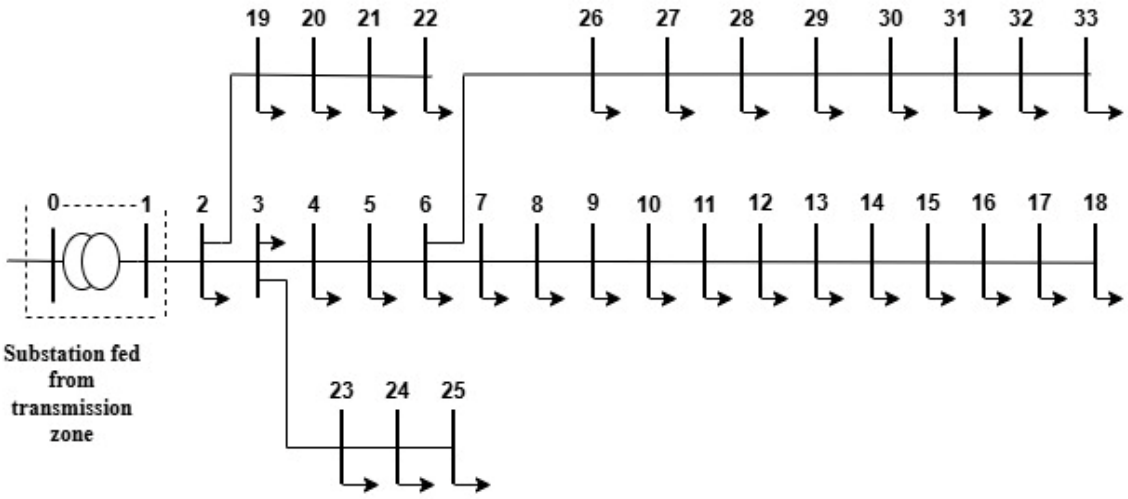


Figure 8: Single line diagram of IEEE-33 Bus System

3.3.2 OpenDSS

OpenDSS is an open-source electric power distribution system simulator developed by EPRI. It is used for simulating and analyzing electric power distribution networks, including various components like distributed energy resources, energy storage, and smart grid technologies. OpenDSS is known for its flexibility, customization, and ability to handle complex analyses required for modern distribution systems. OpenDSS offers a wide range of analysis capabilities, including power flow, harmonic analysis, and time-series simulations, allowing users to assess the impact of different operating conditions and technologies on the distribution network. It is used for power flow analysis, short-circuit studies, harmonic analysis, time-series simulations, and evaluating the impact of distributed energy resources like solar, wind, batteries, and electric vehicles. OpenDSS is highly flexible, supports integration with tools like MATLAB and Python, and can handle large, complex networks. Its main advantages are that it is cost-free, customizable, capable of modeling modern smart grids, and efficient for both steady-state and dynamic studies, making it popular among researchers, utilities, and engineers for distribution planning and renewable energy integration. OpenDSS is used more as a simulation tool for power distribution system projects because it can accurately model and simulate how batteries interact with the power distribution network under different operating conditions. It allows engineers to study battery charging and discharging behavior, assess its impact on voltage

regulation, peak load reduction, power loss minimization, and renewable energy integration. OpenDSS also supports time-series simulations, enabling analysis of battery performance over hours, days, or years, which is essential for evaluating energy storage strategies. Its flexibility, ability to integrate with control algorithms via MATLAB or Python, and suitability for large-scale distribution network studies make it an ideal tool for BESS simulation. OpenDSS also has a report generator that will export the data into a Microsoft Excel (CSV) formatted file if the user would prefer to present the data using Excel. This is a feature that doesn't require the use of the COM interface.

OpenDSS is an open-source distribution system simulation tool that is freely available for anyone to use, modify, and distribute, which makes it especially valuable for researchers, engineers, and educators in the field of power systems. Because it is open source, users can examine and customize the code according to their specific research or practical requirements, encouraging transparency, flexibility, and innovation. OpenDSS is specifically designed for the simulation and analysis of electric power distribution networks, which deliver electricity from substations to end users. It accurately models the electrical behaviour of distribution systems, including feeders, transformers, loads, capacitors, distributed generation, and energy storage devices. This strong focus on distribution-level analysis makes OpenDSS particularly suitable for studying voltage profiles, power losses, power quality issues, and the impact of technologies such as renewable energy sources and Battery Energy Storage Systems (BESS) in realistic distribution network environments. [3]

3.3.3 Python

The optimization framework for BESS placement was implemented using Python by integrating OpenDSS with a Particle Swarm Optimization (PSO) model. System data for the IEEE 33-bus network, including line parameters, load information, and bus coordinates, were first imported from Excel files using the pandas library. These datasets were automatically converted into OpenDSS script files, enabling dynamic construction of the distribution network model within the simulation environment. The Python OpenDSS interface (py_dss_interface) was then used to compile and solve the power flow for both

the base case (without BESS) and multiple candidate cases where a BESS unit was sequentially placed at each potential bus location.

For each candidate bus, the model computed key performance indices, namely total active power loss and voltage deviation index, which were stored and later used as inputs to the PSO formulation. The PSO problem, developed using the PuLP optimization library, an objective function is formulated to minimize system power losses and voltage deviation while satisfying network constraints. After solving the PSO, the optimal BESS location was identified, and a final OpenDSS simulation was executed to validate the improvement.

3.3.4 Draw.io

For the visualization of the project methodology, including block diagrams, flowcharts, and architectural overviews, draw.io (also known as diagrams.net) was used. This free, open-source diagramming tool provides an intuitive interface with extensive shape libraries, templates, and real-time collaboration features, making it ideal for creating diagrams to illustrate complex processes and system structures in the project documentation.

3.3.5 Microsoft Excel

While Python and OpenDSS are both capable of presenting and graphing the recorded data, Excel was chosen for analyzing the data. OpenDSS can export its data into an Excel spreadsheet, we found this program would be the best method in analyzing our results. Excel is more flexible in organizing the data exported from OpenDSS and can easily be used to create graphs for visual representation of the data.

3.4 Simulation and Result Validation

The python software is used for carrying out simulation. Here we analyze the system without and with using battery. The base case without BESS shows poor voltage regulation at end buses and higher power losses. With BESS integration, voltage profiles improve and power losses reduce.

CHAPTER FOUR: RESULT AND DISCUSSION

4.1 Base Case Analysis

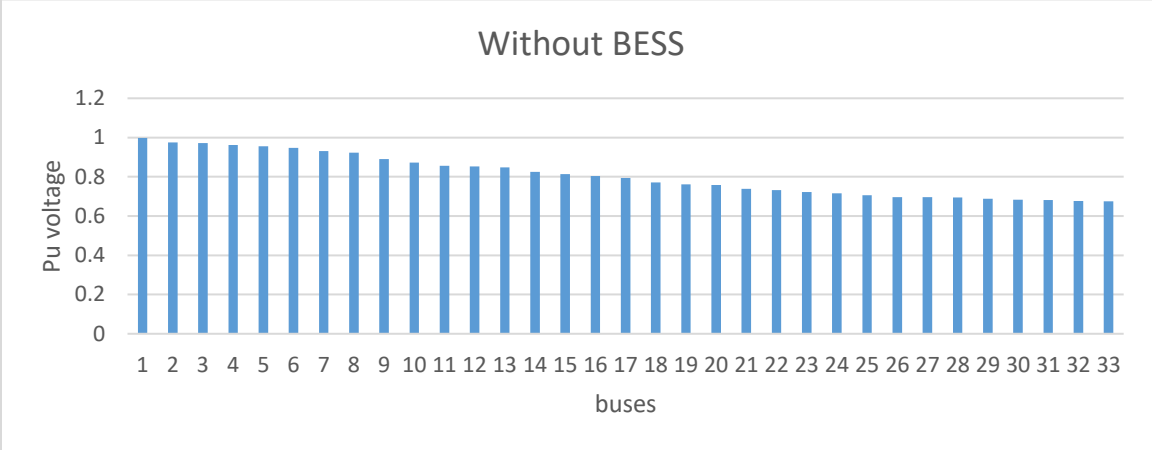


Figure 9: Voltage profile without using BESS

Figure 9 illustrates the voltage profile of the distribution network in the base case where no Battery Energy Storage System (BESS) is installed. From the graph, it can be observed that the voltage gradually decreases as power flows from the source bus to the end buses of the feeder. This voltage drop mainly occurs due to high R/X ratio, longer feeder length, and peak load demand between buses in the distribution network. The end buses experience the lowest voltage levels, indicating poor voltage regulation and higher system losses. This condition highlights the need for an additional support system such as BESS to improve voltage stability and maintain better power quality in the network.

4.2 Random placement of BESS

Voltage profiles after placing BESS at different buses are presented below. The battery is placed at various candidate buses to compare and identify the location that provides better performance. A hit and trial method is adopted for this analysis. Five different cases are analyzed: Case 1 represents 50Kw, Case 2 represents 75 Kw, Case 3 represents 100 Kw, Case 4 represents 200 Kw, Case 5 represents 250 Kw.

4.2.1 BESS placement at bus no. 33

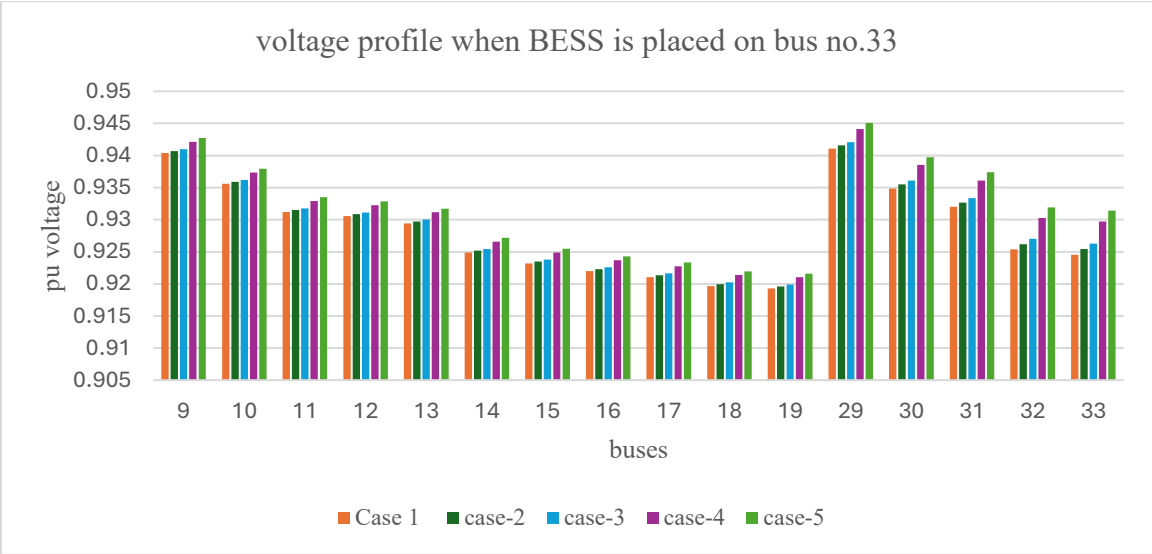


Figure 10: Voltage Profile When BESS is placed on Bus no.33

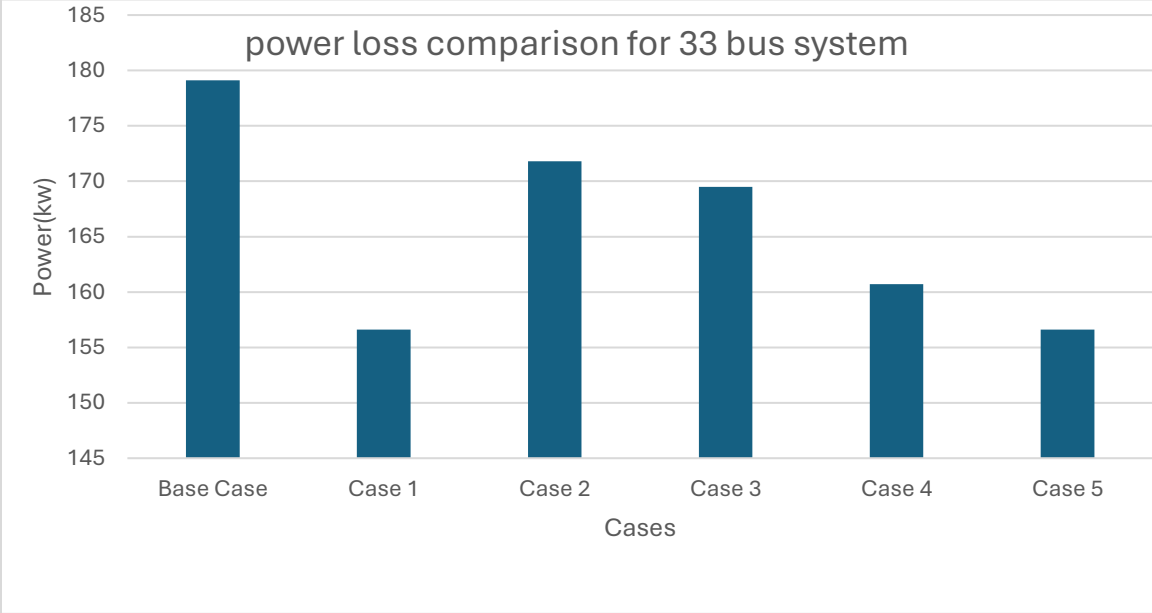


Figure 11: Power loss Comparison for 33 Bus System

Figure 10 represents the voltage profile of the IEEE 33-bus distribution system after integrating the Battery Energy Storage System with different battery capacities. Before the installation of the Battery Energy Storage System (BESS), the network experienced a significant voltage drop, with the minimum voltage reaching near 0.67 p.u., indicating poor

voltage regulation. After placing the BESS at Bus 33, the voltage profile improved considerably, with the minimum voltage rising to approximately 0.945 p.u. The presence of BESS helps supply power locally and supports the system during peak demand, which reduces the burden on distribution lines. As a result, voltage drops along the feeder become smaller and the voltage profile becomes smoother and more stable. This demonstrates that BESS can play an important role in enhancing voltage regulation in larger distribution networks.

Figure 11 presents the analysis of power losses in the 33-bus distribution system under different BESS capacities. The graph shows that the total power loss in the network decreases as the battery capacity increases. This happens because BESS provides part of the required power locally, which reduces the current flowing through the distribution lines. Since line losses are proportional to the square of the current (I^2R losses), reducing the current significantly lowers the overall power loss in the system. This indicates that proper sizing of BESS can greatly improve system efficiency.

4.2.2 BESS placement at bus no. 15

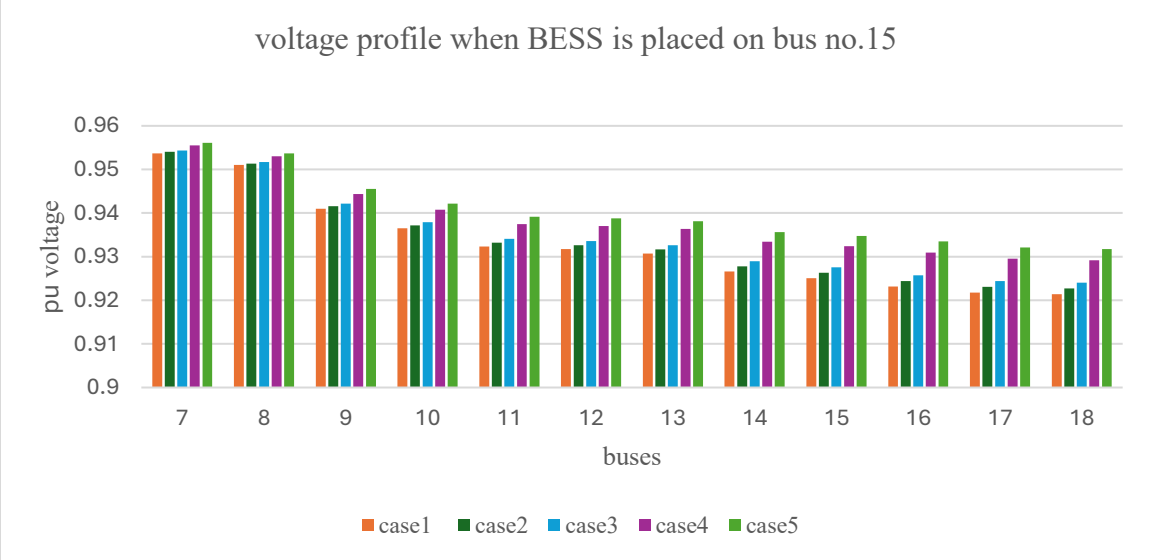


Figure 12: Voltage Profile When BESS is placed on Bus no.15

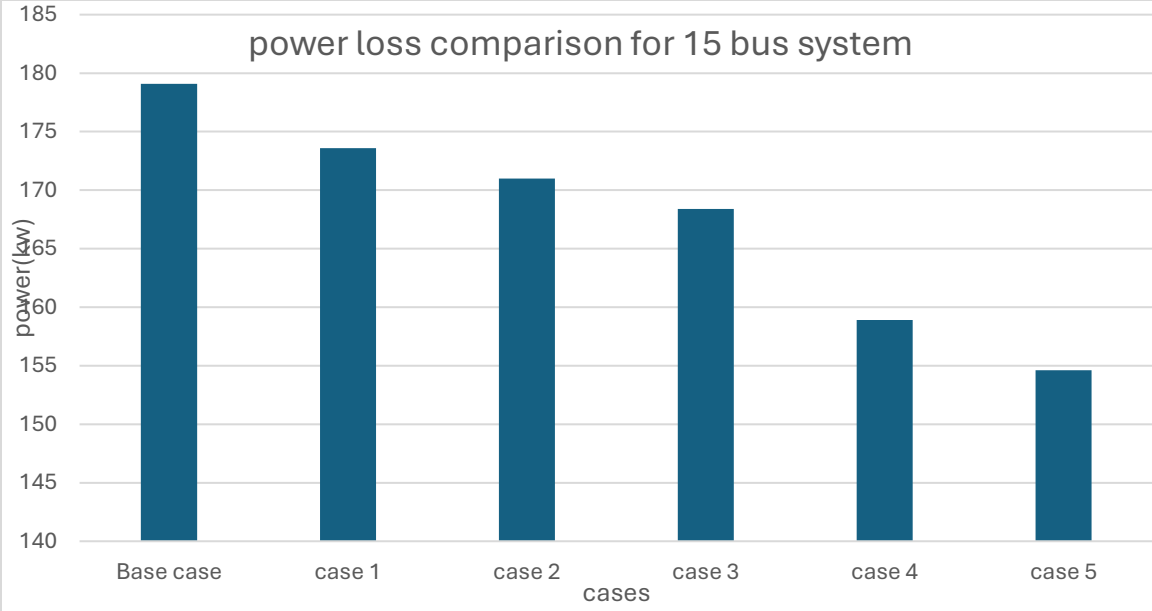


Figure 13: Power loss Comparison for 15 Bus System

Figure 12 shows the voltage profile of the 15-bus distribution system with the integration of BESS. Since the 15-bus system is relatively small and has shorter feeder lengths, the voltage drop in the base case is already limited. However, the installation of BESS still contributes to maintaining a more consistent voltage level across the buses. The battery provides additional support to the system by supplying power locally, which helps reduce minor voltage fluctuations and improves the overall voltage stability of the network.

Figure 13 illustrates the power loss analysis for the 15-bus system with BESS integration. The reduction in power losses is relatively smaller compared to larger systems because the 15-bus network already has shorter feeder lengths and lower initial losses. However, the installation of BESS still contributes to improving system efficiency by slightly reducing the current flow in the lines and supporting the network during higher load conditions.

4.2.3 BESS placement at bus no. 18

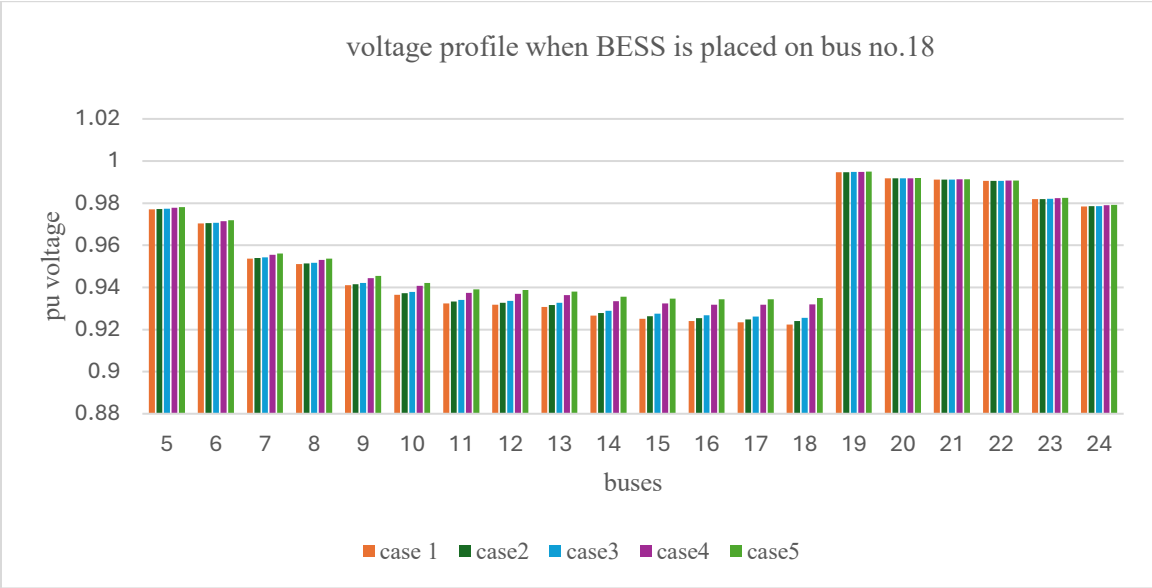


Figure 14: Voltage Profile When BESS is placed on Bus no.18

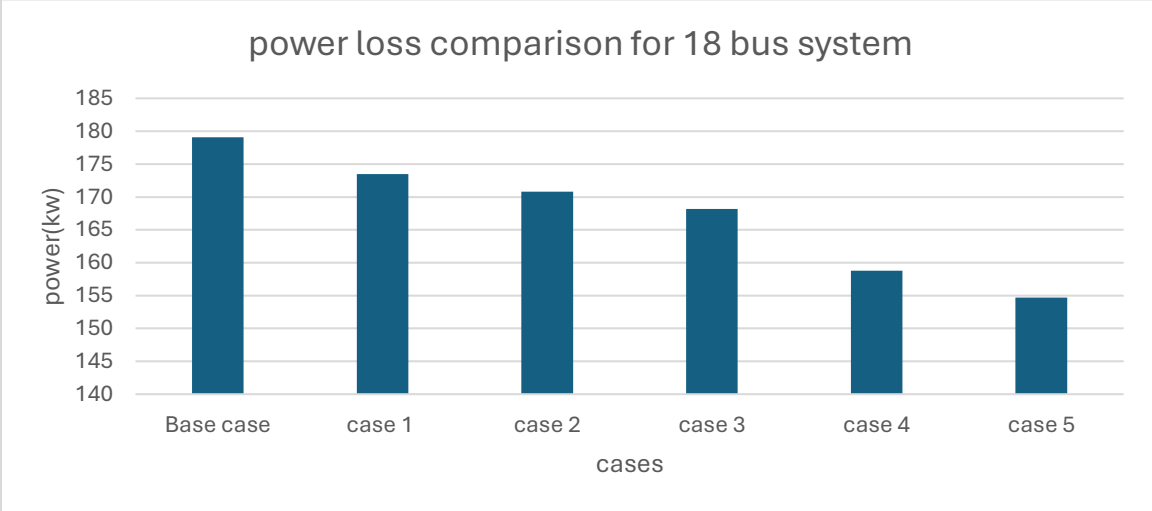


Figure 15: Power loss Comparison for 18 Bus System

Figure 14 represents the voltage profile of the 18-bus distribution system when BESS is incorporated into the network. The figure indicates that the voltage levels at several buses increase compared to the system without battery support. The improvement occurs because BESS injects power into the network when required, which reduces the load on

transmission lines and minimizes voltage drops. As a result, the overall voltage profile becomes more balanced and stable, ensuring better performance of the distribution system.

Figure 15 shows the power loss performance of the 18-bus distribution system when BESS is installed. The results indicate that the integration of BESS leads to a noticeable reduction in power losses compared to the base case. By supplying power closer to the load points, the battery reduces the need for power transmission through long lines. Consequently, the current flowing through the lines decreases, which helps minimize energy losses and improves the overall efficiency of the distribution network.

4.2.4 BESS placement at bus no. 33 and bus no.15

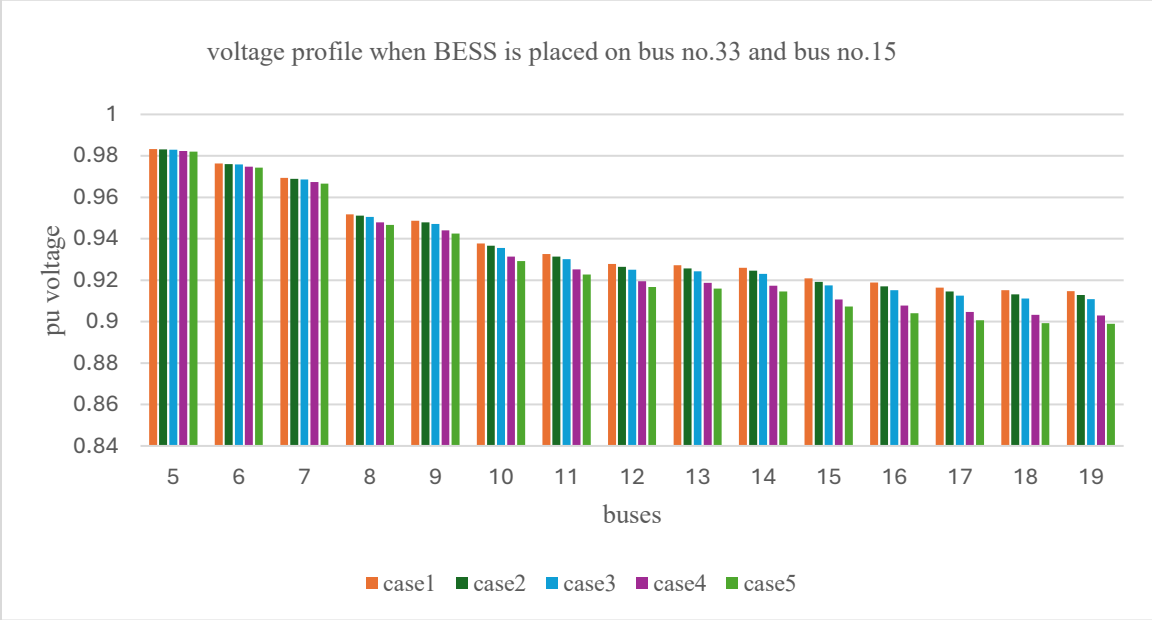


Figure 16: Voltage Profile When BESS is placed on Bus no.33 and Bus no.15

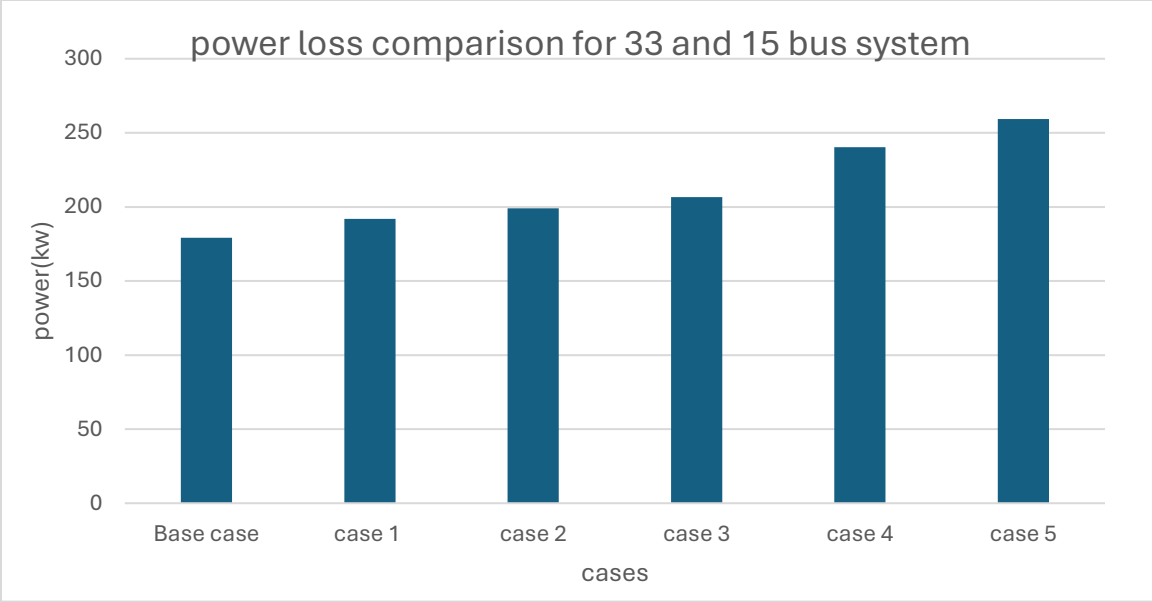


Figure 17: Power loss Comparison for 33 and 15 Bus System

Figure 16 compares the voltage profiles of the 33 bus and 15 bus systems after the installation of BESS. From the comparison, it is evident that the 33-bus system shows a more significant improvement in voltage levels than the 15-bus system. This is mainly because larger systems usually experience higher voltage drops due to longer feeders and greater load demand. The integration of BESS helps compensate for these issues by providing local energy support, thereby improving the voltage profile and enhancing the stability of the system.

Figure 17 compares the power losses of the 33 bus and 15 bus distribution systems when BESS is implemented. According to the figures, the reduction in power losses is much greater in the 33-bus system. This is because larger distribution systems generally experience higher losses due to longer transmission paths and heavier load demand. BESS helps mitigate these losses by providing energy locally, thereby reducing the amount of power that needs to travel through the entire network.

4.2.5 BESS placement at bus no. 15 and bus no.18

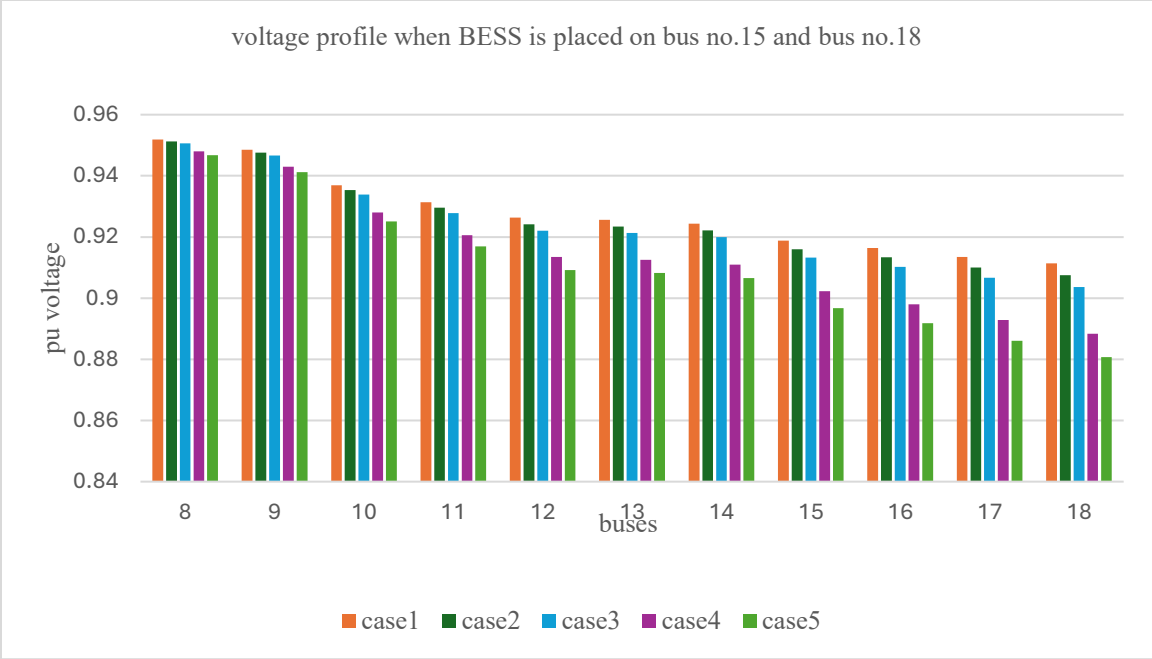


Figure 18: Voltage Profile When BESS is placed on Bus no.15 and Bus no.18

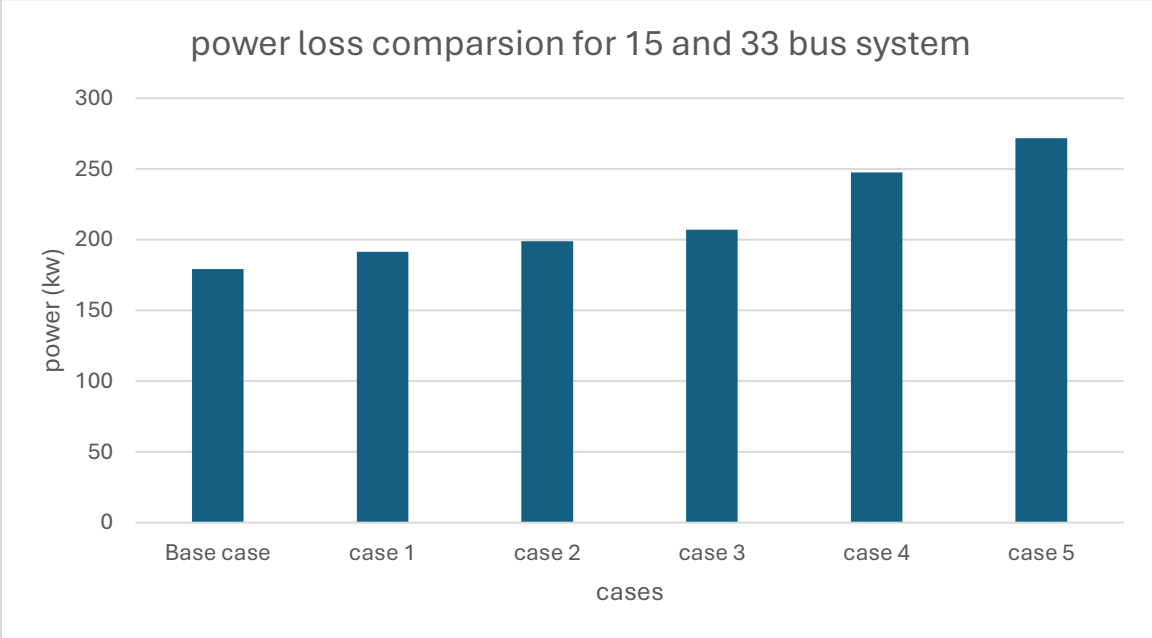


Figure 19: Power loss Comparison for 33 and 15 Bus System

Figure 18 illustrates the comparison of voltage profiles between the 15 bus and 18 bus distribution systems with BESS integration. Both systems show an improvement in voltage levels after the battery is introduced. However, the improvement is slightly more noticeable

in the 18-bus system because it experiences relatively higher voltage drops in the base condition. This comparison demonstrates that BESS is particularly effective in systems where voltage instability or drops are more prominent.

Figure 19 presents a comparison of power losses between the 15-bus and 18-bus distribution systems with BESS integration. The graph shows that the 18-bus system achieves slightly greater loss reduction than the 15-bus system. This occurs because the 18-bus network initially experiences higher line losses, which allows BESS to have a more noticeable impact. Overall, the figure demonstrates that the effectiveness of BESS in reducing power losses depends on the size and load characteristics of the distribution system.

After performing particle swarm optimization, voltage profile is improved and the losses are reduced which is illustrated below.

4.3 Validation of the Result

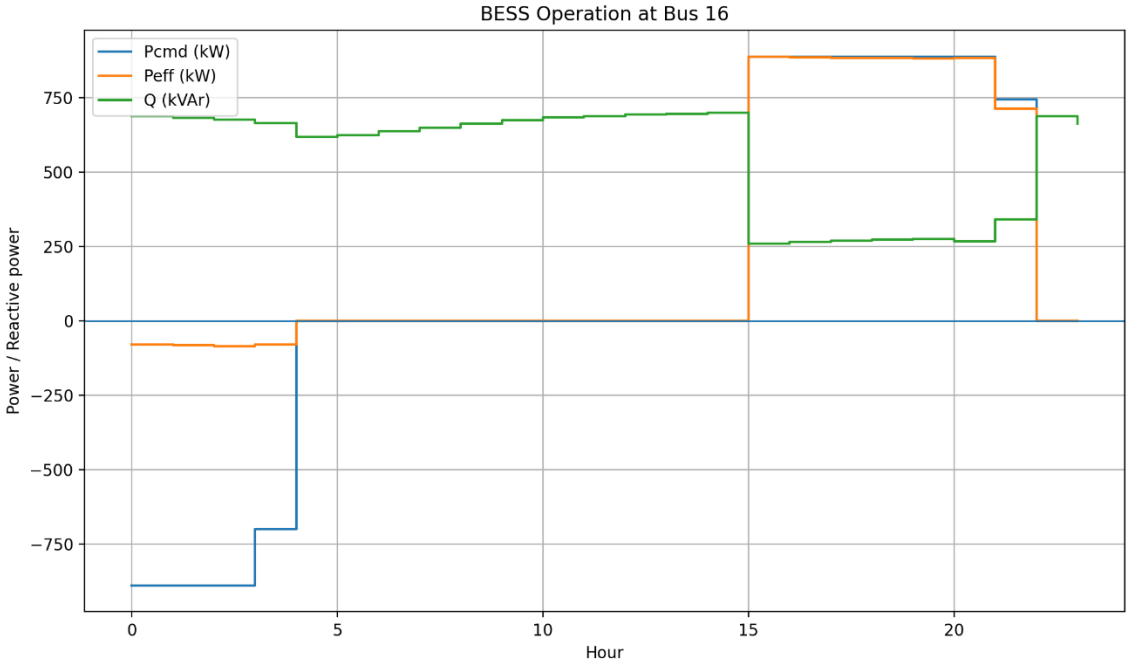


Figure 20: P/Q Schedule of Bus 16 When n=1

During the early hours (0–4 hours), the BESS operates in charging mode. From 5 to 14 hours, both Pcmd and Peff remain at zero, meaning the BESS is idle in terms of real power

exchange; however, it continues supplying reactive power, which gradually increases to maintain system stability. During the peak demand period 15–21 hours, the BESS shifts to discharging mode effectively supplying power to the grid. In this period, reactive power decreases slightly, as the primary focus is on real power delivery. Finally, in the late hours the BESS returns to an idle state with zero real power while reactive power rises again to support voltage. Overall, the graph demonstrates efficient BESS operation, where energy is stored during low-demand periods, supplied during peak demand, and reactive power support is maintained throughout the day for voltage regulation.

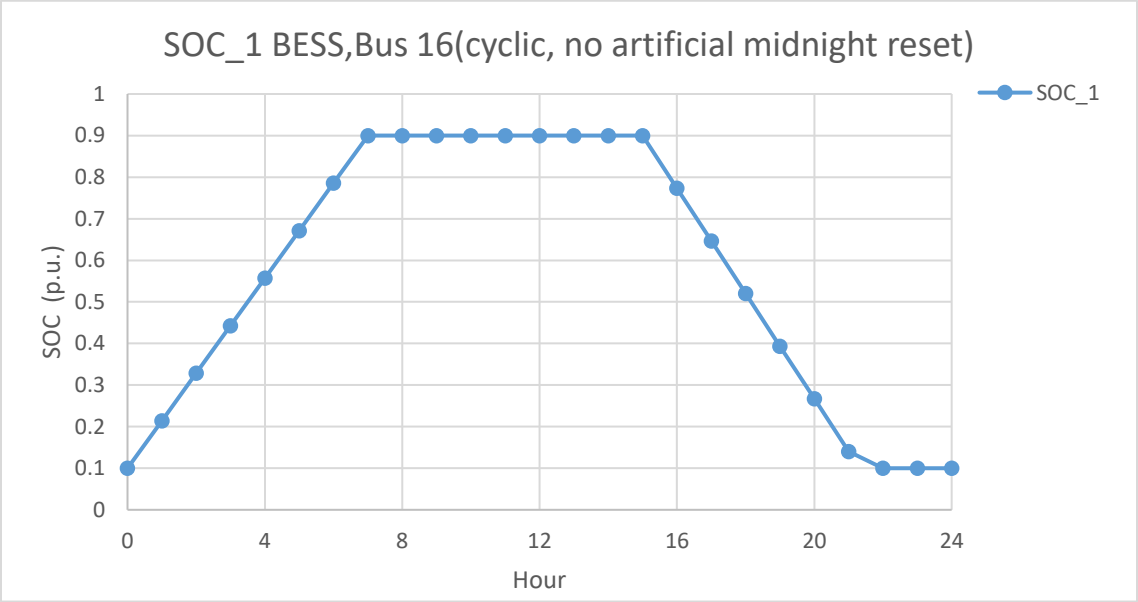


Figure 21: State of Charge of Bus 16 When n=1

This figure shows the State of Charge variation of the BESS at Bus 16 over a 24-hour period under a cyclic operation. Initially, the SOC starts at a low value of around 0.1 p.u. at hour 0 and steadily increases during off-peak hours (0–7 h), reaching approximately 0.9 p.u., indicating the battery is in charging mode. From hour 7 to 15, the SOC remains nearly constant at its maximum level, suggesting the battery is fully charged and idle. After that, during peak demand hours (16–22 h), the SOC gradually decreases, dropping back to about 0.1 p.u., which reflects discharging to support the grid. Finally, the SOC stabilizes at the minimum level toward the end of the day, completing the cycle without any artificial reset.

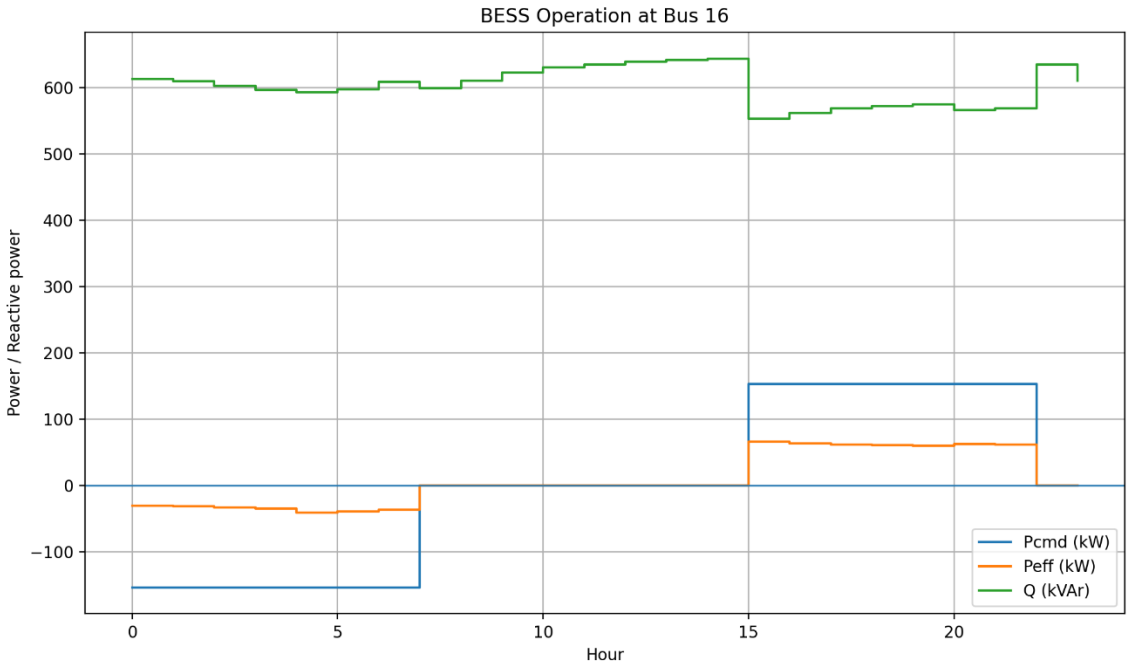


Figure 22: P/Q Schedule of Bus 16 When n=2

During the early hours (0–1 hour), the BESS operates in a light charging mode, as indicated by small negative values of Pcmd and Peff, meaning it absorbs a limited amount of power from the grid. After this period, from hour 2 to 14, both Pcmd and Peff remain at zero, indicating that the BESS is idle in terms of real power exchange. Around the peak period (15–16 hours), the BESS briefly switches to discharging mode, after hour 16, the system returns to an idle state with zero real power output. Meanwhile, reactive power shows a slight dip during the discharge period and then gradually increases again, indicating its continued role in voltage regulation. Overall, the graph demonstrates that this BESS configuration is primarily used for reactive power support and minor peak shaving, with only limited charging and discharging activity over the day.

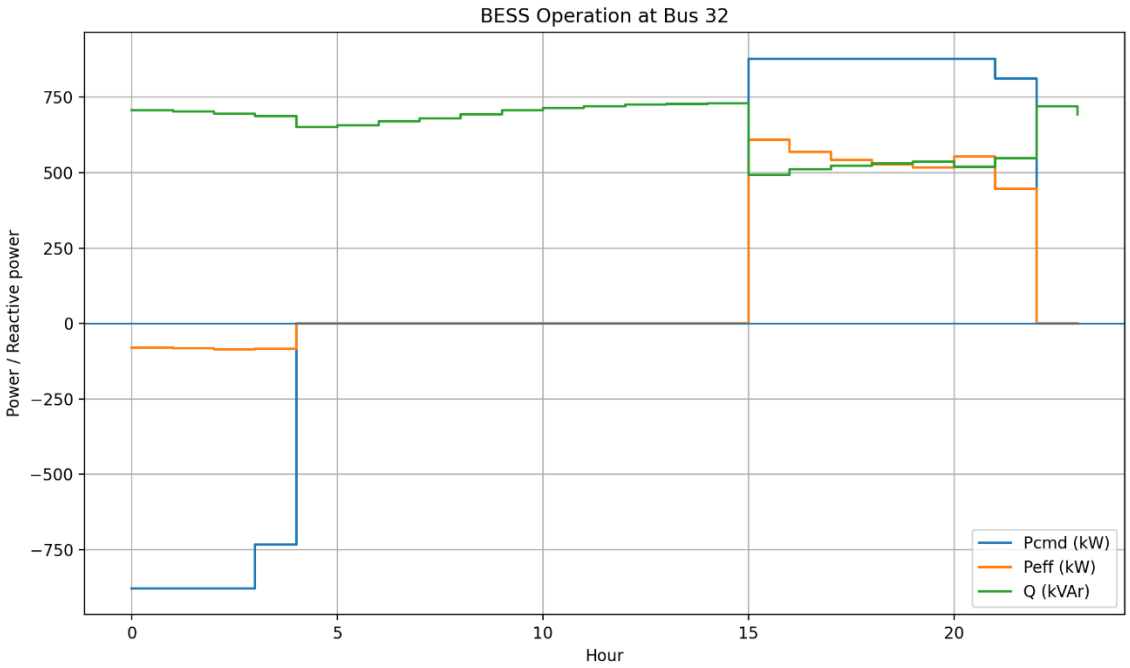


Figure 23: P/Q Schedule of Bus 32 When n=2

This graph shows the operation of a Battery Energy Storage System on Bus 32 over a 24-hour period**.** At 0 to 5 hours, the battery is in charging mode. During this time, Q remains high (around 600 kVAr), meaning the BESS is also supporting the system with reactive power for voltage stability. From hour 5 to 15**,** both Pcmd and Peff become zero, indicating that the battery is idle**.** However, the reactive power Q is still positive, which shows that even when the battery is not exchanging active power, it continues to provide reactive power support to the grid. From hour 15 to 22**,** the battery is now in discharging mode. Finally, after hour 22**,** both Pcmd and Peff return to zero**,** so the battery becomes idle again.

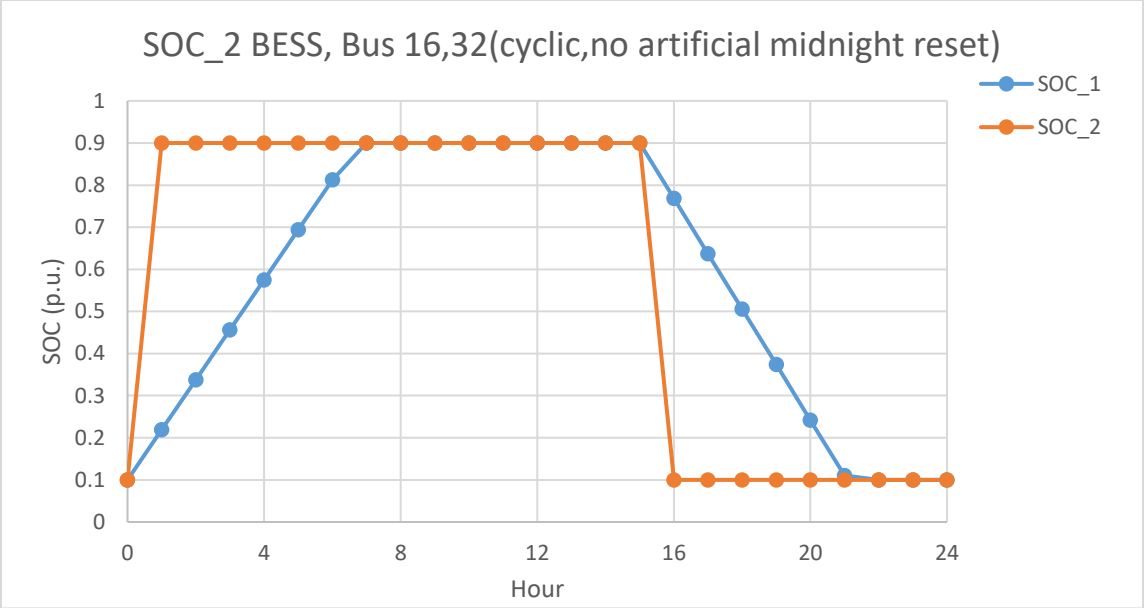


Figure 24: State of Charge of Bus 32 and 16 When n=2

The figure illustrates the State of Charge profiles of the Battery Energy Storage System installed at Bus 16 and Bus 32 over a 24-hour period. Both buses follow a cyclic operating strategy; however, their discharge behaviour differs due to their location and system impact. At Bus 16, the SOC increases from approximately 0.1 p.u. to 0.9 p.u. during off-peak hours (0–7 h), indicating active charging. It then remains nearly constant at the maximum level until around hour 15, after which the SOC gradually decreases during peak hours (16–22 h), reaching back to 0.1 p.u**.** In contrast, at Bus 32, the SOC remains at a lower level (around 0.1 p.u.) for most of the day and shows limited or minimal variation, especially during peak hours. This suggests that the BESS at Bus 32 is either less utilized or operates with lower discharge contribution compared to Bus 16. Since Bus 32 is typically located toward the end of the feeder, its impact is more localized, and the optimization results indicate that it may not require significant energy exchange for overall system improvement.

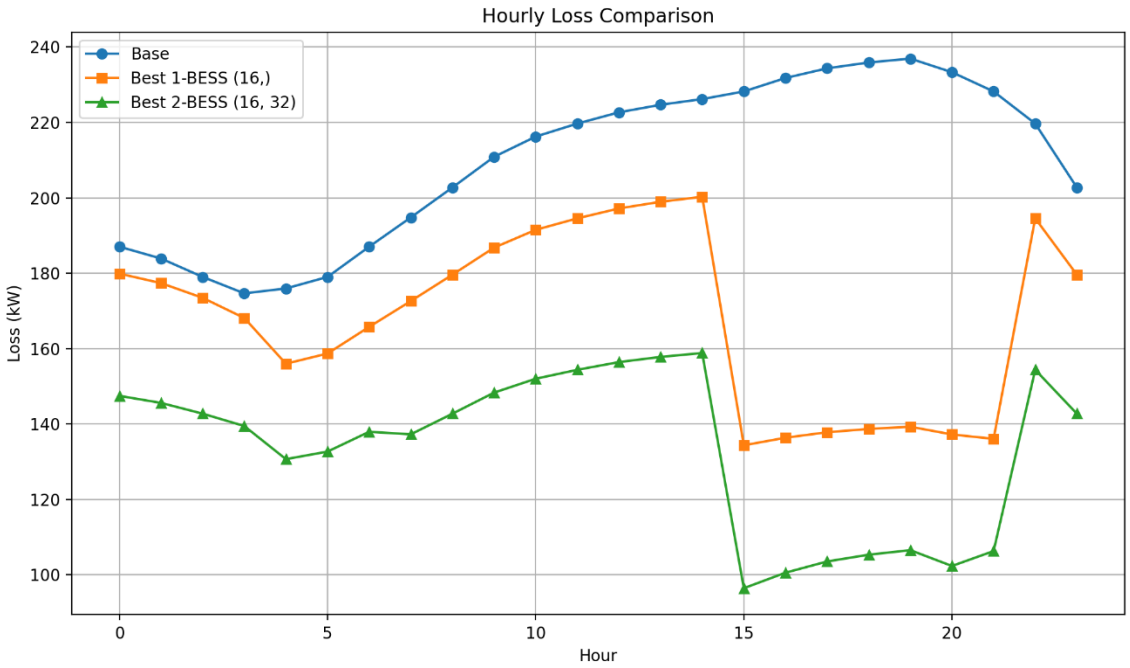


Figure 25: Comparative Analysis of Hourly Losses in BESS Configuration for BESS-1 and BESS-2

This graph illustrates how placing Battery Energy Storage Systems reduces electrical power losses over a 24-hour period. The Blue lines represent the original system without storage, showing a steady climb in losses that peak around hour 19. When the Best 1-BESS and Best 2-BESS configurations are applied, they both successfully lower these losses across most of the day improvement occurs between hours 15 and 21, where the batteries are likely discharged to offset high demand, causing the loss levels to plummet from nearly 140 kW down to approximately 100 kW. Overall, the data demonstrates that while one battery unit provides significant benefits, the two-battery system is the most efficient, consistently maintaining the lowest power losses and most effectively "shaving" the peak energy waste during high-use hours.

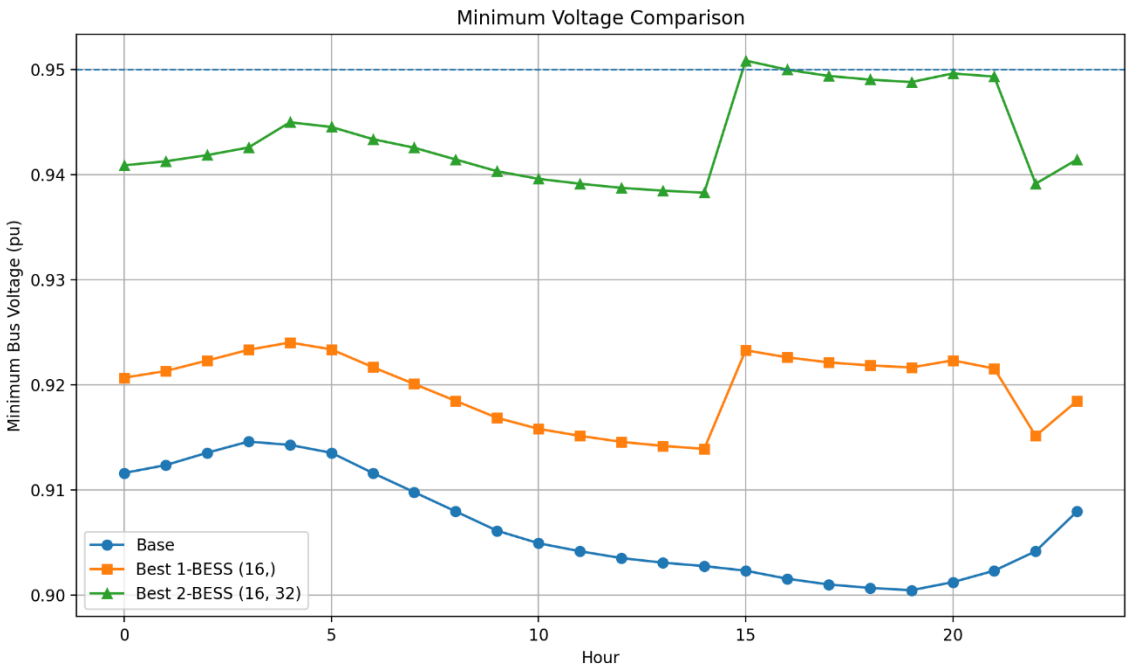


Figure 26: Comparative Analysis of Minimum voltage in BESS Configuration for BESS-1 and BESS-2

This graph illustrates how adding BESS helps maintain healthy voltage levels across an electrical grid throughout a 24-hour day. The Base scenario shows that without any batteries, the grid minimum voltage stays well below this safety limit all day, dipping to its lowest point of approximately 0.906 pu during the peak demand hour 19. While the addition of one battery Best 1-BESS significantly lifts the voltage profile, it still fails to reach the 0.95 pu target. In contrast, the two-battery configuration Best 2-BESS provides the most effective support, successfully pushing the minimum voltage up to or above the 0.95 pu threshold for several hours, particularly after a sharp improvement at hour 15. Ultimately, the graph proves that the two-battery system is the only configuration shown that brings the grid close to ideal operating standards, ensuring the system remains stable and efficient even during high-load periods.

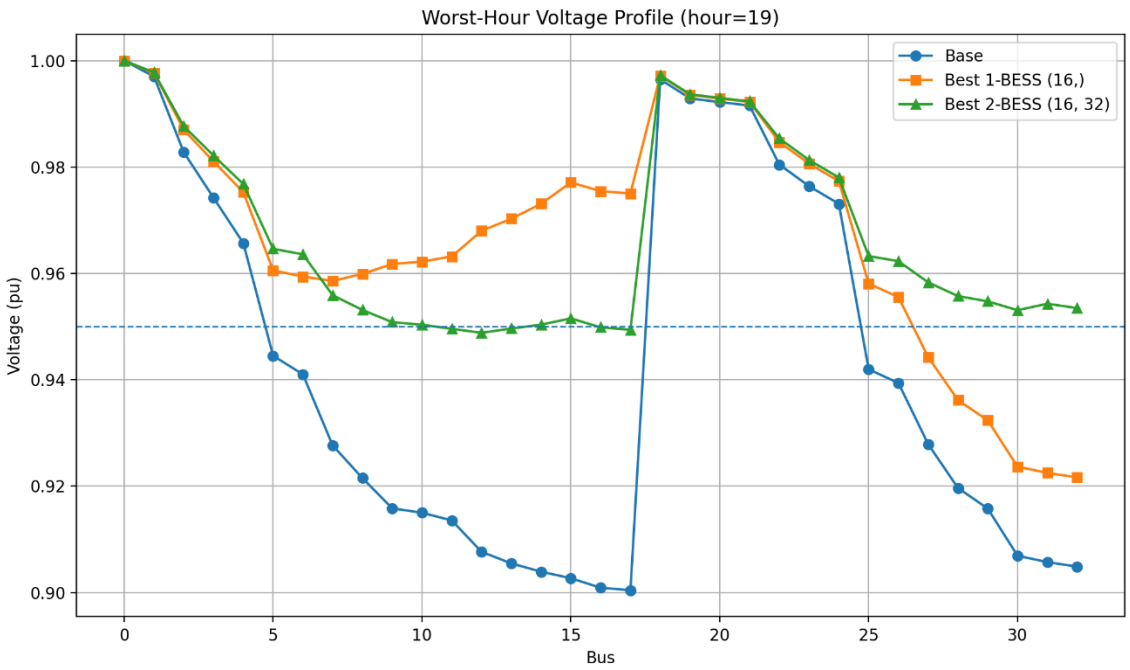


Figure 27: Comparative Analysis of Voltage Profile Across Buses (Worst Base Hour 19)

The graph illustrates how the addition of battery systems helps maintain stable voltage levels across different points in an electrical grid during the hour of highest demand. The base shows that without any batteries, the voltage drops significantly below this safety limit at many locations, reaching as low as 0.91 near bus 17. When the Best 1-BESS is added at bus 16, it successfully boosts the voltage, pulling it backs up toward the 0.95 limit and showing a clear peak where the battery is located. The Best 2-BESS, which adds a second battery at bus 32, provides the most stable performance by keeping the voltage levels higher and more consistent across the entire grid, especially at the far end buses 25–33. Ultimately, the graph demonstrates that while one battery helps, the two-battery configuration is the most effective solution for preventing power dips and ensuring the grid stays within healthy operating limits during peak hours.

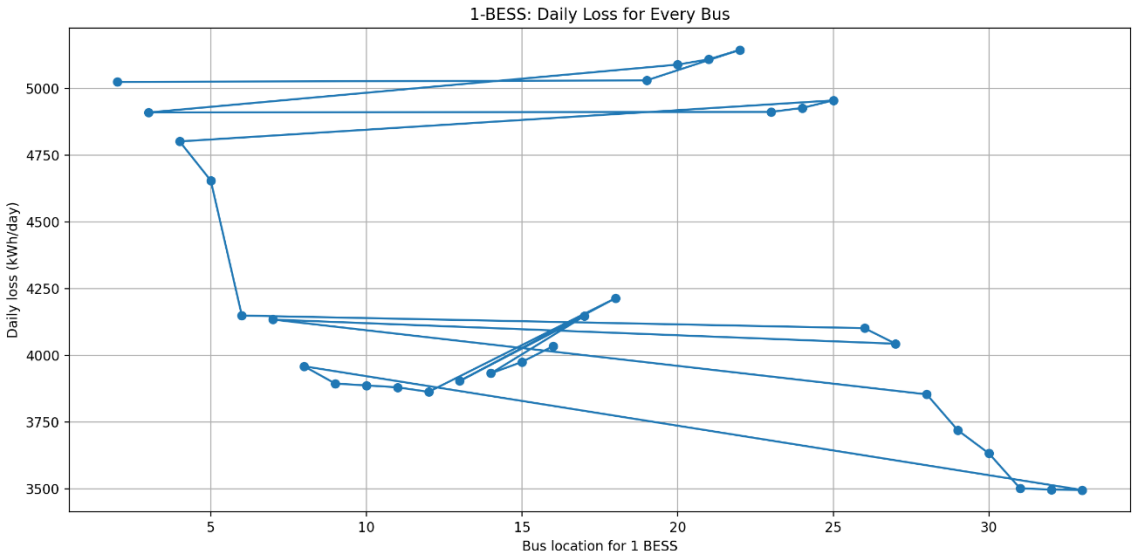


Figure 28: Daily loss for Every Bus for BESS-1

The figure shows daily power losses for different bus locations when a single BESS is installed. Losses are higher (around 5000–5150 kWh/day) when the BESS is placed near the starting buses. As the placement shifts toward the middle buses, the losses decrease to about 3900–4200 kWh/day. The lowest losses are observed near the end buses (around bus 30–33), reaching approximately 3500 kWh/day. This indicates that placing the BESS closer to the feeder end is more effective. Overall, the graph highlights the importance of optimal BESS location for loss reduction.

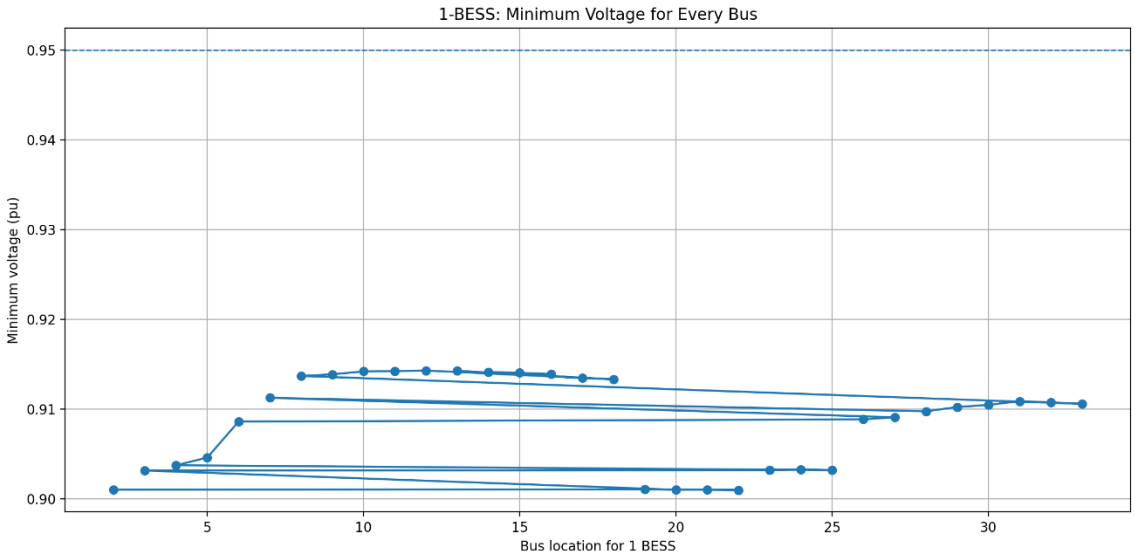


Figure 29: Minimum voltage for Every Bus BESS-1

This graph illustrates the minimum voltage levels at various bus locations in an IEEE 33-bus power distribution system when a single Battery Energy Storage System (BESS) is installed. Each point on the graph shows the minimum voltage observed at that bus while the BESS is operating in the system. The presence of the BESS helps improve voltage stability by supporting buses that might otherwise experience voltage drops. This information is vital as voltage levels below certain limits can lead to poor power quality and equipment malfunction. Therefore, monitoring the minimum voltage at each bus helps ensure the system runs reliably and efficiently.

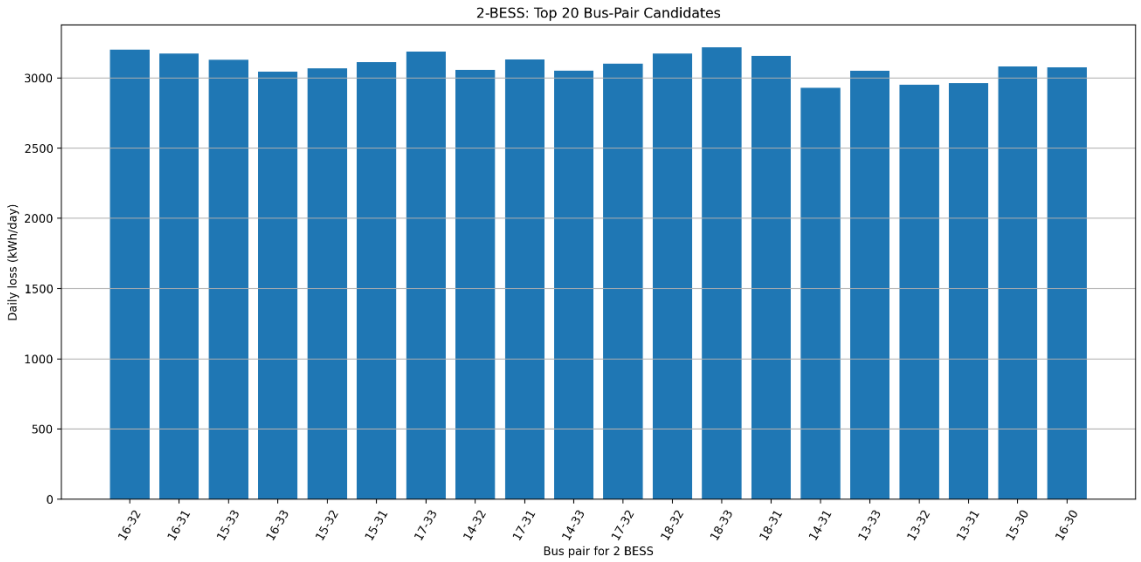


Figure 30: Top 20 Bus-pair candidates for BESS-2

This graph illustrates the top 20 pairs of buses in the IEEE 33-bus electrical distribution system that are the best candidates for installing two Battery Energy Storage Systems (2-BESS) based on their impact on reducing daily energy losses. Each bus pair is labeled, such as "16-32," "15-33," or "18-31," indicating the two locations within the network where the storage units could be placed. Alongside these bus pairs, the graph displays the corresponding daily energy loss values in kilowatt-hours per day (kWh/day), which range from zero up to approximately 3500 kWh/day. Lower energy loss values indicate more efficient energy distribution and less wasted power. The key takeaway is that installing two BESS at the bus pairs listed can significantly reduce daily power losses, thereby improving the overall efficiency of the system. By identifying and targeting these top bus pairs, utility operators can optimize the placement of energy storage systems to minimize losses, enhance reliability, and support better load management in the distribution network.

4.4 Optimal location and Sizing

4.4.1 One BESS unit (n=1)

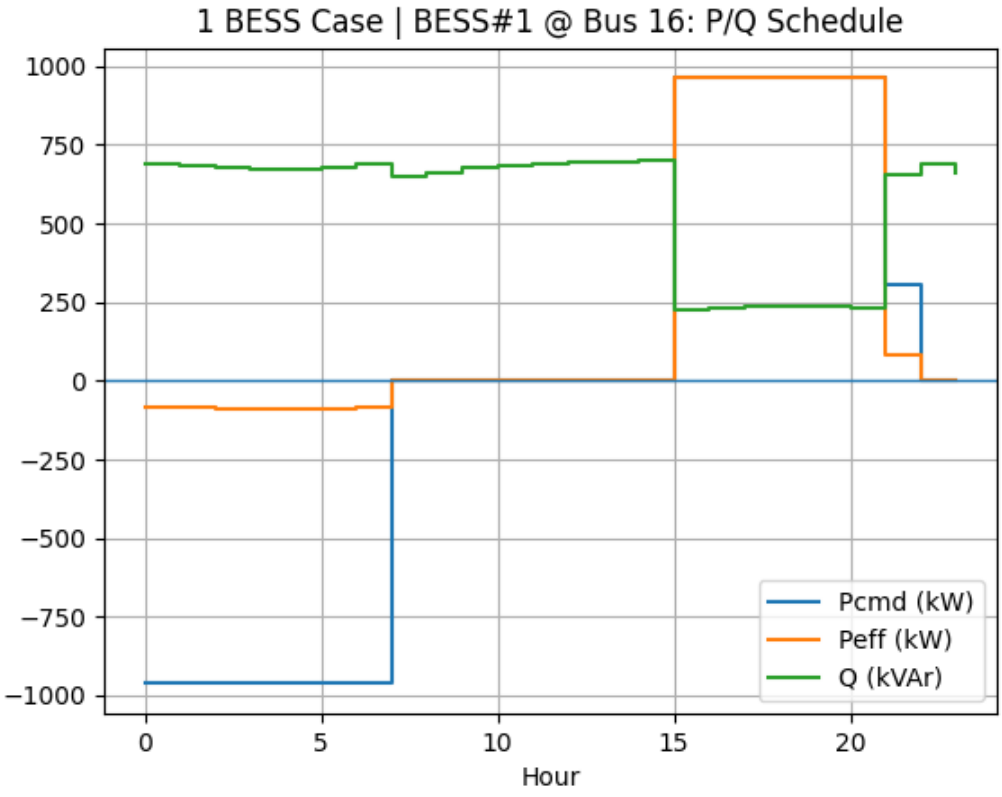


Figure 31: P/Q Schedule of Bus 16 When n=1

This figure shows how Battery Energy Storage System works on Bus 16 over a 24-hour period for a single unit (n=1). It explains how real power (P) and reactive power (Q) change during the day. The blue lines represent the commanded active power (Pcmd) which means what the battery is told to do, orange line represents the effective power output (Peff) means what the battery really does, and green line represents the reactive power. The active power, showing a significant charge command of -1000 kW during the early morning and a discharge command of 1000 kW in the evening. The Peff shows the batteries actual performance; while it follows the general command, its magnitude is much lower during the initial charging phase (around -100 kW), suggesting physical constraints or limits on how much power the battery could take in. Between 7 and 15 hours, the battery remains idle.

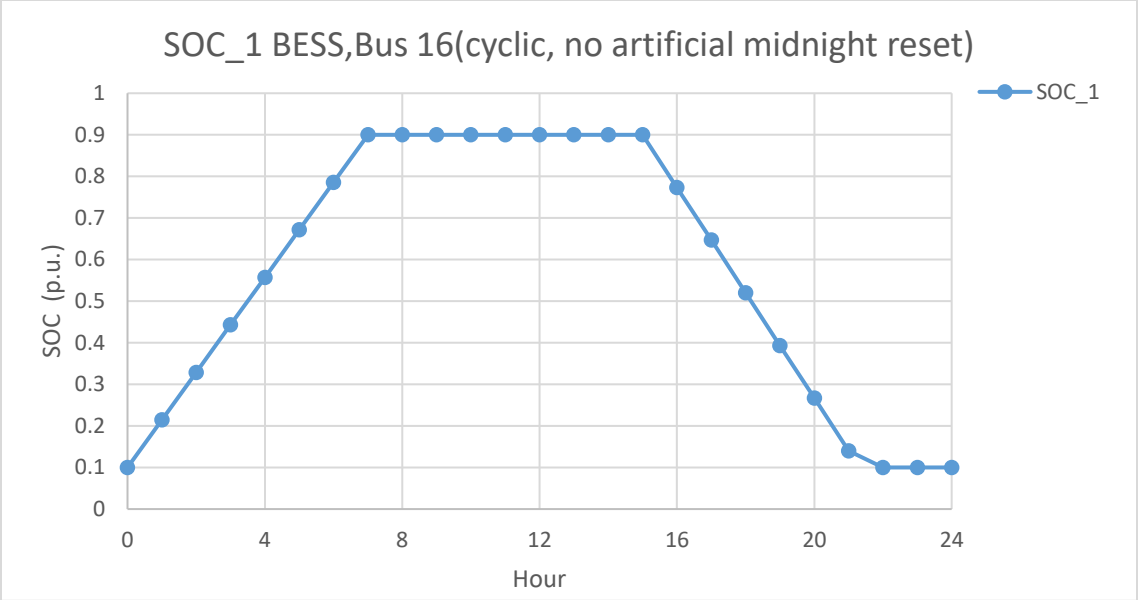


Figure 32: State of Charge of Bus 16 When n=1

This figure shows the State of Charge (SOC) variation of the BESS at Bus 16 over a 24-hour period under a cyclic operation. Initially, the SOC starts at a low value of around 0.1 p.u. at hour 0 and steadily increases during off-peak hours (0–7 h), reaching approximately 0.9 p.u., indicating the battery is in charging mode. From hour 7 to 15, the SOC remains nearly constant at its maximum level, suggesting the battery is fully charged and idle. After that, during peak demand hours (16–22 h), the SOC gradually decreases, dropping back to about 0.1 p.u., which reflects discharging to support the grid. Finally, the SOC stabilizes at the minimum level toward the end of the day, completing the cycle without any artificial reset.

4.4.2 Two BESS units (n=2)

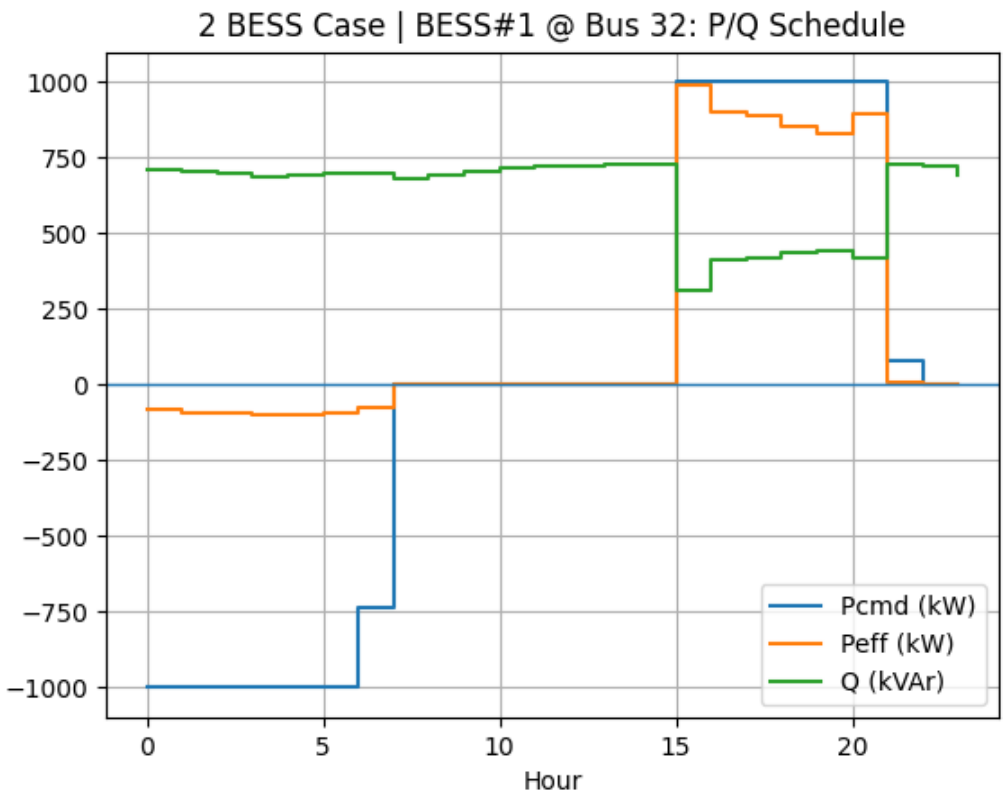


Figure 33: P/Q Schedule of Bus 32 When n=2

This graph displays the scheduled performance of a BESS at Bus 32 in a scenario involving two battery units. The blue line represents the power command sent to the battery, showing a request to charge at roughly -1000 kW during the first 7 hours and discharge at approximately 1000 kW between hours 15 and 21. The orange line represents the actual effective power delivered, which is notably lower than the command; it stays around -140 kW during the morning charging phase and reaches about 980 kW during the evening. The green line maintains a steady level near 740 kVAr for most of the day but drops to roughly 270 kVAr during the heavy discharge period to manage voltage stability.

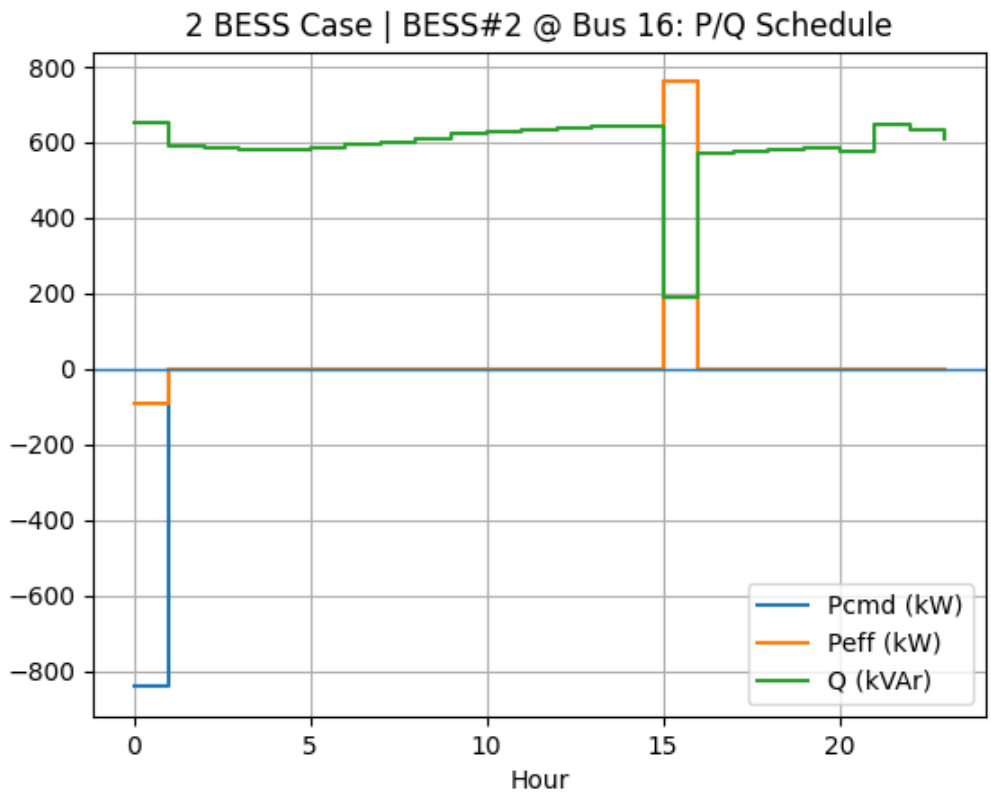


Figure 34: P/Q Schedule of Bus 16 When n=2

This graph shows the active and reactive power schedule for the second battery unit in a two-battery system located at bus 16 over a 24hour period. The active power, which shows very small requests to charge at roughly negative 800 Kw during the first two hours and discharge between hours 15 and 17. The orange line illustrates the actual effective power produced, which mirrors the commanded pattern but at even lower levels, indicating that the battery is providing very little active energy movement. The green line represents reactive power, which stays consistently high between 200 and 620 Kvar for the duration of the day. This high level of reactive power suggests that the primary role of this second battery unit is to maintain voltage stability on the grid rather than performing heavy charging or discharging tasks.

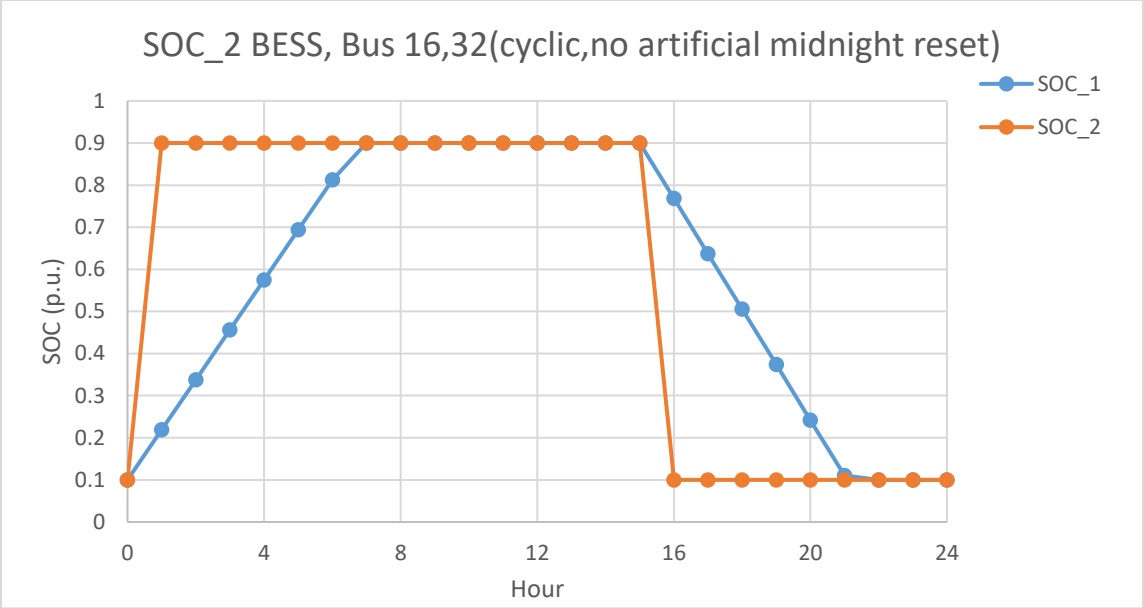


Figure 35: State of Charge of Bus 32 and 16 When n=2

The figure illustrates the State of Charge (SOC) profiles of the Battery Energy Storage System (BESS) installed at Bus 16 and Bus 32 over a 24-hour period. Both buses follow a cyclic operating strategy; however, their discharge behaviour differs due to their location and system impact. At Bus 16, the SOC increases from approximately 0.1 p.u. to 0.9 p.u. during off-peak hours (0–7 h), indicating active charging. It then remains nearly constant at the maximum level until around hour 15, after which the SOC gradually decreases during peak hours (16–22 h), reaching back to 0.1 p.u**.** In contrast, at Bus 32, the SOC remains at a lower level (around 0.1 p.u.) for most of the day and shows limited or minimal variation, especially during peak hours. This suggests that the BESS at Bus 32 is either less utilized or operates with lower discharge contribution compared to Bus 16. Since Bus 32 is typically located toward the end of the feeder, its impact is more localized, and the optimization results indicate that it may not require significant energy exchange for overall system improvement.

4.4.3 Three BESS units (n=3)

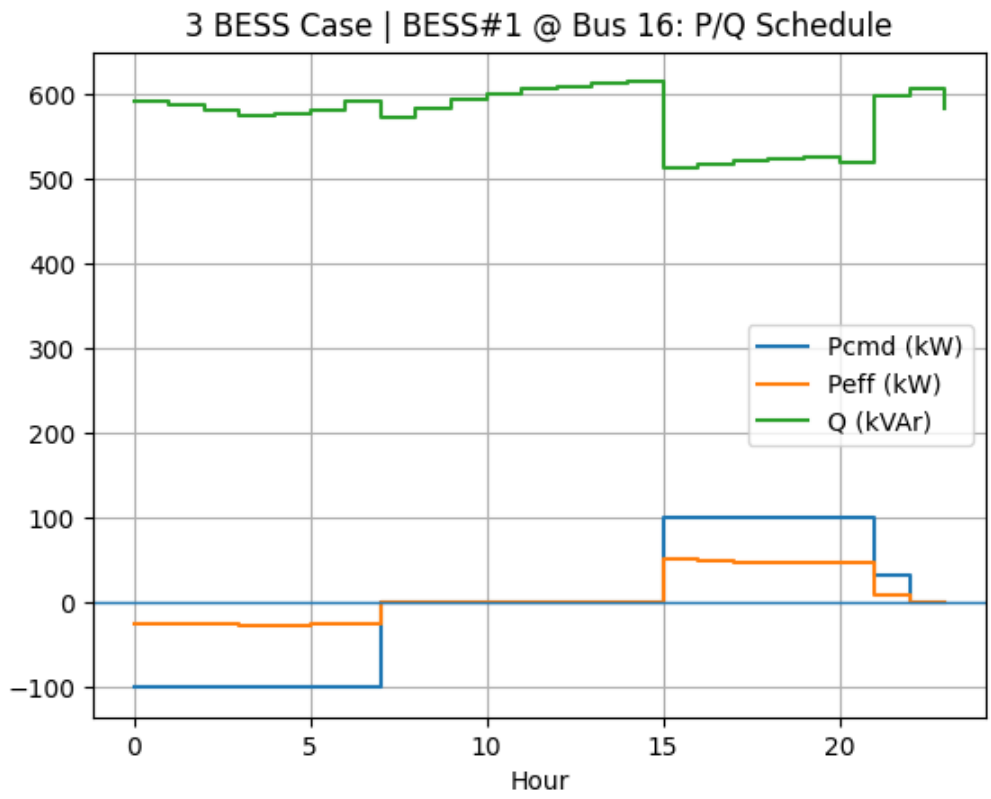


Figure 36: P/Q Schedule of Bus 16 When n=3

This graph illustrates the power schedule for a BESS at Bus 16 in a scenario involving three battery units over a 24-hour cycle. The blue line represents the commanded power, showing a request for the battery to charge at -100 kW during the early morning (hours 0–7) and to discharge at 800 kW during the evening (hours 15–21). The orange line shows the actual effective power produced; while it perfectly matches the 800 kW command during the evening discharge phase, it falls significantly short during the morning charging phase, only reaching about -30 kW. This indicates that the battery can fully meet discharge demands but is constrained by physical or system limits when trying to charge. The green line (Q) represents reactive power in kVAr, which remains positive at approximately 580 to 620 kVAr for most of the day, showing how the system adjusts its voltage support strategy depending on whether it is idle or active.

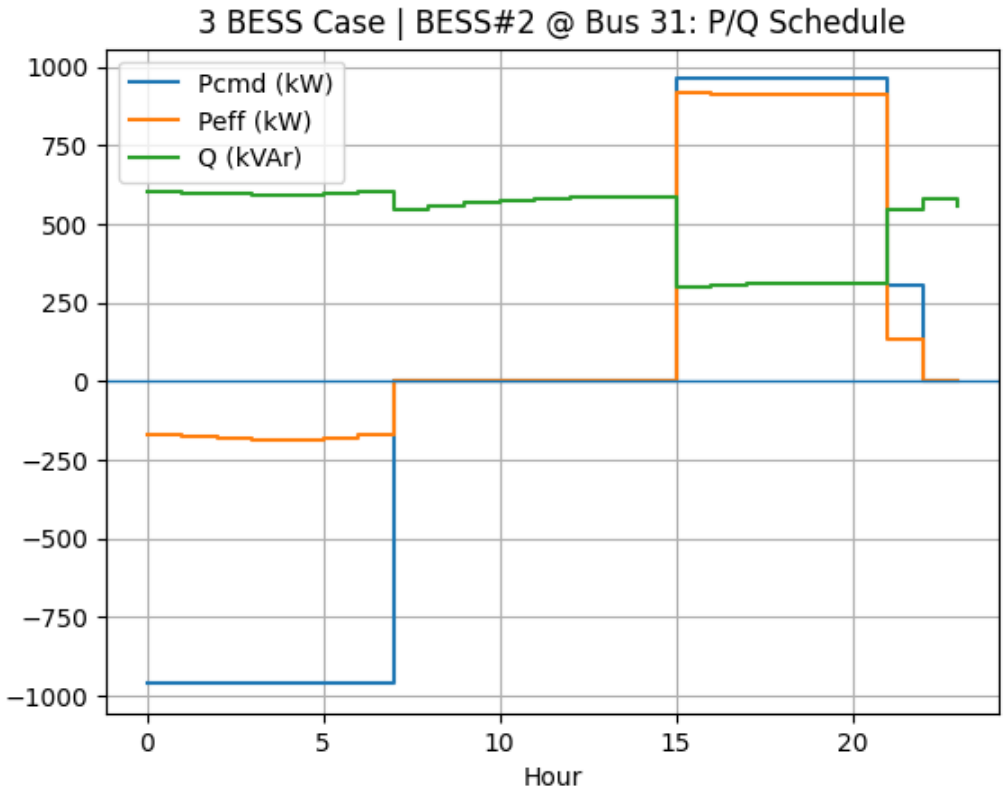


Figure 37: P/Q Schedule of Bus 31 When n=3

This graph presents the power schedule for the second battery unit at bus 31 within a three-battery system setup. The blue line represents the commanded active power, which shows a request for the battery to charge at -1000Kw from the start of the day until hour 7, and then to discharge 1000 Kw between hours 15 and 21. The orange line shows the actual effective power delivered, while the battery successfully meets the full discharge command of 800KW in the evening, its charging performance in the morning is limited to only about -150 Kw, suggesting a system or capacity restriction during that time. The green line tracks the reactive power in Kvar, which remains steady around 600 to 650 kilovars for the first 15 hours before dropping nearly to zero during the peak discharge window, illustrating how the battery prioritizes active power delivery over voltage support during high-demand periods.

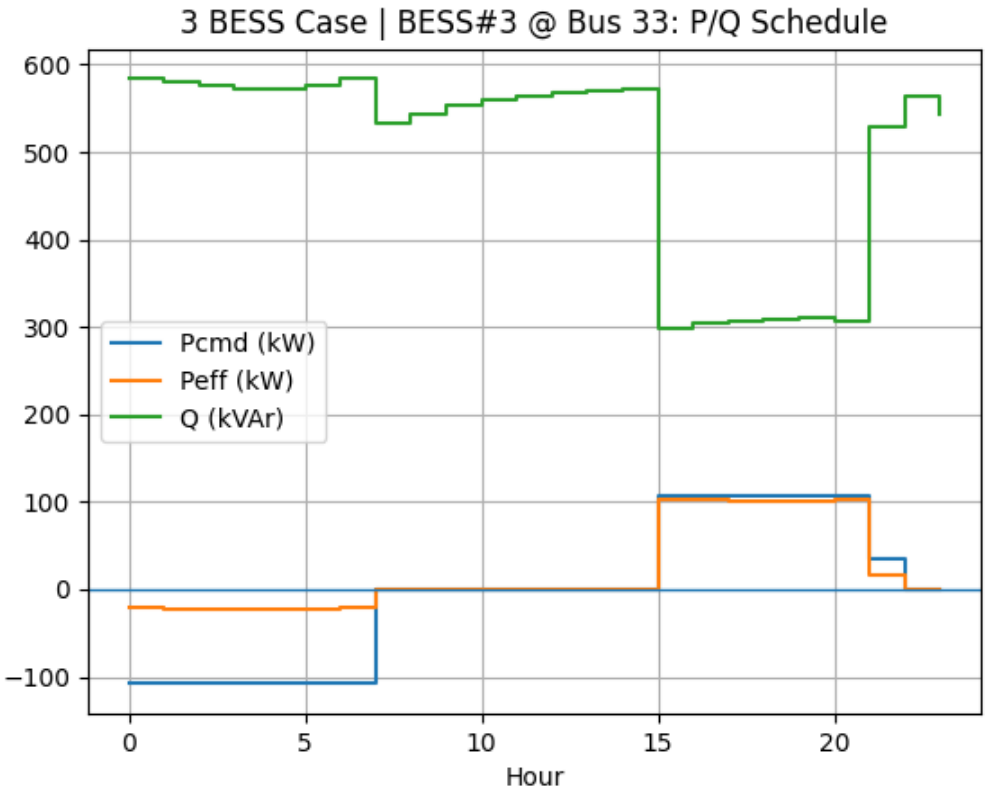


Figure 38: P/Q Schedule of Bus 33 When n=3

This graph shows the BESS operation at Bus 33 over 24 hours. From hour 0–7, it charges at about –100 kW (Pcmd) with effective power around –30 to 100 kW (Peff), while providing 600–500 kVAr reactive power. From hour 7–15, active power is nearly 0 kW, and reactive power stays around 530–560 kVAr. During peak hours (15–21), it discharges up to ~800 kW (Peff), with reactive power dropping to ~160–180 kVAr. After hour 21, discharge reduces to about 40–60 kW, and reactive power increases again to 550–590 kVAr, supporting voltage.

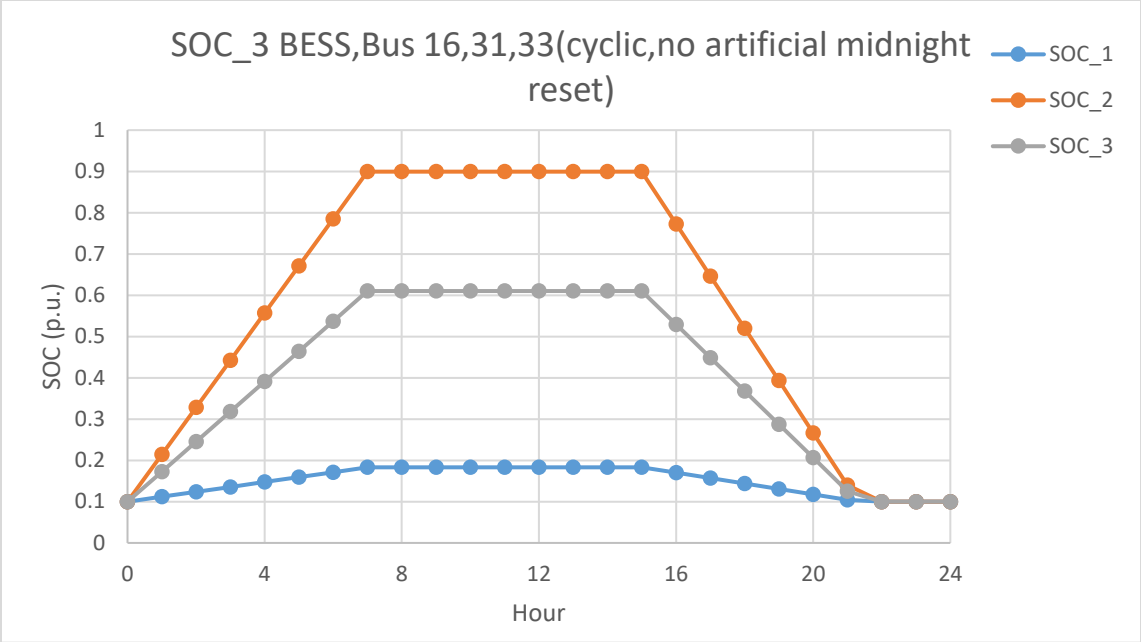


Figure 39: State of Charge of Bus 16, 31 and 33 When n=3

This graph illustrates the state of charge for three battery energy storage systems at buses 16, 12, and 33 over a 24-hour period, measured in per-unit values. All three batteries begin the day at a minimum charge level of 0.1 and experience a steady charging phase during the first 7 hours. The first battery, shown by the blue line, reaches a peak capacity of approximately 0.199, while the second and third batteries, reach almost 0.6 and 0.9 respectively. From hour 7 to hour 15, all systems remain in an idle state, holding their stored energy at a constant level without further charging or discharging. Starting at hour 15, the batteries begin to discharge their energy back into the grid, causing a steady decline in the charge levels until they all return to the baseline of 0.1 by hour twenty-two. For the final hours of the day, the charge remains flat at the minimum limit, completing a full cyclic operation that ensures the batteries end the day at the same state they started.

4.4.4 Four BESS units (n=4)

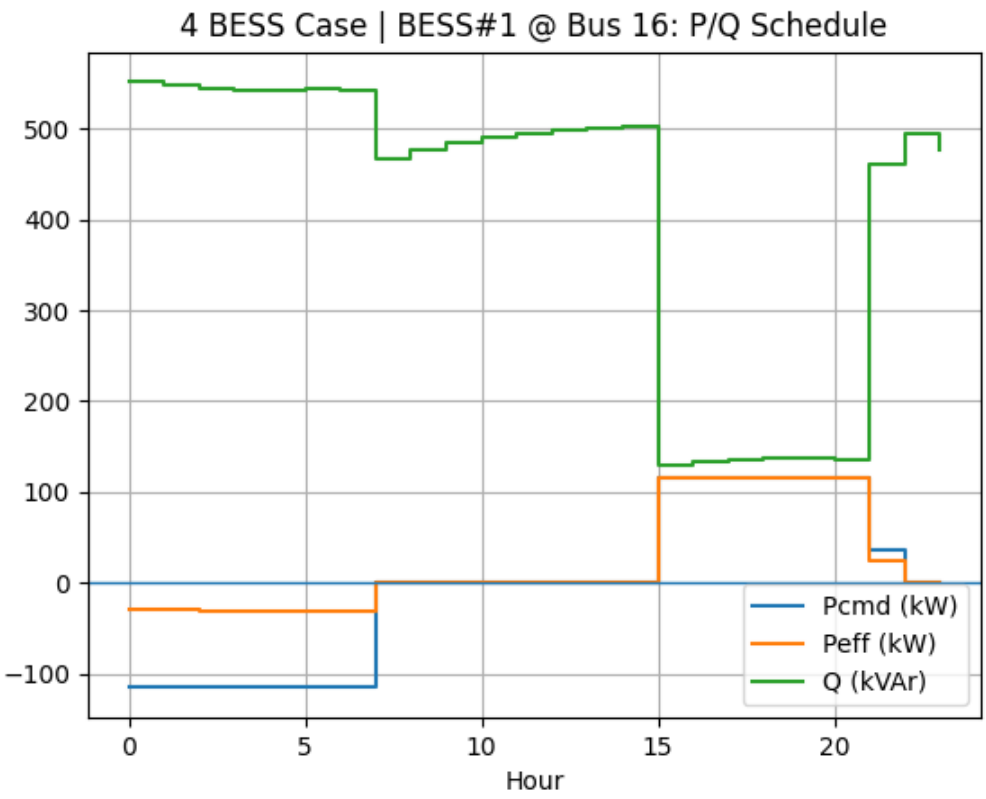


Figure 40: P/Q Schedule of Bus 16 When n=4

This graph shows the scheduled operation of a BESS at Bus 16 in a scenario involving four battery units. The blue line, showing a request for the battery to charge at -110 kW during the early morning (hours 0–7) and discharge at 110 kW during the evening (hours 15–21). The orange line perfectly meets the discharge command of 120 kW, but it is significantly limited during the morning charging phase, only reaching about -40 kW. This indicates a physical or system constraint that prevents the battery from charging at the requested rate despite being able to discharge at full capacity. The green line maintains a steady level around 550 kVAr for most of the day but drops significantly to about 490 kVAr during the peak discharge period, demonstrating how the system balances its voltage support with its active power output.

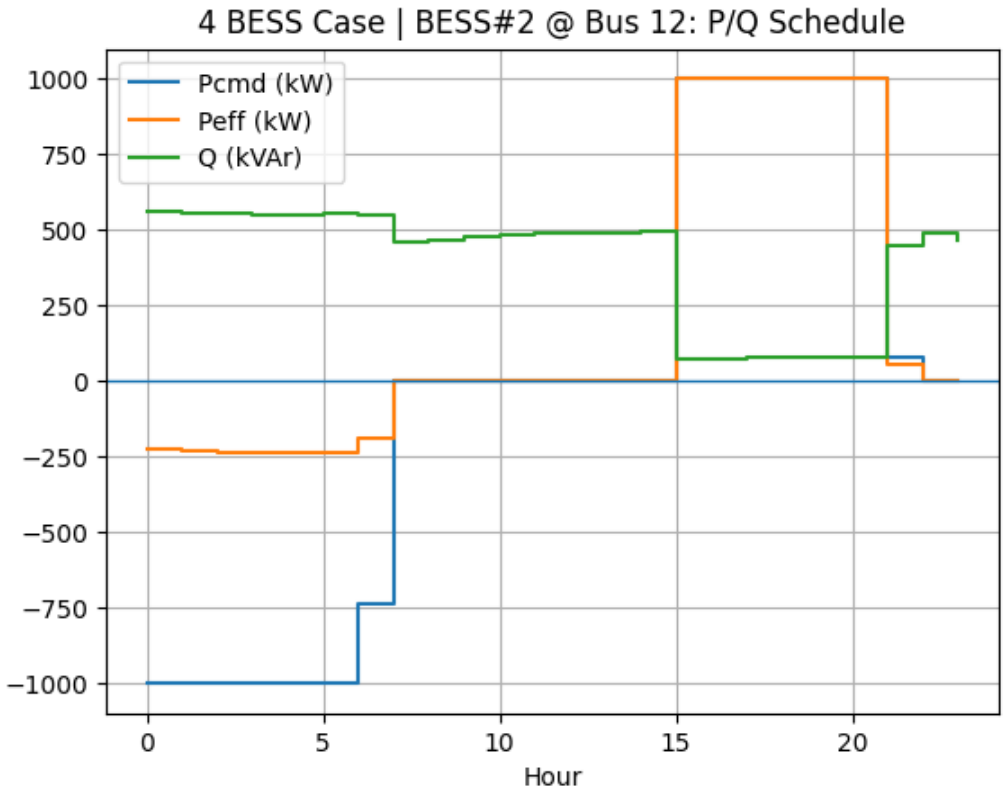


Figure 41: P/Q Schedule of Bus 12 When n=4

This figure shows the daily P/Q schedule of BESS at Bus 12 for four battery units. From hour 0–7, the battery charges with a commanded power of about –1000 kW, while the effective power is around –230 to –170 kW, and it provides high reactive power support of 500–520 kVAr. Between hours 7–15, the active power remains approximately 0 kW, and reactive power slightly decreases to 480–500 kVAr. During peak hours (15–21), the battery discharges with an effective power of about 1000 kW, while reactive power drops significantly to 40–60 kVAr. After hour 21, the discharge reduces to around 30–50 kW, and reactive power increases again to 440–500 kVAr, indicating a shift back toward voltage support.

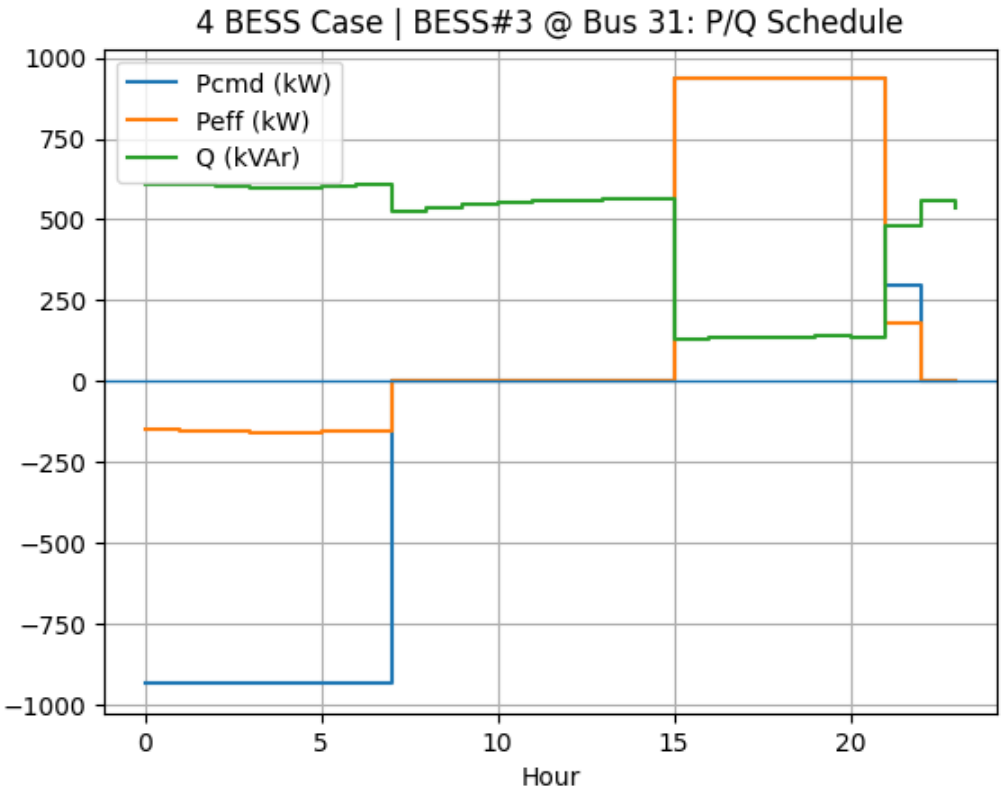


Figure 42: P/Q Schedule of Bus 31 When n=4

This figure presents the P/Q schedule of BESS-3 at Bus 31 over 24 hours. From hour 0–7, the battery charges with a commanded power of about –900 kW, while the effective charging power is around –150 to –200 kW, and it supplies 660–690 kVAr reactive power. Between hours 7–15, active power remains near 0 kW, and reactive power slightly decreases to 530–550 kVAr. During peak hours (15–21), the battery discharges at a high level of about 950 kW (Peff), while reactive power drops sharply and even become slightly negative at around 160 to 170 kVAr, indicating reactive power absorption. After hour 21, the discharge reduces to about 190–230 kW, and reactive power increases again to 490–540 kVAr, returning to voltage support mode.

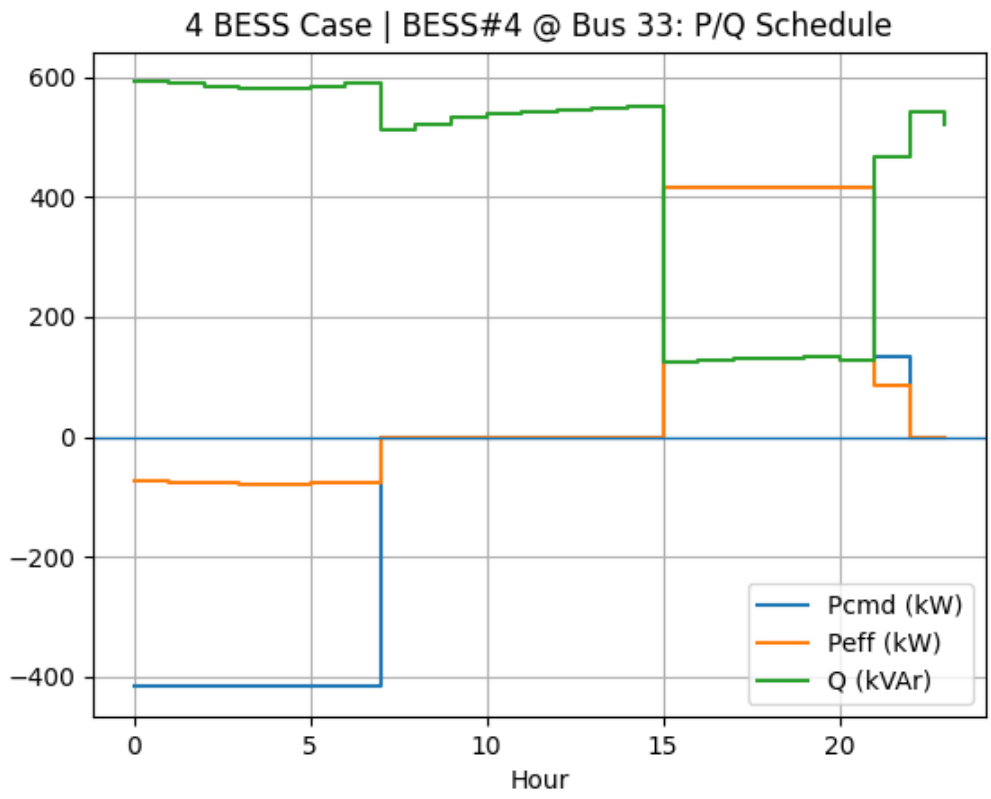


Figure 43: P/Q Schedule of Bus 33 When n=4

This graph illustrates the operational schedule for the fourth battery unit at bus 33 within a system of four battery energy storage systems over a 24-hour period. The blue line represents the commanded active power, which requests the battery to charge at -410 kW during the first seven hours and discharge at 410 kW between hours 15 and 21. The orange line shows the actual effective power delivered; while it perfectly achieves the 410 kW target during the evening discharge phase, it is significantly limited during the morning charging phase, only reaching about -60 kW. The green line tracks the reactive power in kVAr, which maintains a positive level near 600 units for the first part of the day.

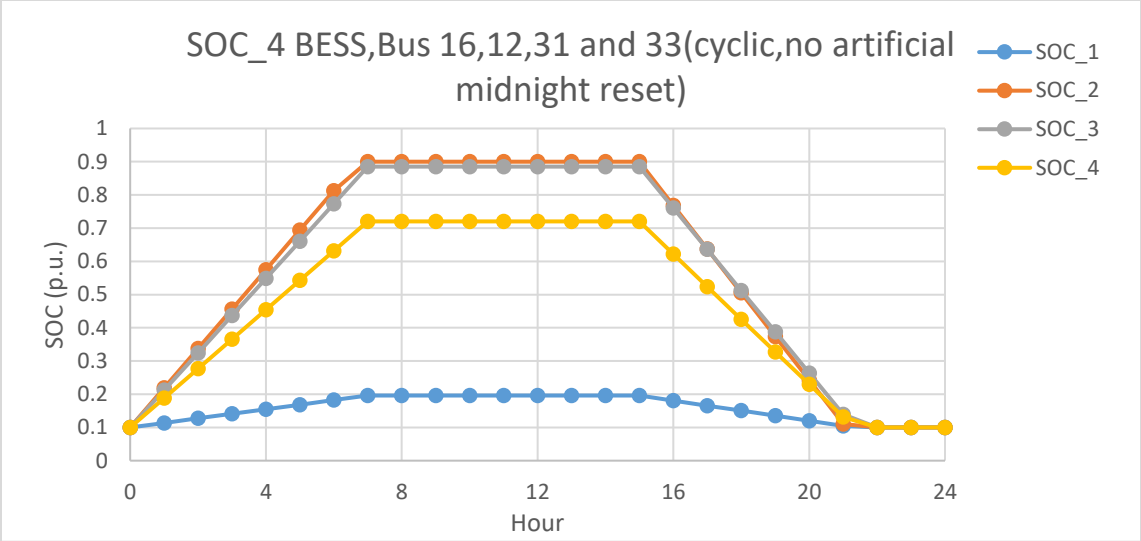


Figure 44: State of Charge of Bus 16, 12, 31 and 33 When n=4

This graph presents the state of charge for four battery energy storage systems located at buses 33, 8, 12, and 15 over a 24-hour cycle. Each battery starts the day at a minimum charge level of 0.1 per unit. During the first 7 hours, all four batteries undergo a steady charging process as they pull energy from the grid. The first three batteries, represented by the yellow, orange, and grey lines, reach a peak capacity of about 0.86, while the fourth battery, shown in blue, reaches a slightly lower peak of roughly nearly 0.2. Between 7 and 15 hours, the charge levels remain flat, indicating an idle period where the stored energy is held in reserve. Starting at hour 15, the batteries begin to discharge their power back into the system, causing the charge levels to drop until they return to the initial 0.1 baseline by hour 22. The batteries remain at this minimum level for the rest of the night, completing a full operational cycle that ensures the system starts and ends the day in the same state.

4.4.5 Overall Comparative Analysis in BESS Configuration

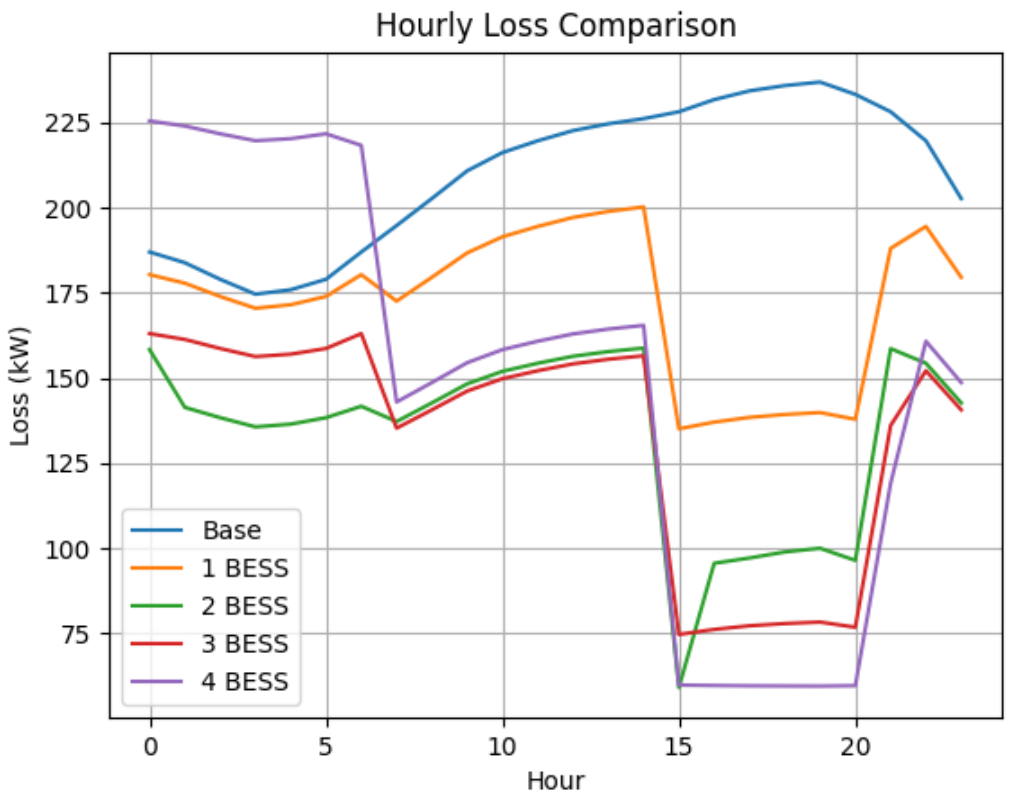


Figure 45: Comparative Analysis of Hourly Losses in BESS Configuration

This graph compares the hourly power losses in a grid system across five different scenarios, starting with a base case and then adding between one and four battery energy storage systems. The blue line represents the base case without any batteries, showing a steady loss that gradually increases and peaks around hour 19. During the first seven hours of the day, the scenarios with three and four batteries show much higher losses than the base case, which occurs because the batteries are actively charging and increasing the total load on the system. However, a major shift happens between 15 and 21 hours, where the cases with more batteries show a significant drop in losses compared to the base case. This improvement happens because the batteries are discharging their stored energy during peak times, which reduces the amount of power that needs to be transported across the grid. Overall, while adding more batteries increases losses during the charging phase, it successfully provides a large reduction in losses during the peak evening hours.

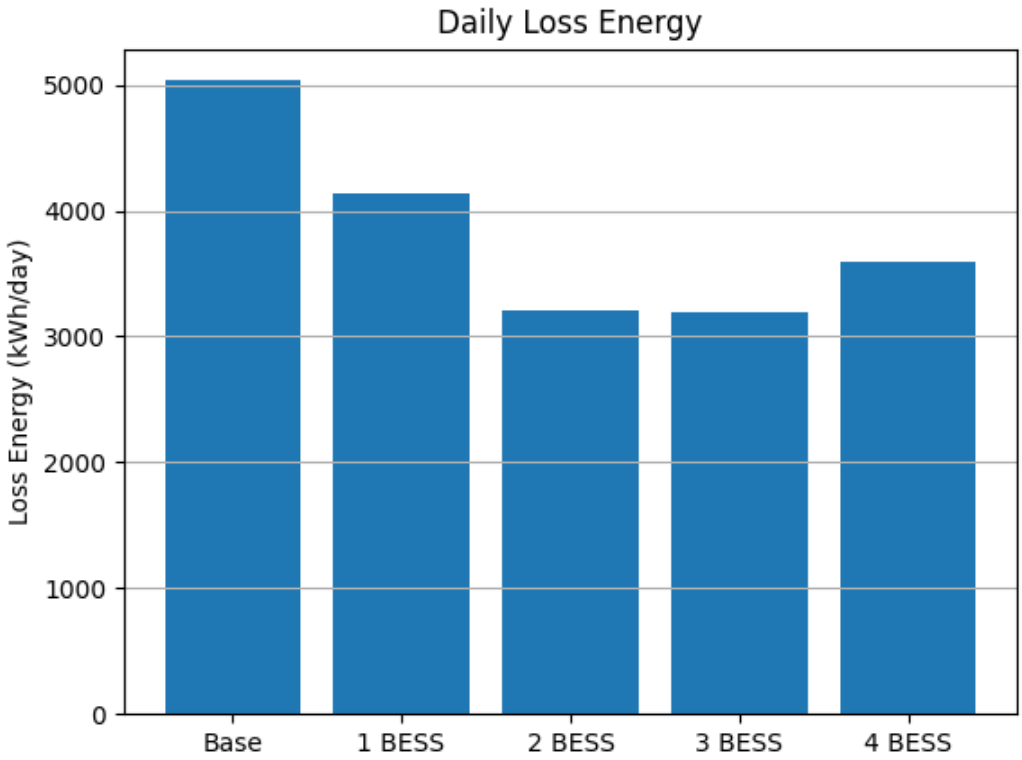


Figure 46: Comparative Analysis of Daily Loss Energy in BESS Configuration

This chart compares the total daily energy losses in a grid system across five different configurations, measured in kilowatt-hours per day. The leftmost bar represents the base case without any battery storage, which shows the highest energy loss at approximately 5100 kWh per day. When one battery energy storage system is added, the losses drop significantly to around 4150 kWh. The most efficient scenario is the two-battery case, which achieves the minimum system losses at roughly 3220 kWh per day. However, the graph shows a trend of diminishing returns as more units are added; with three batteries, the losses decrease to about 3100 kWh, and with four batteries, the losses increase back to nearly 3500 kWh. This indicates that while adding some battery storage is highly beneficial for reducing system inefficiency, there is an optimal limit, after which the extra equipment increases the total energy lost due to the power required for charging and operation.

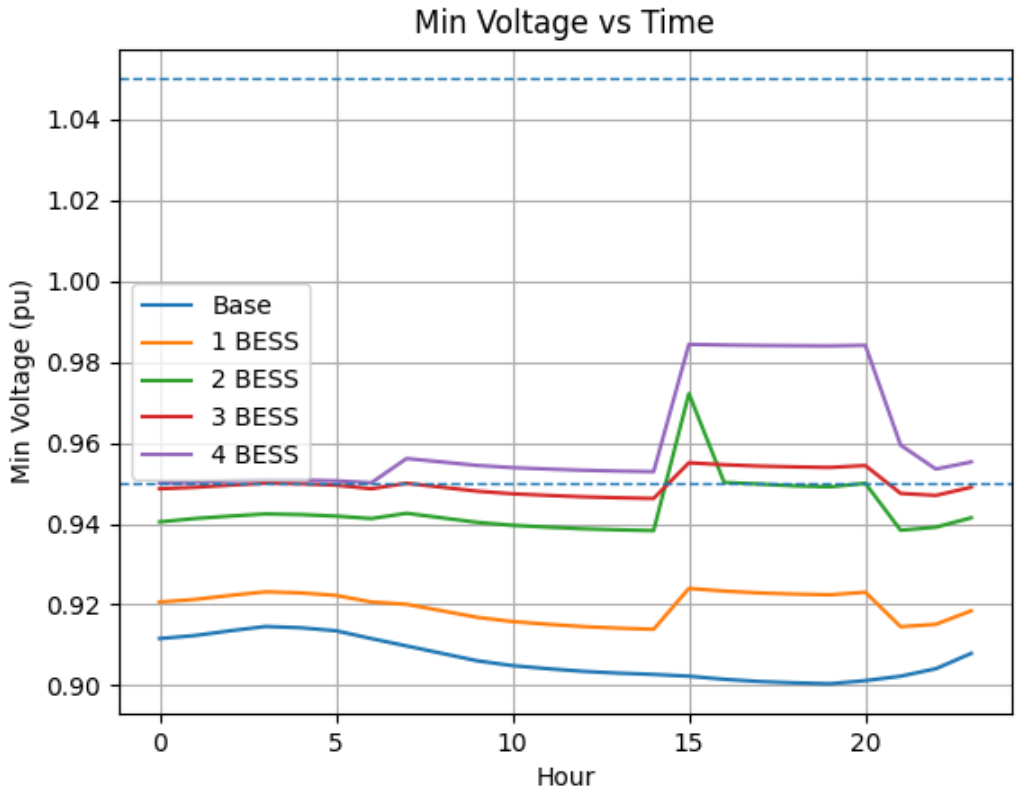


Figure 47: Comparative Analysis of Min Voltage vs Time

This graph shows the minimum voltage of the grid over a 24-hour period across five different scenarios, comparing a base case to systems with one to four batteries. The vertical axis measures voltage in per-unit values, where the horizontal dashed lines at 0.95 and 1.05 represent the standard operating safety limits for the system. The blue line representing the base case shows the lowest voltage levels, staying significantly below the 0.95 threshold for the entire day, which indicates poor grid stability without any support. As more batteries are added, the minimum voltage across the system increases, with the purple line for the four-battery scenario showing the highest overall improvement. A major jump in voltage occurs between 15 hours and 21 when the batteries are discharging energy; during this peak time, the three and four battery scenarios successfully raise the minimum voltage above the 0.95 safety limit. This demonstrates that adding more battery units effectively strengthens the grid by preventing deep voltage drops, particularly during high-demand periods.

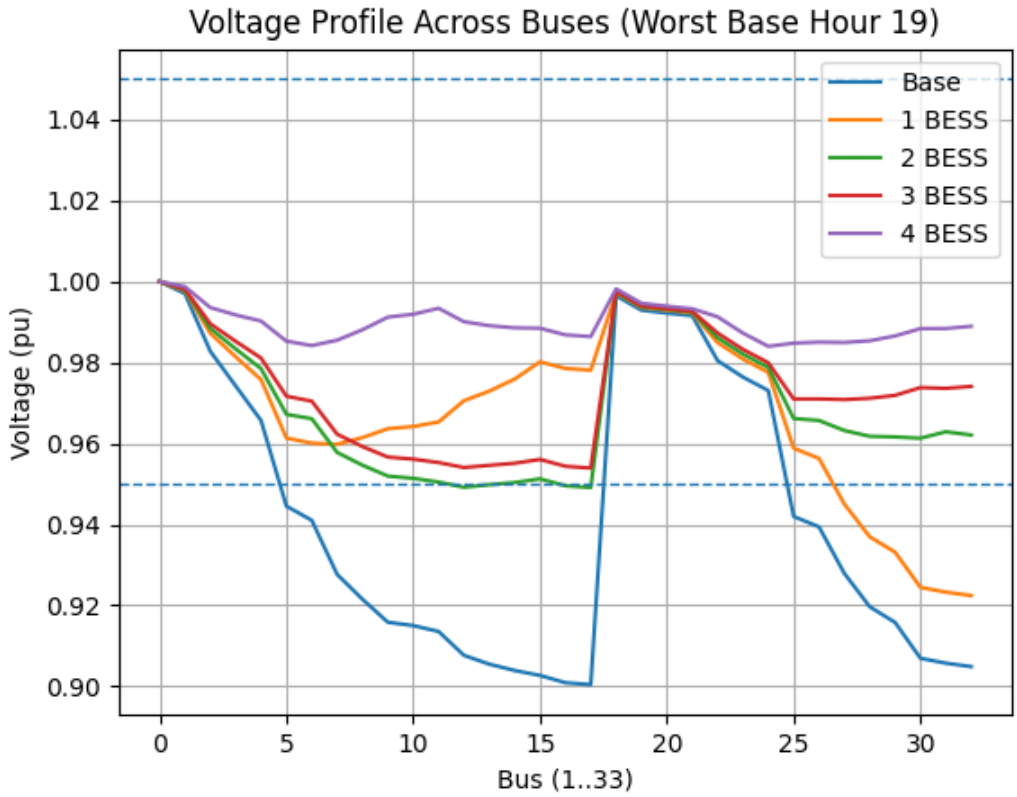


Figure 48: Comparative Analysis of Voltage Profile Across Buses (Worst Base Hour 19)

This graph shows the voltage profile across 33 different buses in a power grid during hour 19, which is the highest stress for the system. The vertical axis measures voltage in per-unit values, and the dashed horizontal line at 0.95 represents the lower safety limit for stable operation. In the base case, the voltage drops sharply and remains significantly below the safety threshold for many buses, reaching its lowest point between buses 10 and 17. As battery energy storage systems are added to the grid, represented by the other colored lines, the voltage profile visibly improves and moves closer to the ideal 1.0 level. The scenarios with three and four batteries, shown in red and purple, are the most effective at correcting these drops, successfully bringing the voltage of almost every bus back above the 0.95 limit. Overall, the graph demonstrates that increasing the number of batteries in the system

helps maintain a much more stable and safe voltage level across the entire network during peak demand periods.

4.5 Comparison Table

Table 2: Comparative analysis table in BESS configuration with base case

Parameters	Base case(No BESS)	1 BESS location	2 BESS location	3 BESS location	4 BESS location
Optimal BESS location	No	16	32 ,16	16,31,33	16,12,31,33
Loss(kWh/day)	5035.2939	4140.0929	3201.4468	3199.189	3596.5
Vmin(pu)	0.900435894	0.913894467	0.938288746	0.946249287	0.950001853
Vmax(pu)	1	1	1	1	1
Peak branch loading (pu)	0.849294542	0.764888919	0.716230547	0.70015684	0.741353921
Battery size(kwh)	-	8000	8000	8000	8000
			1000	8000	8000
				1406.46	7915
					4475
Payback period (yr)	-	0.9777	1.0295	1.9415	1.5596
Loss Reduction %	-	17.77	36.41	36.46	28.57

CHAPTER FIVE: CONCLUSION

The main objective of this project is to improve power quality in the distribution feeder by optimally locating Battery Energy Storage Systems (BESS) for enhanced voltage regulation, reduced power losses, and increased system efficiency. The optimal location and sizing of the BESS, determined by using an optimization technique such as Particle Swarm Optimization (PSO), play a crucial role in achieving maximum benefits. The results confirm that the strategic placement of BESS significantly reduces real power losses, enhances voltage profiles, and mitigates voltage fluctuations within the network. Additionally, the ability of the BESS to operate in both charging and discharging modes allows it to effectively manage peak load conditions and maintain system stability under varying operating scenarios. The integration of BESS in distribution feeders proves to be a reliable and efficient solution for addressing power quality issues and supporting the growing demand for stable and sustainable power systems. The integration of BESS in distribution feeders proves to be a reliable and efficient solution for addressing power quality issues and supporting the growing demand for stable and sustainable power systems. Initially the simulation of BESS was done by using IEEE-33 bus system by implementing hit and trial methods. The hit and trial methods were practically not possible, so particle swarm optimization algorithm was carried out in python. The BESS was placed on different buses and the simulation was done. From the simulation following conclusions are drawn:

1. The voltage profile remains within acceptable operating limits even after BESS integration. This ensures stable and secure system operation without causing voltage violations.

2. Peak branch loading ratio decreases as the number of battery energy storage systems increases, which enhances the system's reliability.

3. Simulation results demonstrate that properly placed and sized BESS can significantly reduce losses, improve voltage regulation and stability.

4. The study confirms that optimal multi-location placement of BESS enhances efficiency, reduces energy losses, improves voltage stability, decreases line loading, and strengthens the reliability of the distribution system.

CHAPTER SIX: RECOMMENDATION

The future implementation should focus on incorporating stochastic modeling to capture the uncertainty of solar generation and EV charging. It should also explore hybrid optimization methods (e.g., PSO-GWO) to improve convergence and solution quality. Expanding the study to looped and multi-feeder networks will help assess scalability in more complex systems. Additionally, integrating battery degradation and State of Health (SOH) into the objective function is essential for realistic long-term analysis. Finally, Hardware-in-the-Loop (HIL) testing should be conducted to validate performance under real-world conditions, including thermal effects and communication delays.

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ANNEX

Load Name	Phases	B1	Voltage (Kv)	Model	Active Power (Kw)	Reactive Power(Kvar)	Vmin(pu)	Vmax(pu)
Load 1	3	2	13.8	1	100	60	0.01	1.50
Load 2	3	3	13.8	1	90	40	0.01	1.50
Load 3	3	4	13.8	1	120	80	0.01	1.50
Load 4	3	5	13.8	1	60	30	0.01	1.50
Load 5	3	6	13.8	1	60	20	0.01	1.50
Load 6	3	7	13.8	1	200	100	0.01	1.50
Load 7	3	8	13.8	1	200	100	0.01	1.50
Load 8	3	9	13.8	1	60	20	0.01	1.50
Load 9	3	10	13.8	1	60	20	0.01	1.50
Load 10	3	11	13.8	1	45	30	0.01	1.50
Load 11	3	12	13.8	1	60	35	0.01	1.50
Load 12	3	13	13.8	1	60	35	0.01	1.50
Load 13	3	14	13.8	1	60	80	0.01	1.50
Load 14	3	15	13.8	1	120	10	0.01	1.50
Load 15	3	16	13.8	1	60	20	0.01	1.50
Load16	3	17	13.8	1	60	20	0.01	1.50
Load 17	3	18	13.8	1	60	40	0.01	1.50
Load 18	3	19	13.8	1	90	40	0.01	1.50
Load 19	3	20	13.8	1	90	40	0.01	1.50
Load 20	3	21	13.8	1	90	40	0.01	1.50

Load 21	3	22	13.8	1	90	40	0.01	1.50
Load 22	3	23	13.8	1	90	40	0.01	1.50
Load 23	3	24	13.8	1	90	50	0.01	1.50
Load 24	3	25	13.8	1	420	200	0.01	1.50
Load 25	3	26	13.8	1	420	200	0.01	1.50
Load 26	3	27	13.8	1	60	25	0.01	1.50
Load 27	3	28	13.8	1	60	25	0.01	1.50
Load 28	3	29	13.8	1	60	20	0.01	1.50
Load 29	3	30	13.8	1	120	70	0.01	1.50
Load 30	3	31	13.8	1	200	600	0.01	1.50
Load 31	3	32	13.8	1	150	70	0.01	1.50
Load 32	3	33	13.8	1	210	100	0.01	1.50

Line Name	B1	B2	Length	Units	R(Ω)	X(Ω)	Current Capacity (A)
L1	1	2	1	km	0.092	0.048	500
L2	2	3	1	km	0.493	0.251	500
L3	3	4	1	km	0.366	0.186	500
L4	4	5	1	km	0.381	0.194	500
L5	5	6	1	km	0.819	0.707	500
L6	6	7	1	km	0.187	0.619	500
L7	7	8	1	km	1.711	1.235	500
L8	8	9	1	km	1.03	0.74	500
L9	9	10	1	km	1.04	0.74	500

L10	10	11	1	km	0.197	0.065	500
L11	11	12	1	km	0.374	0.124	500
L12	12	13	1	km	1.468	1.155	500
L13	13	14	1	km	0.542	0.713	500
L14	14	15	1	km	0.591	0.526	500
L15	15	16	1	km	0.746	0.545	500
L16	16	17	1	km	1.289	1.721	500
L17	17	18	1	km	0.732	0.574	500
L18	2	19	1	km	0.164	0.157	500
L19	19	20	1	km	1.504	1.355	500
L20	20	21	1	km	0.41	0.478	500
L21	21	22	1	km	0.709	0.937	500
L22	3	23	1	km	0.451	0.308	500
L23	23	24	1	km	0.898	0.709	500
L24	24	25	1	km	0.896	0.701	500
L25	6	26	1	km	0.203	0.104	500
L26	26	27	1	km	0.284	0.145	500
L27	27	28	1	km	1.059	0.934	500
L28	28	29	1	km	0.804	0.701	500
L29	29	30	1	km	0.508	0.259	500
L30	30	31	1	km	0.974	0.963	500
L31	31	32	1	km	0.311	0.362	500
L32	32	33	1	km	0.341	0.53	500

Transformer name	Bus 1	Bus 2	Phases	kV 1	kV 2	kVA	%R	%X	Hconn	Lvconn	X0	R0
Substation tr	0	1	3	132	13.8	3000	1.1	1.72	Delta	Wye	9.16	0